Preface

1. The bar construction on algebraic curves

The bar construction of a graded algebra is classical: it lives on the cofree tensor coalgebra $T^c(s^{-1}\bar{A})$ and its differential encodes the multiplication. Three extensions of this classical theory are the subject of this monograph. First, the multiplication of a chiral algebra on a curve X is mediated by a logarithmic propagator $d\log(z_i - z_j)$, and the bar differential becomes an integral transform on the Fulton–MacPherson compactification of configuration spaces. Second, the bar coalgebra carries two coproducts: one cocommutative (recording unordered collisions on $\mathrm{Ran}(X)$, producing the factorization coalgebra $\bar{B}^{\Sigma}(\mathcal{A})$) and one non-cocommutative (recording ordered collisions along $\mathrm{Conf}_n^{<}(\mathbb{R})$, producing the tensor coalgebra $\bar{B}^{\mathrm{ord}}(\mathcal{A})$). The non-cocommutative coproduct is the level-2 ordered source, and its Koszul dual is a Yangian. Third, on curves of genus $g \geq 1$ the propagator deforms via the prime form, the bar differential acquires curvature $\kappa(\mathcal{A}) \cdot \omega_g$ from the Hodge bundle, and the universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ of the modular convolution algebra controls the genus tower.

The 1-form

Let X be a smooth algebraic curve over $\mathbf{C}$. On the configuration space $C_2(X) = X \times X \setminus \Delta$, define the logarithmic 1-form

$$\eta_{12} \;=\; d\log(z_1 - z_2) \;=\; \frac{dz_1 - dz_2}{z_1 - z_2}\,.$$

This form has a simple pole along the diagonal $\Delta \subset X \times X$ with residue 1. The residue is intrinsic: it depends on no coordinate, no metric, no level. A double pole $dz/(z-w)^2$ transforms under coordinate change and has no coordinate-independent residue. A regular 1-form on $X \times X$ has residue zero along Δ and extracts nothing from the collision. The logarithmic derivative is the unique 1-form on $C_2(X)$ with a first-order pole along Δ and conformally invariant residue.

On $C_3(X)$, the three pullbacks $\eta_{12}, \eta_{23}, \eta_{31}$ satisfy the *Arnold relation*

$$\eta_{12} \wedge \eta_{23} \;+\; \eta_{23} \wedge \eta_{31} \;+\; \eta_{31} \wedge \eta_{12} \;=\; 0\,. \tag{0.0.1}$$

The identity restates, in differential forms, the partial-fractions relation

$$\frac{1}{(z_1 - z_2)(z_2 - z_3)} \;+\; \frac{1}{(z_2 - z_3)(z_3 - z_1)} \;+\; \frac{1}{(z_3 - z_1)(z_1 - z_2)} \;=\; 0\,,$$

which holds because the left side, viewed as a rational function of z_1 with z_2, z_3 fixed, has simple poles at z_2 and z_3 whose residues sum to zero, hence vanishes. Arnold proved that (0.0.1), replicated across all triples $\{i, j, k\} \subset \{1, \ldots, n\}$, generates the *complete* ideal of relations in $H^*(\mathrm{Conf}_n(\mathbf{C}), \mathbf{C})$.

Under a coordinate change $z \mapsto w(z)$, the propagator transforms as $\eta_{ij} \mapsto \eta_{ij} + d \log \frac{w(z_i)-w(z_j)}{z_i - z_j}$. Near the diagonal this becomes $d \log w'(z_i)$: the standard cocycle for $\mathrm{Diff}(X)$. The propagator is a BRST ghost for diffeomorphisms.

Chiral algebras on a curve

Let X be a smooth algebraic curve over $\mathbb{C}$. Write $\mathcal{D}_X$ for the sheaf of differential operators on X; a $\mathcal{D}_X$-module is a sheaf on which $\mathcal{D}_X$ acts, equivalently a sheaf with a flat connection. A *chiral algebra* $\mathcal{A}$ on X is a $\mathcal{D}_X$-module equipped with a *chiral bracket*

$$\mu^{\mathrm{ch}} \; : \; j_* j^*(\mathcal{A} \boxtimes \mathcal{A}) \;\longrightarrow\; \Delta_!(\mathcal{A}),$$

where $j \colon X \times X \setminus \Delta \hookrightarrow X \times X$ is the inclusion of the complement of the diagonal and $\Delta_! \mathcal{A}$ is the !-pushforward along the diagonal embedding $\Delta \colon X \hookrightarrow X \times X$ (the $\mathcal{D}$-module direct image; concretely, $\Delta_! \mathcal{A} \cong \mathcal{A} \otimes \delta_\Delta$, sections of $\mathcal{A}$ supported on the diagonal via the δ-function). The source $j_* j^*(\mathcal{A} \boxtimes \mathcal{A})$ is the $\mathcal{D}$-module of sections of the external product $\mathcal{A} \boxtimes \mathcal{A}$ with poles of arbitrary order along Δ; the chiral bracket extracts the singular part and projects it onto the diagonal.

In local coordinates, the chiral bracket encodes the *operator product expansion*: for sections $a, b \in \mathcal{A}$,

$$a(z)\, b(w) \;\sim\; \sum_{n \geq 0} \frac{a_{(n)} b(w)}{(z-w)^{n+1}} \, .$$

The n-th products $a_{(n)} b \in \mathcal{A}$ are the *Fourier modes* of the bracket, extracted by the residue formula $a_{(n)} b = \mathrm{Res}_{z=w}[a(z)\, b(w)\, (z-w)^n]$. The zeroth product $a_{(0)} b$ is the Lie-type bracket (first-order pole); the first product $a_{(1)} b$ contributes the second-order pole, carrying the symmetric/metric data. All higher products ($n \geq 2$) are determined by the $\mathcal{D}_X$-module structure, i.e. by translation covariance.

The *Borcherds identity* couples all pole orders into a single algebraic constraint:

$$\boxed{\sum_{j \geq 0} \binom{m}{j} (a_{(n+j)} b)_{(m+k-j)} c \;=\; \sum_{j \geq 0} (-1)^j \binom{n}{j} \Big(a_{(m+n-j)} (b_{(k+j)} c) - (-1)^n b_{(n+k-j)} (a_{(m+j)} c) \Big).}$$

At $n = 0$: the Jacobi identity for the Lie bracket $a_{(0)}$. At $m = 0$: the commutator formula. At $n = 1$, $m = k = 0$: the derivation property of $a_{(0)}$ with respect to $a_{(1)} b$. The Borcherds identity is the complete axiom system: locality, Jacobi, and associativity in a single formula.

The bar complex on X

Place sections $a_1, \ldots, a_n \in \mathcal{A}$ at distinct points $z_1, \ldots, z_n \in X$. Assign to each ordered pair (i, j) the logarithmic 1-form

$$\eta_{ij} \;:=\; d \log(z_i - z_j) \;=\; \frac{dz_i - dz_j}{z_i - z_j} \, .$$

For a spanning tree T of the complete graph on $\{1, \ldots, n\}$, with edge set $E(T)$, form the decorated tensor

$$\omega_T(a_1, \ldots, a_n) \;:=\; a_1(z_1) \otimes \cdots \otimes a_n(z_n) \;\otimes \bigwedge_{(i,j)\in E(T)} \eta_{ij}\,.$$

The form $\bigwedge_{(i,j)\in E(T)} \eta_{ij}$ is a section of $\Omega^{n-1}_{\log}$ on the configuration space $\mathrm{Conf}_n(X)$; its degree is $|E(T)| = n - 1$ since T is a tree.

The Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n(X)$ resolves $\mathrm{Conf}_n(X)$ by iterated real oriented blowup along all diagonal strata. This is a *blowup*, not the complement of the diagonal: points of $\overline{\mathrm{FM}}_n(X) \setminus \mathrm{Conf}_n(X)$ record the limiting direction and relative scales of approach as points collide. The boundary is a manifold with corners; the codimension-one faces are indexed by subsets $S \subseteq \{1, \ldots, n\}$ with $|S| \geq 2$, recording which points have collided.

The *chiral bar complex* on X:

$$\bar{B}_X(\mathcal{A}) \;=\; \bigoplus_{n\geq 1} (s^{-1}\bar{\mathcal{A}})^{\otimes n}_{\Sigma_n} \;\otimes\; \Gamma\big(\overline{\mathrm{FM}}_n(X),\; \Omega^{n-1}_{\log}\big).$$

Here $\bar{\mathcal{A}} = \ker(\varepsilon\colon \mathcal{A} \to \omega_X)$ is the augmentation ideal (the kernel of the vacuum map), and s^{-1} denotes *desuspension*: the shift $|s^{-1}a| = |a| - 1$ in cohomological degree. Throughout this monograph, cohomological conventions are used: $|d| = +1$. The result is a graded coalgebra: the *deconcatenation coproduct*

$$\Delta[s^{-1}a_1|\cdots|s^{-1}a_n] \;=\; \sum_{i=0}^{n} [s^{-1}a_1|\cdots|s^{-1}a_i] \;\otimes\; [s^{-1}a_{i+1}|\cdots|s^{-1}a_n]$$

splits an ordered sequence at every position. It is equipped with a differential $d_{\bar{B}}$ that decomposes into three pieces:

$$d_{\bar{B}} \;=\; d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}}\,.$$

The three components:

- d_{int}: the internal differential of $\mathcal{A}$ (the translation operator, acting on each tensor factor).
- d_{res}: the *residue differential*. For a codimension-one boundary face D_S of $\overline{\mathrm{FM}}_n(X)$ corresponding to the collision $\{z_i\}_{i\in S} \to z_0$, the residue along D_S extracts the OPE coefficient: the iterated contraction $a_{i_1(n_1)}(a_{i_2(n_2)}(\cdots))$ at the pole orders dictated by the form $\bigwedge \eta_{ij}$. In formulas: for $S = \{i, j\}$,

  $$d^{(ij)}_{\mathrm{res}}\big(a_1 \otimes \cdots \otimes a_n \otimes \eta_{ij} \wedge \omega'\big) \;=\; \pm\, a_1 \otimes \cdots \otimes \widehat{a_i} \otimes \cdots \otimes \widehat{a_j} \otimes \cdots \otimes a_{i(p)}a_j \;\otimes\; \omega',$$

  where p is the pole order read from the form-theoretic coefficient, and hats denote omission.
- d_{dR}: the de Rham differential on the logarithmic forms on $\overline{\mathrm{FM}}_n(X)$.

Does $d^2_{\bar{B}} = 0$?

The proof that $d^2_{\bar{B}} = 0$

Expand $(d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}})^2$ into nine terms. The diagonal terms: $d^2_{\mathrm{int}} = 0$ (since $\mathcal{A}$ is a cochain complex), $d^2_{\mathrm{dR}} = 0$ (de Rham), and d^2_{res} must be controlled. The cross-terms $d_{\mathrm{int}}\,d_{\mathrm{dR}} + d_{\mathrm{dR}}\,d_{\mathrm{int}} = 0$ by the Leibniz rule. The remaining five terms pair into three cancellation mechanisms, governed by the combinatorics of index overlap:

(a) *Disjoint supports.* When $\{i,j\} \cap \{k,l\} = \emptyset$, the boundary faces $D_{\{i,j\}}$ and $D_{\{k,l\}}$ are transverse in $\overline{\mathrm{FM}}_n(X)$. The corresponding residue operators commute, so the cross-terms cancel: $d_{\mathrm{res}}^{(ij)}\, d_{\mathrm{res}}^{(kl)} + d_{\mathrm{res}}^{(kl)}\, d_{\mathrm{res}}^{(ij)} = 0$. Operadic associativity of disjoint contractions.

(b) *Shared index.* When $\{i,j\} \cap \{k,l\} = \{j = k\}$, the triple collision $z_i = z_j = z_l$ lies in a codimension-two stratum. The form-theoretic contribution involves $\eta_{ij} \wedge \eta_{jl}$. The *Arnold relation*

$$\eta_{ij} \wedge \eta_{jl} + \eta_{jl} \wedge \eta_{li} + \eta_{li} \wedge \eta_{ij} \;=\; 0$$

identifies this with a cyclic sum over the three ordered approaches to the triple-collision point. On the algebraic side, the three terms produce $(a_{i(m)}a_j)_{(n)}a_l$, $(a_{j(m)}a_l)_{(n)}a_i$, $(a_{l(m)}a_i)_{(n)}a_j$ (with appropriate pole orders), whose sum vanishes by the Borcherds identity. The Arnold relation kills the form; the Borcherds identity kills the algebra. Both are needed, and both are consumed.

(c) *Same pair.* $d_{\mathrm{res}}^{(ij)}\, d_{\mathrm{res}}^{(ij)} = 0$ by the sign rule on desuspensions: extracting a residue twice at the same diagonal is zero for degree reasons.

The result: $d_{\bar{B}}^2 = 0$ if and only if the n-th products satisfy the Borcherds identity. The bar construction accepts exactly chiral algebras.

All of this is genus 0. At genus $g \geq 1$, the propagator acquires quasi-periodicity corrections and $d_{\mathrm{fib}}^2 \neq 0$: the curvature $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ produces a single scalar $\kappa(\mathcal{A})$ controlling the genus expansion. Genus 0 is uncurved algebra; genus ≥ 1 is curved geometry.

The categorical logarithm

The bar construction is a categorical logarithm: it sends tensor products to direct sums.

$$B(\mathcal{A} \otimes \mathcal{A}') \;\simeq\; B(\mathcal{A}) \oplus B(\mathcal{A}').$$

The logarithmic form $\eta_{ij} = d\log(z_i - z_j)$ is the integral kernel. The Arnold relation

$$\eta_{ij} \wedge \eta_{jk} + \eta_{jk} \wedge \eta_{ki} + \eta_{ki} \wedge \eta_{ij} \;=\; 0$$

is the cocycle condition for log: the Arnold relations make $d\log$ well-defined on products.

The cobar functor $\Omega(\bar{B}(\mathcal{A}))$ is the exponential. The domain on which the exponential inverts the logarithm, the *radius of convergence*, is the Koszul locus: the full subcategory of chiral algebras for which

$$\varepsilon\colon \Omega(\bar{B}(\mathcal{A})) \;\xrightarrow{\sim}\; \mathcal{A}.$$

Outside the Koszul locus, the cobar-bar resolution is infinite: higher homotopies measure the failure of convergence.

The cobar functor recovers the original algebra $\mathcal{A}$ from its bar complex. A different operation, linear duality of bar cohomology, produces a different algebra: the Koszul dual $\mathcal{A}^!$.

Two duals of the bar complex

The *Koszul dual algebra* $\mathcal{A}^!$ is obtained by a two-step algebraic procedure. First, take bar cohomology:

$$\mathcal{A}^{\text{¡}} \;:=\; H^*\big(B(\mathcal{A})\big),$$

a graded coalgebra (the coproduct of $B(\mathcal{A})$ descends to cohomology). Second, apply graded linear duality:

$$\mathcal{A}^! \;:=\; \left(\mathcal{A}^{\mathrm{i}}\right)^{\vee}.$$

The result is the Koszul dual algebra, not the cobar. $\Omega(B(\mathcal{A})) \simeq \mathcal{A}$ is bar-cobar *inversion*, recovering the original algebra; $\mathcal{A}^!$ is obtained by *linear duality of bar cohomology*, producing a different algebra except when $\mathcal{A} \cong \mathcal{A}^!$ under Koszul duality (e.g. $\mathrm{Vir}_{c=13}$).

A separate, geometric operation also acts on the bar coalgebra: Verdier duality on $\mathrm{Ran}(X)$ sends $\bar{B}_X(\mathcal{A})$ to the *homotopy* Koszul dual $\mathcal{A}^!_\infty$:

$$\mathbb{D}_{\mathrm{Ran}}\, \bar{B}_X(\mathcal{A}) \;\simeq\; \mathcal{A}^!_\infty.$$

Verdier duality converts the coalgebra structure of $\bar{B}_X(\mathcal{A})$ into an algebra structure, so $\mathcal{A}^!_\infty$ is a factorization *algebra*. It is richer than $\mathcal{A}^!$: it retains full chain-level data, while $\mathcal{A}^!$ remembers only cohomology. That the cohomology of $\mathcal{A}^!_\infty$ recovers $\mathcal{A}^!$ on the Koszul locus is the content of Theorem A (the Verdier intertwining theorem), not a tautology.

When $\mathcal{A}$ is chirally Koszul, $B(\mathcal{A})$ has cohomology concentrated in a single row, $\mathcal{A}^{\mathrm{i}}$ is a coalgebra with a unique grading, and $\mathcal{A}^!$ governs a linear (quadratic) resolution of $\mathcal{A}$. The bar-to-cobar spectral sequence collapses at E_2. Classical Koszul duality (Priddy, Beilinson–Ginzburg–Soergel, Loday–Vallette) embeds via $X = \mathbb{A}^1$: the affine line is contractible, so $\overline{C}_n(\mathbb{A}^1)$ deformation-retracts onto a point and the FM chain cooperad becomes quasi-isomorphic to the associative cooperad. The retraction is additional data (a homotopy equivalence, not a canonical identification), and the resulting bar construction recovers the classical bar complex of the underlying Lie algebra only after specifying this transfer. On $\mathbb{P}^1$ the global topology (Fulton–MacPherson compactification, Arnold relations in $H^*(\overline{C}_n)$) has no counterpart over a bare point.

The Koszul dual $\mathcal{A}^!$ arises from the holomorphic (symmetric) bar complex. The bar construction has a second, topological (ordered) wing, producing a second Koszul dual.

Two Koszul dualities

The bar complex of a chiral algebra $\mathcal{A}$ carries two operadic colours, and each colour has its own Koszul duality.

The ordered bar before averaging. The classical bar construction of Stasheff assigns to any A_∞-algebra the cofree *tensor* coalgebra $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ equipped with the *deconcatenation* coproduct

$$\Delta[s^{-1}a_1|\cdots|s^{-1}a_n] \;=\; \sum_{i=0}^{n}[s^{-1}a_1|\cdots|s^{-1}a_i] \;\otimes\; [s^{-1}a_{i+1}|\cdots|s^{-1}a_n].$$

This is an E_1-coalgebra: the coproduct is coassociative but *not* cocommutative, reflecting the linear ordering of the tensor factors. It is the level-2 chain object before averaging. On the chiral side, $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ is built on ordered configurations $\mathrm{Conf}^<_n(\mathbb{R})$ parametrising the positions of line operators on an interval; the linear ordering on $\mathbb{R}$ is retained, with no Σ_n-quotient. Its Koszul dual $\mathcal{A}^!_{\mathrm{line}}$ governs the category of line operators: when $\mathcal{A}$ is affine Kac–Moody, $\mathcal{A}^!_{\mathrm{line}}$ is the Yangian $Y(\mathfrak{g})$ (the quantum group that deforms the enveloping algebra $U(\mathfrak{g}[z])$ of polynomial currents).

The symmetric bar (coinvariant quotient). The *symmetric* bar complex $\bar{B}^{\Sigma}(\mathcal{A}) = \mathrm{Sym}^c(s^{-1}\bar{\mathcal{A}})$ is the Σ_n-coinvariant quotient of the ordered bar. It carries the *coshuffle* coproduct: at tensor degree n, the coshuffle sums over all 2^n bipartitions of an *unordered* set, in contrast to the $n+1$ consecutive splittings of the deconcatenation. This is an E_∞-coalgebra: coassociative and cocommutative. Geometrically, it is built on the Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n(\mathbb{C})$ with Σ_n-coinvariants, and it is the natural coalgebra on $\mathrm{Ran}(X)$: the factorization coalgebra of Beilinson and Drinfeld. Its Koszul dual $\mathcal{A}^!_{\mathrm{ch}}$ is the chiral Koszul dual described above, the Francis–Gaitsgory construction via Verdier duality on $\mathrm{Ran}(X)$.

The R-matrix as cross-colour data. The averaging map $\mathrm{av}\colon \bar{B}^{\mathrm{ord}}(\mathcal{A}) \to \bar{B}^{\Sigma}(\mathcal{A})$ takes Σ_n-coinvariants. For a chiral algebra with OPE poles, this descent is *not* naive: it requires the spectral R-matrix $R(z)$ as twisting datum. The R-matrix is the degree-2 component of the ordered Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}}$: it is the collision residue

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}\big(\Theta^{E_1}_{\mathcal{A}}\big),$$

a matrix-valued rational function of the spectral parameter z encoding the full binary OPE data. Applying the averaging map at degree 2 collapses the z-dependent profile to a single scalar:

$$\mathrm{av}\big(r(z)\big) \;=\; \kappa(\mathcal{A}) \quad \text{for abelian and scalar families,}$$

while non-abelian affine Kac–Moody satisfies

$$\mathrm{av}\big(r(z)\big) + \dim(\mathfrak{g})/2 \;=\; \kappa(V_k(\mathfrak{g})).$$

the modular characteristic of Theorem D. More generally, each degree-n component of the ordered Maurer–Cartan element projects under av to the degree-n shadow of the modular obstruction tower $\Theta_{\mathcal{A}}$: at degree 3, the KZ associator projects to the cubic shadow C; at degree 4, the Yangian higher coproduct projects to the quartic resonance class Q.

The five main theorems A–D and H are the invariants of the ordered Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}}$ that survive averaging. The modular characteristic κ, the shadow obstruction tower, the complementarity pairing, and the chiral Hochschild cohomology are all Σ_n-coinvariant projections of the richer E_1-data. The ordered story is the primitive; the modular story is its shadow.

The R-matrix is derived from the local OPE for vertex algebras (E_∞-chiral), and is independent input for nonlocal quantum vertex algebras (E_1-chiral, in the sense of Etingof–Kazhdan).

The Heisenberg atom

The Heisenberg chiral algebra $\mathcal{H}_k$ is generated by a single field α of conformal weight 1 with OPE

$$\alpha(z)\,\alpha(w) \;\sim\; \frac{k}{(z-w)^2}\,.$$

Only the first product is nonzero: $\alpha_{(1)}\alpha = k\cdot\mathbf{1}$ and $\alpha_{(n)}\alpha = 0$ for $n \neq 1$. No first-order pole: $\alpha_{(0)}\alpha = 0$. The Borcherds identity is satisfied trivially: all triple products vanish since the only nontrivial product is $\alpha_{(1)}\alpha \in \Bbbk\cdot\mathbf{1}$, and $\mathbf{1}_{(n)} = 0$ for $n \geq 0$.

The bar complex at tensor degree n:

$$\bar{B}^n_X(\mathcal{H}_k) \;=\; (s^{-1}\alpha)^{\otimes n}_{\Sigma_n} \;\otimes\; \Gamma\big(\overline{\mathrm{FM}}_n(X),\; \Omega^{n-1}_{\log}\big).$$

Tensor degree 1: the space is $s^{-1}\alpha$, one-dimensional, with $d_{\bar{B}} = d_{\mathrm{int}} = 0$.

Tensor degree 2: the bar component is $(s^{-1}\alpha)^{\otimes 2} \otimes \langle \eta_{12} \rangle$. The bar differential extracts the full chiral product: $d_{\text{res}}(s^{-1}\alpha \otimes s^{-1}\alpha \otimes \eta_{12}) = \alpha_{(1)}\alpha = k \cdot \mathbf{1}$. The double-pole mode is extracted by $d_{\text{curvature}}$ (Proposition **??**); since $\alpha_{(0)}\alpha = 0$, the bracket component vanishes. The level k is visible already at genus 0; at genus 1 the curvature acquires a topological component $d_{\text{fib}}^2 = \kappa(\mathcal{H}_k) \cdot \omega_1$ (Example **??**).

Tensor degree $n \geq 3$: every residue at a codimension-one collision stratum $D_{\{i,j\}}$ extracts $\alpha_{(1)}\alpha = k \cdot \mathbf{1}$ on the pair (i, j). The remaining $n-2$ factors are α; since $\mathbf{1}_{(p)}\alpha = 0$ for $p \geq 0$, no further contractions occur. All structure is captured at tensor degree 2: the bar complex is *quadratic*. The Heisenberg algebra is chirally Koszul.

The modular characteristic:

$$\kappa(\mathcal{H}_k) \;=\; k.$$

The Koszul dual is the *curved* symmetric chiral algebra

$$\mathcal{H}_k^! \;=\; \big(\mathrm{Sym}^{\mathrm{ch}}(V^*),\; m_0 = -k\omega\big),$$

where V^* is the $\Bbbk$-linear dual of the generating space and $m_0 = -k\omega$ is the curvature: a degree-2 element satisfying the curved A_∞ relation

$$m_1^2(a) \;=\; [m_0, a]$$

(commutator, with the minus sign). The differential m_1 does not square to zero; its failure is controlled by the curvature. The curvature m_0 is the bar-dual of the OPE coefficient k: the algebra side has a scalar OPE; the dual side has a scalar curvature.

The Heisenberg has only a second-order pole: the bar complex is quadratic. The first example with both first- and second-order poles is the affine Kac–Moody algebra.

The E_1 face. The ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{H}_k)$ carries the R-matrix

$$r(z) \;=\; \frac{k}{z},$$

a scalar rational function with a single simple pole at the origin. This is the collision residue of the ordered Maurer–Cartan element at degree 2: the second-order OPE pole $k/(z-w)^2$ drops by one order under the logarithmic kernel $d\log(z-w)$ (the bar differential absorbs one power of $(z-w)$), producing the first-order r-matrix pole k/z. Applying the averaging map:

$$\mathrm{av}\big(r(z)\big) \;=\; k \;=\; \kappa(\mathcal{H}_k).$$

For the Heisenberg algebra, the R-matrix *is* the modular characteristic: averaging loses nothing, because $r(z) = k/z$ is already Σ_2-invariant (a scalar, not a matrix). For the affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level k, the R-matrix is $r(z) = k\,\Omega/z$ where $\Omega = \sum_a J^a \otimes J_a$ is the Casimir tensor: matrix-valued, carrying the full Lie-algebraic structure. Averaging collapses the Casimir to its trace: $\mathrm{av}(k\,\Omega/z) = k\dim(\mathfrak{g})/(2h^\vee) = \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k)$, and the full $\kappa(\widehat{\mathfrak{g}}_k)$ adds the Sugawara shift $\dim(\mathfrak{g})/2$. The Casimir tensor structure, invisible to the modular characteristic, is the data that builds the Yangian.

The Kac–Moody OPE

The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ at level k is generated by currents J^a, $a = 1, \ldots, \dim \mathfrak{g}$, with OPE

$$J^a(z)\, J^b(w) \;\sim\; \frac{f^{ab}{}_c\, J^c(w)}{z-w} \;+\; \frac{k\,\kappa^{ab}}{(z-w)^2}\,,$$

where $f^{ab}{}_c$ are the structure constants of the simple Lie algebra $\mathfrak{g}$ and κ^{ab} is the Killing form normalised so that long roots have squared length 2. Both poles are present: the first-order pole carries the Lie bracket $J^a_{(0)}J^b = f^{ab}{}_c J^c$; the second-order pole carries the invariant inner product $J^a_{(1)}J^b = k\,\kappa^{ab} \cdot \mathbf{1}$. The bar complex of $\widehat{\mathfrak{g}}_k$ therefore has nontrivial contributions at all tensor degrees: the first-order pole generates structure constants at tensor degree n from tree-level OPE contractions, and the second-order pole contributes the metric/curvature data.

The bracket and curvature decomposition:

$$\mu^{\mathrm{ch}} \;=\; \underbrace{f^{ab}{}_c\, J^c \cdot \eta_{12}}_{\text{Lie bracket (first-order pole)}} \;+\; \underbrace{k\,\kappa^{ab} \cdot \mathbf{1} \cdot d\eta_{12}}_{\text{inner product (second-order pole)}} \;.$$

The Lie bracket contributes the tree-level bar differential (combinatorics of the Chevalley–Eilenberg complex); the inner product contributes the curvature κ. The coexistence of both poles, absent in the pure second-order-pole Heisenberg case, makes $\widehat{\mathfrak{g}}_k$ the first nontrivial example.

Remark 0.0.1 (The zero-dimensional shadow). When $X = \operatorname{Spec} \Bbbk$ is a point, configuration spaces collapse, logarithmic forms vanish, and the chiral bar complex recovers the classical bar construction of an associative algebra A. This identification requires specifying the retraction data: vertex algebras live on the formal disk D, and the passage from D to a bare point discards the thickening data (completion, growth conditions) on which the $\mathcal{D}$-module structure depends. A *quadratic algebra* $A = T(V)/(R)$ has quadratic dual $A^! = T(V^*)/(R^\perp)$; when A is Koszul, the bar cohomology concentrates in diagonal degrees and $H^*(B(A)) \cong (A^!)^\vee$. The operadic instances $\mathrm{Com}^! = \mathrm{Lie}$ and $\mathrm{Sym}^! = \Lambda$ recover the Chevalley–Eilenberg complex as a restricted bar construction. When Koszulness fails, the cobar-bar resolution $\Omega(B(A)) \xrightarrow{\sim} A$ is the universal replacement.

Everything above takes place at genus 0: the propagator $\eta_{12} = d\log(z_1 - z_2)$ is globally defined on $\mathbb{P}^1$, and the Arnold relation holds without correction. On a curve of genus $g \geq 1$, the function $z_1 - z_2$ has no global meaning: the curve is compact, and any meromorphic function with a single simple zero must also have a simple pole. The Arnold relation acquires a defect, the bar differential no longer squares to zero, and the defect is controlled by a single scalar.

2. Curvature and the genus tower

The genus-g propagator

At genus $g \geq 1$, the difference $z_1 - z_2$ has no global meaning. The replacement is the *prime form* $E(z_1, z_2)$: a section of $K^{-1/2} \boxtimes K^{-1/2}$ on $X \times X$, vanishing to first order along the diagonal Δ, with no other zeros. The bundle is $K^{-1/2} \boxtimes K^{-1/2}$, *not* $K^{+1/2} \boxtimes K^{+1/2}$: the prime form is a $(-\frac{1}{2}, -\frac{1}{2})$-differential. In a local coordinate where the diagonal is $z_1 = z_2$,

$$E(z_1, z_2) \;=\; (z_1 - z_2) \cdot \big(1 + O(z_1 - z_2)\big) \cdot (dz_1)^{-1/2}(dz_2)^{-1/2}.$$

The prime form has nontrivial monodromy. Analytic continuation around the B-cycle B_j in the variable z_1:

$$E(z_1 + B_j, z_2) \;=\; -\exp\bigl(-\pi i \Omega_{jj} - 2\pi i \textstyle\int_{z_2}^{z_1} \omega_j\bigr) \cdot E(z_1, z_2),$$

where $\Omega = (\Omega_{jk})$ is the period matrix of X ($\Omega_{jk} = \oint_{B_k} \omega_j$) and $\omega_1, \dots, \omega_g$ are the holomorphic differentials normalised by $\oint_{A_j} \omega_k = \delta_{jk}$. The A-cycle monodromy is trivial: $E(z_1 + A_j, z_2) = E(z_1, z_2)$.

The logarithmic derivative

$$\omega(z_1, z_2) \;:=\; d_{z_1} \log E(z_1, z_2)$$

is the fundamental meromorphic differential of the second kind: symmetric, a double pole on Δ, vanishing A-periods. The *genus-g propagator*:

$$\eta_{12}^{(g)} \;=\; d \log E(z_1, z_2) \;+\; \pi \sum_{j,k=1}^{g} (\operatorname{Im} \Omega)^{-1}_{jk} \operatorname{Im}\Bigl(\int_{z_2}^{z_1} \omega_j\Bigr)\, \omega_k(z_1).$$

The period correction $(\operatorname{Im}\Omega)^{-1}_{jk}$ compensates the B-cycle monodromy, rendering $\eta_{12}^{(g)}$ single-valued on X. The price: $\eta_{12}^{(g)}$ is no longer holomorphic: the correction term is a $(1,0)$-form in z_1 multiplied by a real-analytic function of both variables. The propagator couples to the Hodge bundle $\mathbb{E} \to \overline{\mathcal{M}}_g$ through the period matrix Ω.

Curvature

The Arnold relation at genus g acquires a defect. At genus 0,

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} \;=\; 0$$

holds identically. At genus $g \geq 1$, the period corrections produce a residual $(1,1)$-form: the three-term sum does not vanish. The fibrewise bar differential satisfies

$$d_{\text{fib}}^2 \;=\; \kappa(\mathcal{A}) \cdot \omega_g, \tag{0.0.2}$$

where ω_g is the Arakelov $(1,1)$-form on the fibre of the universal curve $\pi\colon \mathcal{C}_g \to \overline{\mathcal{M}}_g$ (the family of genus-g curves parametrised by the Deligne–Mumford moduli space $\overline{\mathcal{M}}_g$), and $\kappa(\mathcal{A}) \in \Bbbk$ is the scalar *modular characteristic*, determined entirely by the leading OPE singularity.

For the Heisenberg algebra: the only OPE datum entering the bar differential is $\alpha_{(1)}\alpha = k$. Every residue at a codimension-one collision stratum extracts this coefficient. The defect of the Arnold relation at genus g is

$$d_{\text{fib}}^2\big|_{\mathcal{H}_k} \;=\; k \cdot \omega_g, \qquad \text{so} \quad \kappa(\mathcal{H}_k) = k.$$

For the Kac–Moody algebra: the bracket $J^a_{(0)} J^b = f^{ab}{}_c J^c$ and the inner product $J^a_{(1)} J^b = k\kappa^{ab}$ both contribute. The curvature receives the trace over the adjoint representation:

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \frac{(k + h^\vee)\dim \mathfrak{g}}{2h^\vee}.$$

The numerator $k + h^\vee$ is the *shifted level*; $h^\vee$ is the dual Coxeter number. At $k = -h^\vee$ (the critical level), $\kappa = 0$ and the bar complex is *uncurved*: the Sugawara construction is undefined (the central charge $c = k \dim \mathfrak{g}/(k + h^\vee)$ has a pole), but the curvature parameter vanishes.

For the Virasoro algebra at central charge c: the stress tensor T has OPE $T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}$. The bar propagator $d \log E(z, w)$ absorbs one pole order (the $d \log$ measure sends z^{-n} to $z^{-(n-1)}$), so the collision residue $r(z) = (c/2)/z^3 + 2T/z$ has pole orders one less than the OPE. The curvature computation gives

$$\kappa(\mathrm{Vir}_c) \;=\; \frac{c}{2}\,.$$

The Koszul dual is Vir_{26-c}, with $\kappa(\mathrm{Vir}_{26-c}) = (26 - c)/2$: the two curvatures sum to 13, not zero. At $c = 26$ the dual is uncurved; at $c = 0$ the algebra itself is uncurved; at $c = 13$ the algebra is self-dual under Koszul duality.

Period integrals restore nilpotence

The curvature (0.0.2) obstructs $d_{\mathrm{fib}}^2 = 0$, but not the total $D_g^2 = 0$. Integration against A-cycle periods defines a connection on the family of bar complexes parameterised by $\overline{\mathcal{M}}_g$. The corrected total differential

$$D_g \;=\; d_{\mathrm{fib}} + \nabla^{\mathrm{GM}}$$

incorporates the Gauss–Manin connection (the canonical flat connection on the cohomology bundle $R^1\pi_*\mathbb{C}$ of the universal family, encoding how the periods of the curve vary with moduli) on the Hodge bundle, and $D_g^2 = 0$: the $(1,1)$-curvature of the fibrewise differential is absorbed by the $(0,1)$-part of the base connection.

The mechanism: the fibrewise curvature $\kappa \cdot \omega_g$ is a class in $H^{1,1}(C_g)$; the period map $\overline{\mathcal{M}}_g \to \mathcal{A}_g$ (to the moduli of abelian varieties) identifies this class with the $(0,1)$-curvature of the Gauss–Manin connection on $R^1\pi_*\mathbb{C}$. The two curvatures cancel:

$$D_g^2 \;=\; d_{\mathrm{fib}}^2 + [d_{\mathrm{fib}}, \nabla^{\mathrm{GM}}] + (\nabla^{\mathrm{GM}})^2 \;=\; \kappa \cdot \omega_g - \kappa \cdot \omega_g + 0 \;=\; 0.$$

The genus tower

The bar complex traces a family of curved cochain complexes over $\overline{\mathcal{M}}_g$. For uniform-weight algebras (single-generator, or multi-generator with all generators of the same conformal weight), the obstruction to flatness at genus g is the characteristic class

$$\mathrm{obs}_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g \;\in\; H^*(\overline{\mathcal{M}}_g), \qquad \lambda_g \;=\; c_g(\mathbb{E}),$$

where $\mathbb{E} = \pi_*\omega_\pi$ is the *Hodge bundle*: the vector bundle on $\overline{\mathcal{M}}_g$ whose fibre over a point $[C]$ is the g-dimensional space $H^0(C, \omega_C)$ of holomorphic differentials on C. At genus 1 this factorization holds unconditionally for all families; for multi-weight families at genus $g \geq 2$ the scalar formula *fails*: a cross-channel correction from mixed-propagator boundary graphs appears (Theorem **??**; see §**??**). Grothendieck–Riemann–Roch applied to the universal curve $\pi\colon C_g \to \overline{\mathcal{M}}_g$ computes the total obstruction: the Chern character of $R\pi_*\omega_\pi$ involves the Todd class of the relative tangent bundle T_π, which is the $\hat{A}$-genus. The generating function:

$$\sum_{g\geq 1} \mathrm{obs}_g(\mathcal{A}) \cdot x^{2g} \;=\; \kappa(\mathcal{A}) \cdot \big(\hat{A}(ix) - 1\big).$$

Since $\hat{A}(ix) = (x/2)/\sin(x/2) = 1 + x^2/24 + 7x^4/5760 + \cdots$ (all positive), the first Faber–Pandharipande coefficients are:

$$F_1(\mathcal{A}) \;=\; \frac{\kappa(\mathcal{A})}{24},$$
$$F_2(\mathcal{A}) \;=\; \frac{7\,\kappa(\mathcal{A})}{5760},$$
$$F_3(\mathcal{A}) \;=\; \frac{31\,\kappa(\mathcal{A})}{967680}.$$

The genus-g coefficient is $\kappa(\mathcal{A}) \cdot \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ (the Faber–Pandharipande λ_g-integral). The $\hat{A}$-genus appears from the bar complex as the connection between chiral algebra and index theory.

The factorization $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \alpha_g$ separates the genus expansion into an algebra-dependent scalar $\kappa(\mathcal{A})$ and a universal sequence $\alpha_g := \lambda_g^{\mathrm{FP}}$ that depends only on the genus. The sequence α_g is not an artifact of a particular family: it is the λ_g-integral on $\overline{\mathcal{M}}_g$, a topological invariant of the moduli space of curves, determined by the Grothendieck–Riemann–Roch theorem applied to the universal curve. That every uniform-weight chiral algebra produces the same sequence is a consequence of scalar saturation (Section 9.1): the degree-2 MC component is the unique shadow that couples a scalar to the Hodge class.

Proposition 0.0.2 (Universality of α_g)**.** *Define $\alpha_g := \int_{\overline{\mathcal{M}}_g} \lambda_g = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$, the Faber–Pandharipande λ_g-integral. For every uniform-weight chirally Koszul algebra $\mathcal{A}$ (single strong generator, or multi-generator with all generators of the same conformal weight):*

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \alpha_g \qquad \textit{for all } g \geq 1$$

The coefficient α_g is independent of the algebra, the level, and the central charge; it depends only on the genus g. The generating function is the $\hat{A}$-genus: $\sum_{g\geq 1} \alpha_g\, x^{2g} = \hat{A}(ix) - 1 = (x/2)/\sin(x/2) - 1$.

The universality of α_g is the genus-tower statement that all uniform-weight chiral algebras share a single modular homotopy type at degree 2, parametrised by the scalar κ. The algebra enters only through κ; the geometry of $\overline{\mathcal{M}}_g$ enters only through α_g. The two factors decouple completely.

The multi-weight genus expansion. For multi-weight algebras (strong generators of distinct conformal weights), the genus expansion acquires a correction (Theorem **??**):

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \;+\; \delta F_g^{\mathrm{cross}}(\mathcal{A}) \qquad \text{(all-weight).}$$

The correction $\delta F_g^{\mathrm{cross}}$ is a graph sum over mixed-channel boundary graphs of $\overline{\mathcal{M}}_{g,0}$: stable graphs in which propagators connect generators of distinct conformal weight. Three structural results constrain the correction:

(i) *Genus-1 universality*: $\delta F_1^{\mathrm{cross}} = 0$ for all $\mathcal{A}$, so $F_1(\mathcal{A}) = \kappa(\mathcal{A})/24$ is unconditional.

(ii) *Uniform-weight universality* (Proposition 0.0.2): if all generators share the same conformal weight, $\delta F_g^{\mathrm{cross}} = 0$ for all $g \geq 1$.

(iii) *Free-field exactness* (Proposition 0.0.4): for free-field algebras (Heisenberg, bc, $\beta\gamma$, lattice), $\delta F_g^{\mathrm{cross}} = 0$ for all $g \geq 1$, even when generators have distinct weights.

The correction is nonzero starting at genus 2 for interacting multi-weight families ($\mathcal{W}_N$ with $N \geq 3$, non-principal $\mathcal{W}$-algebras).

The scalar $\kappa(\mathcal{A})$ is only the leading term. The bar complex carries higher-degree data: a universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ whose projection to degree r yields a shadow invariant $S_r(\mathcal{A})$. On each primary deformation line L, the shadow generating function $H(t) = \sum_{r\geq 2} r S_r\, t^r$ satisfies the algebraic relation

$$H(t)^2 \;=\; t^4\, Q_L(t), \qquad Q_L(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2, \tag{0.0.3}$$

a quadratic polynomial in three genus-0 invariants (κ, α, S_4) (Theorem ??). The infinite tower is algebraic of degree 2: determined, computable, and governed by the single discriminant $\Delta = 8\kappa S_4$.

The shadow obstruction tower is not merely a formal construction. Its projections yield: the Pixton ideal of the modular curve (Theorem ??), the integrable hierarchies of Gelfand–Dickey type (Proposition ??), the Hitchin spectral curve of the shadow connection (Remark ??), the Baxter Q-operator of quantum integrable lattice models (Remark ??), and the Bekenstein–Hawking entropy formula of three-dimensional gravity (Chapter ??). These are not separate results but windows onto a single object.

Complementarity

Verdier duality on the Ran space splits the center local system into two halves, one controlled by $\mathcal{A}$ and one by $\mathcal{A}^!$. The splitting carries a shifted-symplectic structure. The complementary summands:

$$R\Gamma(\overline{\mathcal{M}}_g,\ \mathcal{Z}_{\mathcal{A}}) \;\simeq\; \mathbf{Q}_g(\mathcal{A}) \;\oplus\; \mathbf{Q}_g(\mathcal{A}^!).$$

What $\mathcal{A}$ sees as deformation, $\mathcal{A}^!$ sees as obstruction. The natural pairing

$$\mathbf{Q}_g(\mathcal{A}) \otimes \mathbf{Q}_g(\mathcal{A}^!) \;\longrightarrow\; R\Gamma(\overline{\mathcal{M}}_g,\ \omega_{\overline{\mathcal{M}}_g}) \;\simeq\; \Bbbk[-(3g-3)]$$

is a $(-(3g-3))$-shifted symplectic form, and the two summands are *Lagrangian*: each is isotropic and of half the total dimension. At the scalar level, complementarity reduces to

$$\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) \;=\; 0 \qquad \text{(for Kac–Moody and free-field algebras).}$$

For $\mathcal{W}$-algebras the sum $\kappa + \kappa^!$ is a nonzero constant: 13 for Virasoro, $250/3$ for $\mathcal{W}_3$.

Feigin–Frenkel duality

The Sugawara construction produces the Virasoro field

$$T(z) \;=\; \frac{1}{2(k+h^\vee)} \sum_{a=1}^{\dim \mathfrak{g}} {:}J^a(z)J^a(z){:}\,,$$

with central charge $c = k \dim \mathfrak{g}/(k + h^\vee)$. At the critical level $k = -h^\vee$, the denominator vanishes: $T(z)$ is *undefined*, not "c diverges." The chiral algebra acquires a large center, the Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee})$, isomorphic to the algebra of functions on the space of local ${}^L\mathfrak{g}$-opers on the formal disc.

The Feigin–Frenkel involution

$$k \longleftrightarrow -k - 2h^\vee$$

is Koszul duality on configuration spaces: it interchanges $\mathcal{A}$ and $\mathcal{A}^!$ while preserving the bar-cobar adjunction. At the level of modular characteristics, $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}_{-k-2h^\vee}) = 0$: complementarity is a direct consequence. At the critical level, the Koszul dual is the Langlands-dual critical-level algebra $\widehat{{}^L\mathfrak{g}}_{-{}^Lh^\vee}$, not a renormalised copy of the original.

The total space

The universal configuration fibration

$$\pi_n : \overline{C}_n(C_g) \longrightarrow \overline{\mathcal{M}}_g$$

packages the bar complex into a single global object. Its boundary divisors carry two distinct geometric roles:

- *Collision divisors* (vertical): the fibrewise boundary of $\overline{\mathrm{FM}}_n(X_s)$ over a point $s \in \overline{\mathcal{M}}_g$. These produce the bar differential d_{res}, the OPE residues.
- *Degeneration divisors* (horizontal): the boundary of $\overline{\mathcal{M}}_g$ itself (nodal curves, reducible fibres). These produce the period corrections, the Gauss–Manin connection terms in D_g.

The collision divisors are the "algebraic" boundary (controlled by the Borcherds identity); the degeneration divisors are the "geometric" boundary (controlled by the period map and the clutching morphisms of stable curves). The interaction between them is the curvature κ: the fibrewise bar differential fails to be nilpotent precisely because the vertical and horizontal boundaries do not commute.

3. Modular homotopy theory

The genus tower demands an organising framework. The fibrewise differential d_{fib}, the Gauss–Manin correction ∇^{GM}, and the clutching morphisms at boundary strata satisfy compatibility relations dictated by the stratification of $\overline{\mathcal{M}}_{g,n}$. The governing algebraic structure is the *modular operad*: an operad whose operations are indexed not by trees but by connected stable graphs of arbitrary genus.

The *modular homotopy type* of $\mathcal{A}$ is the formal moduli problem classifying the inverse system of bar complexes $\bar{B}^{(g)}(\mathcal{A})$:

$$\mathcal{M}^{\mathrm{mod}}_{\mathcal{A}}(R) := \mathrm{MC}\big(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \mathbin{\widehat{\otimes}_{\mathfrak{m}}} R\big)/\sim$$

for local Artin rings R. Here $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is the *modular convolution algebra*: a complete filtered cyclic L_∞-algebra (a chain complex with multilinear brackets $\ell_1, \ell_2, \ell_3, \ldots$ satisfying a hierarchy of Jacobi-type identities, generalising Lie algebras to the homotopy-coherent setting). Three inputs enter its construction: a homotopy chiral algebra on the curve (algebraic), a cooperadic chain complex on logarithmic configuration spaces (geometric), and a convolution L_∞-structure mediating between them (homotopical). The bar complex, the Koszul dual, the genus tower, the complementarity splitting, the shadow obstruction tower, and the holographic correspondence of Volume II are all representations of the formal group $\mathcal{M}^{\mathrm{mod}}_{\mathcal{A}}$.

The convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ carries a canonical Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ (section 4), a tangent complex governing deformations (section 5), and an obstruction tower of finite-order projections (section 6).

3.1. The modular operad

At genus 0, the combinatorics of the bar complex is governed by trees. At higher genus, trees are replaced by stable graphs: graphs with genus labels at their vertices, parametrising the boundary strata of the Deligne–Mumford moduli space $\overline{\mathcal{M}}_{g,n}$.

A *modular operad* $\mathcal{O}$ (Getzler–Kapranov [?GeK98]) has operations indexed not by trees but by *connected stable graphs* (graphs of arbitrary genus). A *connected stable graph* Γ has a genus label $g(v) \in \mathbb{Z}_{\geq 0}$ at each vertex satisfying $2g(v) - 2 + \mathrm{val}(v) > 0$ (stability), and total genus $g(\Gamma) = b_1(\Gamma) + \sum_v g(v)$; trees are the genus-0 case. The structure maps of $\mathcal{O}$ are of two kinds: *separating gluing* $\xi_{\mathrm{sep}} : \mathcal{O}(g_1, n_1{+}1) \otimes \mathcal{O}(g_2, n_2{+}1) \to \mathcal{O}(g_1{+}g_2, n_1{+}n_2)$ (joining two components at a node), and *non-separating gluing* $\xi_{\mathrm{ns}} : \mathcal{O}(g, n{+}2) \to \mathcal{O}(g{+}1, n)$ (identifying two marked points on the same component, raising genus by 1). These satisfy equivariance under Σ_n, associativity (iterated gluings are order-independent), and commutativity.

A stable graph records a degeneration of a smooth algebraic curve into a nodal curve: the vertices of Γ are the irreducible components, the edges are the nodes, the genus labels are the genera of the components, and the half-edges are the marked points. The set of stable graphs $\mathsf{Gr}^{\mathrm{st}}_{g,n}$ parametrises the boundary stratification of the Deligne–Mumford moduli space $\overline{\mathcal{M}}_{g,n}$. Every codimension-one boundary divisor is a one-edge degeneration; every codimension-two stratum is the intersection of exactly two codimension-one divisors, appearing with opposite orientations in the boundary-of-boundary complex.

The *Feynman transform* of $\mathcal{O}$, the modular analogue of the operadic bar construction, replaces trees in (0.0.4) by stable graphs:

$$F\mathcal{O}(g, n) \;=\; \bigoplus_{\Gamma \in \mathsf{Gr}^{\mathrm{st}}_{g,n}} \frac{\hbar^{g(\Gamma)}}{|\operatorname{Aut}(\Gamma)|} \bigotimes_{v \in V(\Gamma)} \mathcal{O}\big(g(v), \mathrm{In}(v)\big). \tag{0.0.4}$$

3.2. The Feynman transform

The *differential* of the Feynman transform contracts one edge at a time:

$$d_{F\mathcal{O}}(\gamma_\Gamma) \;=\; \sum_{e \in E(\Gamma)} (-1)^{\epsilon(e)}\, \xi_e(\gamma_{\Gamma/e}),$$

where $\gamma_{\Gamma/e}$ denotes the image of γ_Γ under edge contraction along e, applying the operadic composition ξ_e to the merging vertex factors; ξ_e invokes the gluing map ξ_{sep} or ξ_{ns} of $\mathcal{O}$ at edge e and contracts e from Γ. The sign $(-1)^{\epsilon(e)}$ is determined by the position of e in the ordering of $E(\Gamma)$ recorded by the orientation line $\det(E(\Gamma))$. The identity $d^2 = 0$ is the codimension-2 boundary cancellation: each pair of edges can be contracted in either order, and the two terms cancel by the associativity of $\mathcal{O}$. At genus 0 (trees, $\hbar^0 = 1$), $F\mathcal{O}$ reduces to the operadic bar construction (0.0.4).

When $\mathcal{O} = \mathrm{Com}$ (the *commutative modular operad*, with $\mathrm{Com}(g, n) = \Bbbk$ for every stable (g, n)), each vertex factor in (0.0.4) is one-dimensional, and the Feynman transform becomes the *stable graph* *complex*

$$F\mathrm{Com}(g,n) \;=\; \bigoplus_{\Gamma\in \mathsf{Gr}^{\mathrm{st}}_{g,n}} \frac{\hbar^{g(\Gamma)}}{|\operatorname{Aut}(\Gamma)|}\;\det\big(E(\Gamma)\big)\;:$$

a chain complex spanned by isomorphism classes of stable graphs, graded by edge number, with edge-contraction differential. At genus 0, $F\mathrm{Com}(0,n) = B(\mathrm{Com})(n)$ recovers the tree-level complex of Ginzburg–Kapranov.

3.3. The convolution ladder

Populating the Feynman transform with algebraic data (chiral amplitudes at vertices, propagators at edges) requires a convolution mechanism pairing the cooperadic structure of the skeleton with the operadic structure of the algebra.

At each level of generality, the bar-type construction is a convolution dg Lie algebra whose Maurer–Cartan elements classify algebraic structures:

$$\begin{array}{rll} \text{algebra}: & \operatorname{Hom}\big(B(\mathrm{Ass}),\ \mathrm{End}_A\big) & A_\infty\text{-structures on } A \\ \text{operad}: & \operatorname{Hom}_\Sigma\big(B(\mathcal{P}),\ \mathrm{End}_V\big) & \mathcal{P}_\infty\text{-structures on } V \\ \text{modular}: & \operatorname{Hom}_\Sigma\big(F\mathcal{O},\ \mathrm{End}_{\bar B(\mathcal{A})}\big) & \text{consistent genus towers} \end{array} \tag{0.0.5}$$

The first line is the algebraic Hochschild cochain complex of the compact-generator chart; the second is the deformation complex of Ginzburg–Kapranov; the third, with $\mathcal{O} = \mathrm{Com}$, governs this book. The *Lie bracket* in each case is the convolution bracket: for f, g in the convolution hom,

$$[f,g] \;:=\; \mu\circ(f\otimes g)\circ\Delta \;-\; (-1)^{|f||g|}\,\mu\circ(g\otimes f)\circ\Delta,$$

where Δ is the cooperadic decomposition (decomposing a cooperad element along an edge into a two-vertex composition) and μ is operadic composition (composing the output of one operation with an input of another).

Given a chiral algebra $\mathcal{A}$ with bar complex $\bar B(\mathcal{A})$, the *modular convolution dg Lie algebra*

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \;:=\; \operatorname{Hom}_\Sigma\big(F\mathrm{Com},\ \mathrm{End}_{\bar B(\mathcal{A})}\big) \tag{0.0.6}$$

is the $\mathcal{O} = \mathrm{Com}$ instance of the modular line of (0.0.5). The bar complex is an $F\mathrm{Com}$-algebra; the universal MC element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is bar-intrinsic ($D^2_{\mathcal{A}} = 0$ implies $\Theta_{\mathcal{A}} \in \mathrm{MC}$), and the five main theorems of this monograph are projections of $\Theta_{\mathcal{A}}$ (§??).

The convolution algebra requires two inputs: a cooperadic coefficient system (the geometry of configuration spaces) and an operadic algebra (the chiral algebra on the curve).

The convolution algebra (0.0.6) uses $\mathcal{O} = \mathrm{Com}$, the commutative ($E_\infty$) operad: the stable graphs in $F\mathrm{Com}$ carry no cyclic ordering at their vertices. Replacing Com by Ass (the associative operad) equips each vertex with a cyclic ordering of its half-edges, and the Feynman transform $F\mathrm{Ass}$ sums over *ribbon graphs* (fatgraphs) whose genus is the genus of the thickened embedding surface:

$$\mathfrak{g}^{\mathrm{mod},E_1}_{\mathcal{A}} \;:=\; \operatorname{Hom}_\Sigma\big(F\mathrm{Ass},\ \mathrm{End}_{\bar B^{E_1}(\mathcal{A})}\big).$$

This is the algebraic form of the 't Hooft $1/N$ expansion. Setting $\hbar = 1/N^2$, the MC element $\Theta^{E_1}_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod},E_1}_{\mathcal{A}})$ is the genus expansion $\sum_g N^{2-2g} F_g(\lambda)$ summed over ribbon graphs weighted by the genus of their embedding surface. Planar diagrams ($g = 0$) dominate at large N; the shadow obstruction tower $\Theta^{E_1,\leq r}$ is the truncation to r-point 't Hooft interactions. The coinvariant map $F\mathrm{Ass} \to F\mathrm{Com}$ projects $\Theta^{E_1} \mapsto \Theta$: colour-ordered amplitudes to full amplitudes. On each *primary line* (a one-dimensional subspace of the cyclic deformation space $H^2_{\mathrm{cyc}}(\mathcal{A}, \mathcal{A})$ spanned by a single primary field), the shadow tower is algebraic of degree 2: the entire infinite sequence is the Taylor expansion of the square root of a single quadratic form Q_L, the *shadow metric* (§??).

Remark (Homotopy types in three stages). The three levels of the convolution ladder mirror three levels of homotopy theory. This parallel organizes the entire classification programme of Section 6. *Over a field* (the bar complex of a graded algebra): the bar construction $B(A) = (T^c(s^{-1}\bar{A}), d_B)$ determines the homotopy type of an associative algebra; formality ($m_k = 0$ for $k \geq 3$) corresponds to the algebra being intrinsically formal (e.g. $\Bbbk[x]/(x^2)$); the Massey products are the obstructions. *Over a manifold*: the Sullivan minimal model $(\Lambda V, d)$ determines the rational homotopy type; Kähler manifolds are formal (Deligne–Griffiths–Morgan–Sullivan): $d = 0$ on the minimal model and the rational homotopy type is determined by $H^*(X; \mathbb{Q})$. The Heisenberg algebra is the vertex-algebraic analogue: its shadow obstruction tower terminates at degree 2 (the Gaussian archetype), all higher obstructions vanish, and the modular homotopy type is determined by κ alone. *Over a curve at all genera*: the A_∞-products m_k of the first stage become the modular L_∞-brackets $\ell_k^{(g)}$; formality becomes the Gaussian archetype ($\ell_k^{(g)} = 0$ for $k \geq 3$ at all g); the shadow obstruction tower is the modular Massey product tower, classifying the complexity of the modular homotopy type exactly as the formality level classifies rational homotopy. The shadow depth classification of Section 6 makes this analogy precise: the four classes G, L, C, M correspond to four levels of modular formality, determined by the vanishing pattern of the higher brackets $\ell_k^{(g)}$.

3.4. Homotopy chiral algebras

Fix an affine cover $\mathfrak{U} = \{U_\alpha\}_{\alpha \in I}$ of the curve X. The Čech totalisation

$$A^{\mathrm{ch}}_\infty \;:=\; \check{C}^\bullet(\mathfrak{U}; \mathcal{A}) \;=\; \bigoplus_{p \geq 0} \prod_{\alpha_0 < \cdots < \alpha_p} \mathcal{A}(U_{\alpha_0} \cap \cdots \cap U_{\alpha_p})$$

carries a Ch_∞-algebra structure (Malikov–Schechtman): the chiral bracket of $\mathcal{A}$, restricted to overlaps of affine opens, does not satisfy the Jacobi identity on the nose; it satisfies it up to a coherent system of higher homotopies, the *secondary Borcherds operations*

$$F_n \colon \mathrm{Lie}(n) \otimes \mathcal{Y}(n) \;\longrightarrow\; P_n(\{A^{\mathrm{ch}}_\infty\},\, A^{\mathrm{ch}}_\infty), \qquad n \geq 2,$$

encoding derived OPE data at all pole orders. The operation F_2 is the chiral bracket itself.

The Jacobiator homotopy F_3 arises from the moduli space $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$: its boundary $\partial\overline{\mathcal{M}}_{0,4}$ consists of three points, corresponding to three OPE compositions $(a_{(m)}b)_{(n)}c$, $(b_{(m)}c)_{(n)}a$, $(c_{(m)}a)_{(n)}b$. The chain $C_1(\overline{\mathcal{M}}_{0,4}) \cong \Bbbk$ provides a 1-chain whose boundary is the alternating sum of the three vertices; F_3 is the chain homotopy witnessing their equality up to d.

Higher operations F_n are determined inductively by $C_*(\overline{\mathcal{M}}_{0,n+1})$ through the homotopy transfer theorem applied to the inclusion $H_*(\overline{\mathcal{M}}_{0,n+1}) \hookrightarrow C_*(\overline{\mathcal{M}}_{0,n+1})$. The cellular chain complex of $\overline{\mathcal{M}}_{0,n+1}$

is the chain complex of the associahedron K_n; the transferred operations F_n are the A_∞-corrections at degree n.

Different affine covers yield quasi-isomorphic Ch_∞-algebras: the Čech-to-derived spectral sequence degenerates for $\mathcal{D}$-modules on smooth curves. The strict chiral algebra $\mathcal{A}$ is the H^0 of this structure: the primitive local object is the homotopy chiral algebra; the strict algebra appears after rectification. The rectification theorem (Vallette): the homotopy category of Ch_∞-algebras is equivalent to the homotopy category of strict chiral algebras, via the bar-cobar adjunction at the operadic level.

3.5. Logarithmic Fulton–MacPherson spaces

The classical Fulton–MacPherson compactification records collisions of points on a smooth variety. Its logarithmic extension, due to Mok, records collisions along a divisor and provides the normal-crossings boundary structure on which $D^2 = 0$ depends.

For a simple-normal-crossing pair (X, D) with D a reduced divisor, Mok constructs the *logarithmic Fulton–MacPherson compactification* $\mathrm{FM}_n(X \mid D)$: a smooth variety with corners, compactifying the open configuration space $\mathrm{Conf}_n(X \setminus D)$, whose boundary stratification is indexed by *rigid planted-forest types*.

A planted-forest type ρ consists of:

- a partition of $\{1, \ldots, n\}$ into blocks (the roots of the forest);
- for each block, a rooted tree (the planted tree) recording the hierarchy of collisions;
- a grid-depth assignment $w_v \in \mathbb{Z}_{\geq 0}$ to each vertex, recording the order of tangency with D.

The ordinary Fulton–MacPherson compactification is the case $D = \emptyset$: all grid depths are zero, and the planted trees are the collision trees of the classical theory.

The *degeneration formula*: for a semistable family $W \to B$ with singular fibre $W_0 = \bigcup Y_v$ (components indexed by vertices of the dual graph), and a rigid type ρ,

$$\mathrm{FM}_n(W/B)(\rho) \longrightarrow \prod_{v \in V(S_\rho)} \mathrm{FM}_{I_v}(Y_v \mid D_v)$$

is a proper birational correspondence factoring the global FM geometry into a fibre product of local FM pieces.

The tropicalisation: planted forests are dual graphs of tropical curves. Precisely,

$$\mathrm{Trop}\big(\mathrm{FM}_n(\mathbb{C} \mid D)\big) = G_{\mathrm{pf}},$$

the planted-forest poset with grafting as partial order. Grafting of planted forests corresponds to degeneration of the underlying curve; the poset structure on G_{pf} is the face poset of the boundary stratification, ordered by inclusion of the corresponding FM boundary strata. The planted-forest poset is the combinatorial skeleton of the geometric input, just as the stable graph complex is the skeleton of the Feynman transform.

The *Mok codimension formula*: for a planted-forest type with grid depths $(w_1, \ldots, w_r)$ and tree vertex counts $(|V(T_j)|)_{j=1,\ldots,s}$,

$$\mathrm{codim}(W_\rho) = \textstyle\sum_{i=1}^{r} w_i + \sum_{j=1}^{s} (|V(T_j)| - 1).$$

The two summands: grid-depth contributions (tangency with D) and tree-depth contributions (collision complexity).

For FM_3: seven boundary types. Codimension 1: $\{12\}, \{23\}, \{13\}, \{123\}$ (four strata, recording pairwise and triple collisions). Codimension 2: $\{12{<}123\}, \{23{<}123\}, \{13{<}123\}$ (three strata, recording sequential collisions: first a pair, then the third point joins).

The chain complexes $C^{\log\mathrm{FM}}_{X|D}(n) := C_\bullet(\mathrm{FM}_n(X \mid D))$ carry cooperadic structure: the inclusion of boundary strata defines cocomposition maps. The *graphwise cocomposition* for a rigid stable graph Γ:

$$\Delta^{\log}_\Gamma \;:=\; (\nu_\Gamma)_* \circ \mathrm{Res}_{D^{\log}_\Gamma} \;:\; C^{\log\mathrm{FM}}_{\mathrm{mod}}(g,n) \;\longrightarrow\; \bigotimes_{v\in V(\Gamma)} C^{\log\mathrm{FM}}_{\mathrm{mod}}\big(g(v),\, I_v \sqcup H(v)\big)[-|E_{\mathrm{int}}(\Gamma)|],$$

where ν_Γ is the Mok birational correspondence on the normalised boundary stratum indexed by Γ, and $\mathrm{Res}_{D^{\log}_\Gamma}$ is the residue along the logarithmic boundary divisor. The degree shift $|E_{\mathrm{int}}(\Gamma)|$ accounts for the codimension of the stratum (one for each internal edge).

3.6. The convolution L_∞-algebra

A twisting morphism $\alpha\colon \mathcal{C} \to \mathcal{P}$ between a cooperad and an operad (satisfying the Maurer–Cartan equation $\partial\alpha + \alpha \star \alpha = 0$ in the convolution algebra) determines a convolution sL_∞-algebra $\hom_\alpha(\mathcal{C}, \mathcal{P})$ (Robert-Nicoud–Wierstra, after Vallette). The underlying graded vector space:

$$\hom_\alpha(\mathcal{C},\mathcal{P}) \;=\; \prod_{n\geq 1} \mathrm{Hom}_{\Sigma_n}\big(\mathcal{C}(n),\, \mathcal{P}(n)\big).$$

The L_∞ brackets:

$$\ell_k(f_1,\ldots,f_k)(x) \;=\; \sum \gamma_{\mathcal{P}}\big((\alpha\circ 1)(f_1\otimes\cdots\otimes f_k)\,\Delta^{(k)}_{\mathcal{C}}(x)\big),$$

where $\Delta^{(k)}_{\mathcal{C}}$ is the iterated decomposition of the cooperad and $\gamma_{\mathcal{P}}$ is the operadic composition. The bracket ℓ_1 is the convolution differential (pre- and post-composition with the differentials of $\mathcal{C}$ and $\mathcal{P}$). The bracket ℓ_2 is the Lie bracket of coderivations. Higher brackets ℓ_k, $k \geq 3$, arise from the higher cooperadic decompositions and are nontrivial whenever $\mathcal{C}$ is not concentrated in a single degree.

The sL_∞-algebra $\hom_\alpha(\mathcal{C},\mathcal{P})$ extends to ∞-morphisms in *either slot separately*: given a quasi-isomorphism $\mathcal{C} \xrightarrow{\sim} \mathcal{C}'$ of cooperads (or $\mathcal{P} \xrightarrow{\sim} \mathcal{P}'$ of operads), there is an induced ∞-quasi-isomorphism of convolution algebras. But *not both simultaneously*: the convolution is functorial in each slot, but does not assemble into a bifunctor on the ∞-categorical level. This is a structural constraint: the MC3 categorical lift of bar-cobar duality must proceed one slot at a time.

The bar-cobar adjunction $B_\kappa \dashv \Omega_\kappa$ is a *Quillen equivalence* (Vallette): it induces an equivalence on homotopy categories. Different Ch_∞ presentations yield quasi-isomorphic deformation theories.

3.7. Assembly

The *modular convolution L_∞-algebra*:

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \;:=\; \mathrm{Conv}_\infty\Big(C^{\log\mathrm{FM}}_{\mathrm{mod}},\; \mathrm{End}_{\mathrm{Ch}_\infty}(\mathcal{A}^{\mathrm{ch}}_\infty)\Big).$$

The cooperadic input is the log-FM chain cooperad $C^{\log\mathrm{FM}}_{\mathrm{mod}}$: the modular cooperad of chain complexes on logarithmic FM spaces, with cocomposition given by the graphwise maps $\Delta^{\log}_{\Gamma}$. The operadic input is the Ch_∞-endomorphism operad of the homotopy chiral algebra $\mathcal{A}^{\mathrm{ch}}_\infty$: the operad of multilinear maps respecting the transferred homotopy-chiral structure.

One-slot convention: fix the log-FM cooperad (the "geometric" slot), vary the chiral algebra input (the "algebraic" slot) by ∞-morphisms.

The *strict chart*: let $\mathfrak{U}$ be an affine cover of X. The coderivation dg Lie algebra

$$\mathfrak{g}^{\mathrm{mod},\log}_{\mathcal{A}}(\mathfrak{U}) \;:=\; \mathrm{Conv}_{\mathrm{str}}\big(C^{\log\mathrm{FM}}_{\mathrm{mod}},\ \mathrm{End}_{\mathrm{Ch}_\infty}(\mathcal{A}^{\mathrm{ch}}_\infty)\big)$$

is the *strict model*: classical Lie bracket (commutator of coderivations), no higher L_∞ brackets, but the *same* Maurer–Cartan moduli. MC elements in the strict chart are in natural bijection with MC elements in the full convolution L_∞-algebra. The strict chart suffices for constructing the universal MC element and computing shadows; the full L_∞ structure is needed for transfer, formality, and gauge equivalence.

3.8. The modular bar functor

In log-FM coordinates, the modular bar functor at type (g,n):

$$\mathrm{B}^{\log}_{\mathrm{mod}}(\mathcal{A})(g,n) \;=\; \bigoplus_{\Gamma\in\mathsf{Gr}^{\mathrm{st}}_{g,n}} \left(\bigotimes_{v\in V(\Gamma)} \bar{B}^{\mathrm{ch}}(\mathcal{A}^{\mathrm{ch}}_\infty;\mathrm{In}(v)) \;\otimes\; C_\bullet(\mathrm{FM}^{\log}_{\Gamma}) \;\otimes\; \mathfrak{o}_{E_{\mathrm{int}}(\Gamma)} \right)_{\mathrm{Aut}(\Gamma)} .$$

The sum is over stable graphs Γ of type (g,n): graphs with first Betti number $b_1(\Gamma)+\sum_v g(v) = g$ and n external legs. The ingredients:

- For each vertex $v\in V(\Gamma)$: the transferred Ch_∞-bar contribution $\bar{B}^{\mathrm{ch}}(\mathcal{A}^{\mathrm{ch}}_\infty;\mathrm{In}(v))$, the chiral bar complex on the incoming half-edges of v, computed in the homotopy-chiral framework.
- For each internal edge: the propagator $P_{\mathcal{A}} = H^{-1}_{\mathcal{A}}$, contracting the two half-edges.
- The geometric factor: the chain complex $C_\bullet(\mathrm{FM}^{\log}_{\Gamma})$ on the log-FM stratum indexed by Γ.
- The orientation line $\mathfrak{o}_{E_{\mathrm{int}}(\Gamma)} = \det(\Bbbk E(\Gamma))\otimes\det(H_1(\Gamma;\Bbbk))^{-1}$: signs from edge orientations and loop orderings.
- Automorphism quotient: coinvariants under $\mathrm{Aut}(\Gamma)$.

3.9. The six-component differential

The total differential has six components, whose mutual cancellations constitute $D^2 = 0$:

$$D^{\log}_{\mathrm{mod}} \;=\; \underbrace{d_{\check{C}}}_{\text{Čech}} + \underbrace{d_{\mathrm{Ch}_\infty}}_{\text{transferred chiral}} + \underbrace{d_{\mathrm{coll}}}_{\text{collision}} + \underbrace{d_{\mathrm{sew}}}_{\text{separating}} + \underbrace{d_{\mathrm{pf}}}_{\text{planted-forest}} + \underbrace{\hbar\Delta}_{\text{genus-raising}} .$$

The six components, their geometric sources, and their action on a pure tensor $a\otimes\eta$ indexed by a stable graph Γ:

$d_{\check{C}}$: the Čech differential of the cover $\mathfrak{U}$. Acts on the Čech indices of the vertex labels. Source: the nerve of the cover.

d_{Ch_∞}: the transferred chiral differential on each vertex contribution. Acts on the Ch_∞-bar complex at each vertex. Source: the secondary Borcherds operations F_n.

d_{coll}: the collision (residue) differential. Contracts an internal edge of Γ by extracting the OPE residue at the corresponding codimension-one collision stratum. Source: codimension-one faces of $\overline{\mathrm{FM}}_n(X)$.

d_{sew}, d_{pf}, and $\hbar\,\Delta$: the three positive-genus boundary operators. On a pure tensor $a \otimes \eta$ indexed by Γ:

$$\begin{aligned} d_{\mathrm{sew}}(a \otimes \eta) &= \sum_{e \in E_{\mathrm{sep}}(\Gamma)} \pm\, \mu_e(a) \otimes (\xi_e)_* \operatorname{Res}_{D_e^{\log}}(\eta), \\ d_{\mathrm{pf}}(a \otimes \eta) &= \sum_{F \in \mathsf{PF}^{\mathrm{rig}}(\Gamma)} \pm\, \mu_F(a) \otimes (\xi_F)_* \operatorname{Res}_{D_F^{\log}}(\eta), \\ \hbar\,\Delta(a \otimes \eta) &= \hbar \sum_{e \in E_{\mathrm{ns}}(\Gamma)} \pm\, \mathrm{Tr}_e(a) \otimes (\xi_e)_* \operatorname{Res}_{D_e^{\log}}(\eta). \end{aligned}$$

Separating edges $E_{\mathrm{sep}}(\Gamma)$ (cutting the graph into two connected components) $\longrightarrow d_{\mathrm{sew}}$: the clutching morphism, gluing two lower-genus curves along a node. Non-separating edges $E_{\mathrm{ns}}(\Gamma)$ (whose creation increases b_1 by one) $\longrightarrow \hbar\Delta$: the genus-raising operator, self-gluing a curve at two marked points. Planted-forest strata $\mathsf{PF}^{\mathrm{rig}}(\Gamma) \longrightarrow d_{\mathrm{pf}}$: the log-FM degeneration corrections.

The maps: μ_e contracts the algebra labels at the two half-edges of e using the chiral bracket; Tr_e contracts using the bilinear pairing (supertrace over the two half-edges of e); μ_F applies the tree-level multilinear operation dictated by the planted forest F. The maps $(\xi_e)_*$ and $(\xi_F)_*$ are the pushforward along the normalisation of the boundary stratum.

The planted-forest differential as push–pull:

$$d_{\mathrm{pf}} \;=\; \sum_{\rho \in \mathsf{PF}^{\mathrm{rig}}} \epsilon_\rho\, (\kappa_\rho)_*\, \mathrm{pr}_\rho^* \otimes \mu_\rho\,,$$

where pr_ρ^* pulls back chains to the planted-forest stratum and $(\kappa_\rho)_*$ pushes forward along the Mok birational correspondence. This is Mok's degeneration formula, internalised in the convolution algebra: the push–pull correspondence on log-FM strata becomes a summand of the total differential.

3.10. $D^2 = 0$

The nilpotence $(D^{\log}_{\mathrm{mod}})^2 = 0$ holds by three cancellation mechanisms, each consuming a distinct geometric fact:

(i) *Vertex labels*: $d_{\check{C}}^2 = 0$, $d_{\mathrm{Ch}_\infty}^2 = 0$, and their anticommutator $\{d_{\check{C}}, d_{\mathrm{Ch}_\infty}\} = 0$ by the Ch_∞-algebra axioms on $\mathcal{A}^{\mathrm{ch}}_\infty$.

(ii) *One-edge expansions*: the anticommutator of a vertex differential ($d_{\check{C}}$ or d_{Ch_∞}) with an edge-contraction differential (d_{coll}, d_{sew}, d_{pf}, or $\hbar\Delta$) vanishes by the Leibniz rule for coderivations. Each edge contraction is a coderivation of the coalgebra structure, and the vertex differentials are derivations of the algebra structure.

(iii) *Codimension-two cancellation*: the sum of all products of two distinct edge-contraction operators vanishes. The codimension-two boundary of Mok's log-FM compactification is a normal-crossings divisor (Mok, Theorem 3.3.1), and each codimension-two face appears as the intersection of exactly two codimension-one faces. The two boundary contributions cancel because the face maps compose with opposite orientations. This is $\partial^2 = 0$ for the boundary of a manifold with corners, lifted through the residue correspondence.

The third mechanism is the nontrivial one. The normal-crossings property of Mok's compactification is essential: it guarantees that every codimension-two intersection is transverse and contributes to exactly one cancelling pair. Without normal crossings, the codimension-two faces could have excess intersection, and the cancellation would fail.

4. The universal Maurer–Cartan element

4.1. Γ-amplitudes and Taylor coefficients

Each stable graph produces a multilinear operation on the convolution algebra. The operation is built from three ingredients: chiral operations at vertices, propagators along edges, and geometric factors from the log-FM strata. The Taylor coefficients of the modular L_∞-structure are obtained by summing these Γ-amplitudes over all stable graphs of fixed genus and vertex count.

For a stable graph $\Gamma \in \mathsf{Gr}^{\mathrm{st}}$ with $|V(\Gamma)| = k$ vertices and elements $f_1, \ldots, f_k \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ assigned to those vertices, the Γ-*amplitude* assembles the chiral operations, the propagator, and the modular twisting morphism into a single multilinear operation. Let $\mu_\Gamma\colon \bigotimes_{v\in V(\Gamma)} A^{\otimes n(v)} \to A$ denote the graph composition: at each vertex v of genus $g(v)$ and valence $n(v)$, place the transferred chiral operation $m^{(h)}_{g(v),n(v)}$; along each internal edge, contract with the propagator $P_{\mathcal{A}} = \iota \circ \pi$ (homotopy retraction of the augmentation ideal to its cohomology). Write $\alpha^{\mathrm{mod}}_\Gamma$ for the component of the modular twisting morphism on Γ, and $\Delta^{\log}_\Gamma\colon C^{\log\mathrm{FM}}_{\mathrm{mod}} \to \bigotimes_{v\in V(\Gamma)} C^{\log\mathrm{FM}}_{g(v),n(v)}$ for the cooperadic cocomposition. The Γ-amplitude is:

$$\ell_\Gamma(f_1, \ldots, f_k) \;=\; \mu_\Gamma \circ (\alpha^{\mathrm{mod}}_\Gamma \otimes f_1 \otimes \cdots \otimes f_k) \circ \Delta^{\log}_\Gamma.$$

This is a k-linear map $\ell_\Gamma\colon (\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})^{\otimes k} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ of cohomological degree $2 - 2g(\Gamma) - k$, where $g(\Gamma) = b_1(\Gamma) + \sum_v g(v)$ is the total genus.

The modular L_∞ structure on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ has Taylor coefficients obtained by summing over all stable graphs of fixed genus and vertex count:

$$\ell^{(g)}_k \;=\; \sum_{\substack{\Gamma\in\mathsf{Gr}^{\mathrm{st}},\ |V(\Gamma)|=k \\ b_1(\Gamma)+\sum_v g(v)=g}} \frac{1}{|\operatorname{Aut}(\Gamma)|}\, \ell_\Gamma.$$

Here $\ell^{(g)}_k$ is a symmetric k-linear bracket of genus g. The full modular L_∞ structure is the formal sum $\ell = \sum_{g\geq 0}\sum_{k\geq 1} \hbar^g\, \ell^{(g)}_k/k!$, and the homotopy Jacobi identity is encoded by $D^2 = 0$ on the total convolution algebra.

Explicit low-order Taylor coefficients. The first several brackets admit direct geometric description. At tree level ($g = 0$), the binary bracket is the Lie bracket on cohomology transferred through the homotopy retraction data (π, ι, h):

$$\ell^{(0)}_2(\alpha, \beta) \;=\; \pi\big[m_2\big(\iota(\alpha), \iota(\beta)\big)\big].$$

The ternary bracket is a sum over binary trees with three leaves:

$$\begin{aligned}\ell_3^{(0)}(\alpha,\beta,\gamma) &= \sum_{T\in\mathsf{Trees}_3} |\operatorname{Aut}(T)|^{-1}\,(\pi\circ m_T^{(h)}\circ\iota^{\otimes 3})(\alpha,\beta,\gamma)\\ &= \pi\Big(m_2\big(h\,m_2(\iota\alpha,\iota\beta),\ \iota\gamma\big) + m_2\big(\iota\alpha,\ h\,m_2(\iota\beta,\iota\gamma)\big)\\ &\qquad + m_2\big(h\,m_2(\iota\alpha,\iota\gamma),\ \iota\beta\big)\Big).\end{aligned}$$

There are three binary trees in Trees_3 (one for each channel s, t, u), each with trivial automorphism group. The homotopy h propagates the intermediate state through $P_{\mathcal{A}}$ before re-composing. This is the classical homological perturbation lemma in its L_∞ incarnation.

The quaternary bracket sums over the fifteen labeled trees in Trees_4 (two unlabeled topologies: three balanced $((ab)(cd))$ and twelve caterpillar $(((ab)c)d)$):

$$\ell_4^{(0)}(\alpha,\beta,\gamma,\delta) = \sum_{T\in\mathsf{Trees}_4} \pm|\operatorname{Aut}(T)|^{-1}\,(\pi\circ m_T^{(h)}\circ\iota^{\otimes 4})(\alpha,\beta,\gamma,\delta).$$

Each tree T has two internal edges, hence two propagator insertions. The signs are Koszul signs from the symmetric monoidal structure. The two topologies reflect the two vertices of the graph complex $\mathsf{Gr}_4^{\mathrm{tree}}$.

The first genus-raised bracket is unary:

$$\ell_1^{(1)}(\alpha) \;=\; \operatorname{tr}(P_{\mathcal{A}}\circ\alpha\circ\delta^{\mathrm{ns}}) \;=\; \sum_i \langle e^i,\ \alpha(e_i)\rangle_{\mathrm{Sh}},$$

where $\{e_i\}$, $\{e^i\}$ is a dual basis for the cyclic pairing on $H^*(\bar{\mathcal{A}})$ and δ^{ns} is the non-separating node operator. Geometrically, this is the BV operator: the unique genus-one graph with one vertex and one self-loop. The trace is over the propagator contracting the two half-edges at the non-separating node of $\overline{\mathcal{M}}_{1,1}$.

The genus-one binary bracket accounts for the separating node of $\overline{\mathcal{M}}_{1,2}$:

$$\ell_2^{(1)}(\alpha,\beta) \;=\; \operatorname{tr}\big(P_{\mathcal{A}}\circ m_2(\iota\alpha,-)\circ\delta^{\mathrm{ns}}\circ\iota\beta\big) \;+\; (\alpha\leftrightarrow\beta).$$

These low-order brackets are the only data needed for all shadow computations through degree 4.

4.2. The bar-intrinsic construction

The element $\Theta_{\mathcal{A}}$ requires no equation to solve: it is extracted from a differential that already squares to zero.

The total bar differential $D_{\mathcal{A}}\colon \mathrm{B}^{\mathrm{mod}}(\mathcal{A})\to\mathrm{B}^{\mathrm{mod}}(\mathcal{A})$ acts on the modular bar complex. Decompose $D_{\mathcal{A}} = d_0 + \Theta_{\mathcal{A}}$, where $d_0 = d_{\mathrm{int}} + [\tau,-]$ is the tree-level local part: the internal differential of $\mathcal{A}$ plus the commutator with the tautological twisting morphism τ. The *universal Maurer–Cartan element* is the remainder:

$$\Theta_{\mathcal{A}} \;:=\; D_{\mathcal{A}} - d_0 \;=\; \sum_{\Gamma\in\mathsf{Gr}^{\mathrm{st}}} \frac{W_\Gamma^{\log\mathrm{FM}}}{|\operatorname{Aut}(\Gamma)|}\otimes\Phi_\Gamma^{\mathcal{A}} \;\in\; \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}). \tag{0.0.7}$$

Here $W_\Gamma^{\log \mathrm{FM}} \in H^*(\overline{\mathcal{M}}^{\log}_{g(\Gamma),n(\Gamma)})$ is the log-FM weight (the fundamental class of the boundary stratum indexed by Γ), and $\Phi_\Gamma^{\mathcal{A}} \in \mathrm{Hom}(\bar{\mathcal{A}}^{\otimes n(\Gamma)}, \Bbbk)$ is the chiral amplitude on Γ (propagators on internal edges, operations at vertices, cyclic trace at the output).

The MC property is immediate: $D_{\mathcal{A}}^2 = 0$ by the geometric fact that $\overline{\mathcal{M}}^{\log}_{g,n}$ has normal-crossings boundary; $d_0^2 = 0$ by construction; expanding $(d_0 + \Theta_{\mathcal{A}})^2 = 0$ gives

$$d_0\Theta_{\mathcal{A}} + \Theta_{\mathcal{A}}\, d_0 + \Theta_{\mathcal{A}}^2 = 0 \qquad \Longleftrightarrow \qquad [d_0, \Theta_{\mathcal{A}}] + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

The MC property is a formal consequence of $\partial^2 = 0$ on the moduli space of curves, requiring no simple-Lie restriction, no convergence hypothesis, and no choice of basis.

4.3. $\Theta_{\mathcal{A}}$ as universal modular twisting morphism

The Maurer–Cartan space of the modular convolution L_∞ algebra is identified with the space of modular twisting morphisms:

$$\mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}) \;\simeq\; \mathrm{Tw}_{\alpha^{\mathrm{mod}}}\Big(C^{\log \mathrm{FM}}_{\mathrm{mod}},\ \mathrm{End}_{\mathrm{Ch}_\infty}(A^{\mathrm{ch}}_\infty)\Big).$$

Under this identification, $\Theta_{\mathcal{A}}$ is the *universal modular twisting morphism*, the operadic analogue of the canonical flat connection on a principal bundle. A twisting morphism $\alpha\colon \mathcal{C} \to \mathcal{P}$ between a cooperad and an operad determines a twisted composition product $\mathcal{C} \circ_\alpha \mathcal{P}$ whose differential squares to zero if and only if α satisfies the MC equation. $\Theta_{\mathcal{A}}$ is the unique twisting morphism that simultaneously twists the modular bar complex and intertwines with the log-FM boundary structure.

The principal-bundle analogy is exact. The modular bar bundle $\mathrm{B}^{\log}_{\mathrm{mod}}(\mathcal{A}) \to \overline{\mathcal{M}}_{g,n}$ is the principal bundle; $\Theta_{\mathcal{A}}$ is the connection; the MC equation is the flatness condition; the shadow tower (Section 6) is the Chern–Weil theory. The structure group is the pro-unipotent MC gauge group $\exp(\mathfrak{g}^{\mathrm{mod},0}_{\mathcal{A}})$.

4.4. Tridegree, depth, and convergence

Each coefficient of $\Theta_{\mathcal{A}}$ carries a natural tridegree

$$(g, n, d) \;=\; (\text{loop genus},\ \text{external degree},\ \text{log boundary depth}),$$

where $d(\rho) = \#E_{\mathrm{sep}}(\Gamma_\rho) + \#E_{\mathrm{ns}}(\Gamma_\rho) + \mathrm{depth}_{\mathrm{pf}}(\Gamma_\rho)$ is the codimension of the log-FM stratum indexed by the degeneration pattern ρ. Here E_{sep} counts separating edges, E_{ns} counts non-separating edges, and $\mathrm{depth}_{\mathrm{pf}}$ counts planted-forest depth levels. The total depth equals the codimension of the stratum $D_\rho^{\log}$ in the log-FM compactification.

The depth filtration $F^d\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \bigoplus_{d' \geq d}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})_{*,*,d'}$ makes the convolution algebra complete:

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \;=\; \varprojlim_{d}\ \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}/F^d\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}.$$

The inverse limit $\Theta_{\mathcal{A}} = \varprojlim_r \Theta_{\mathcal{A}}^{\leq r}$ converges unconditionally: no analytic input, no growth estimate, no restriction on the algebra $\mathcal{A}$. The convergence is purely algebraic, guaranteed by the completeness of the depth filtration.

The tridegree decomposition organizes computation. At fixed (g, n), only finitely many stable graph types Γ contribute (by stability: $2g(\Gamma) - 2 + n(\Gamma) > 0$ forces a finite graph set). The depth filtration refines this further: at depth d, the stratum has codimension d in $\overline{\mathcal{M}}^{\log}_{g,n}$, and the amplitude $\Phi^{\mathcal{A}}_{\Gamma}$ has internal-edge count (hence propagator insertions) bounded by d.

4.5. The weight filtration

Decompose the MC equation by internal-edge count (equivalently, by total propagator insertions):

$$\Theta_{\mathcal{A}} = \sum_{r\geq 1} \Theta^{[r]}_{\mathcal{A}}, \qquad \Theta^{[r]}_{\mathcal{A}} = \sum_{\substack{\Gamma \in \mathsf{Gr}^{\mathrm{st}} \\ |E_{\mathrm{int}}(\Gamma)|=r}} \frac{W^{\log \mathrm{FM}}_{\Gamma}}{|\operatorname{Aut}(\Gamma)|} \otimes \Phi^{\mathcal{A}}_{\Gamma}.$$

The MC equation decomposes by weight into an inductive tower:

$$\begin{aligned}
&\text{Weight 1:} && [D_{\mathrm{loc}}, \Theta^{[1]}_{\mathcal{A}}] = 0,\\
&\text{Weight 2:} && [D_{\mathrm{loc}}, \Theta^{[2]}_{\mathcal{A}}] + \tfrac{1}{2}[\Theta^{[1]}_{\mathcal{A}}, \Theta^{[1]}_{\mathcal{A}}] = 0,\\
&\text{Weight 3:} && [D_{\mathrm{loc}}, \Theta^{[3]}_{\mathcal{A}}] + [\Theta^{[1]}_{\mathcal{A}}, \Theta^{[2]}_{\mathcal{A}}] = 0,\\
&\text{Weight } N\text{:} && [D_{\mathrm{loc}}, \Theta^{[N]}_{\mathcal{A}}] + \tfrac{1}{2}\sum_{\substack{i+j=N\\ i,j\geq 1}} [\Theta^{[i]}_{\mathcal{A}}, \Theta^{[j]}_{\mathcal{A}}] = 0.
\end{aligned}$$

At each stage, $[D_{\mathrm{loc}}, \Theta^{[N]}]$ is determined by the quadratic error from previous stages. The obstruction to extending from weight $N-1$ to weight N lies in $H^2(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}/F^{N+1}, D_{\mathrm{loc}})$. The bar-intrinsic construction solves all stages simultaneously.

Weight 1: boundary divisors. The weight-one component $\Theta^{[1]}_{\mathcal{A}}$ lives on boundary divisors of moduli spaces. In degree $(0,4)$: the three boundary points of $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, corresponding to the three channels $(12|34)$, $(13|24)$, $(14|23)$:

$$\Theta^{[1]}_{\mathcal{A}}\big|_{(0,4)} = [\delta_s] \otimes \Phi_s + [\delta_t] \otimes \Phi_t + [\delta_u] \otimes \Phi_u.$$

Here $[\delta_s]$, $[\delta_t]$, $[\delta_u]$ are the fundamental classes of the three boundary points $\{0,1,\infty\} \subset \mathbb{P}^1$, and Φ_s, Φ_t, Φ_u are the corresponding chiral four-point amplitudes (binary composition in channel s, t, u respectively). The weight-one MC equation $[D_{\mathrm{loc}}, \Theta^{[1]}] = 0$ at $(0,4)$ reads:

$$\Phi_s + \Phi_t + \Phi_u = F_3,$$

where F_3 is the Malikov–Schechtman Jacobiator homotopy, the failure of the chiral bracket to satisfy the Jacobi identity on the nose. This is the first Borcherds identity: the Jacobiator is a boundary in the bar complex.

In degree $(1,1)$: the single boundary divisor $\delta_{\mathrm{irr}} \subset \overline{\mathcal{M}}_{1,1}$ (the irreducible nodal cubic):

$$\Theta^{[1]}_{\mathcal{A}}\big|_{(1,1)} = [\delta_{\mathrm{irr}}] \otimes \Delta,$$

where $\Delta = \sum_i e_i \otimes e^i$ is the Casimir element of the cyclic pairing, the non-separating sewing propagator. The weight-one MC equation at $(1, 1)$ encodes the consistency of the one-loop trace with the modular structure.

In degree $(0, 5)$: the weight-one component restricts to the ten boundary divisors of $\overline{\mathcal{M}}_{0,5}$ (the Petersen graph), and the MC equation recovers the pentagon identity for the L_∞ structure. The entire A_∞ hierarchy is the tree-level, genus-zero projection of a single MC element.

Weight 2: the first planted-forest layer. At weight 2, the codimension-two strata of the log-FM compactification enter. These strata have two types: classical (products of two boundary divisors in $\overline{\mathcal{M}}_{g,n}$) and genuinely logarithmic (planted-forest strata of $\mathrm{FM}_n(X \mid D)$ with no counterpart in the classical Deligne–Mumford compactification). Mok's rigid planted-forest types enter at this weight.

The weight-two MC equation is:

$$[D_{\mathrm{loc}}, \Theta_{\mathcal{A}}^{[2]}] \;=\; -\tfrac{1}{2}[\Theta_{\mathcal{A}}^{[1]}, \Theta_{\mathcal{A}}^{[1]}].$$

The right-hand side is a quadratic expression in the weight-one data: the square of the boundary-divisor class. The planted-forest correction $\Theta_{\mathrm{pf}}^{[2]}$ is the new contribution: it records how collision limits in the planted-forest sector of $\mathrm{FM}_n(X \mid D)$ correct the naive square.

For the $\mathcal{W}_3$ central-parameter line, the self-coupling of the unique genus-one tangent direction at weight one is d_0-exact (the quadratic self-interaction is removable by gauge), so any nontrivial genus-two obstruction comes from this planted-forest sector. Cubic gauge triviality (Section 6.3): the weight-2 obstruction vanishes by gauge; the first genuine invariant is at weight 3.

4.6. The all-genus recursion

Project the MC equation to fixed genus g and degree n:

$$d\Theta_{\mathcal{A}}^{(g,n)} + \tfrac{1}{2}\sum_{\substack{g_1+g_2=g\\ n_1+n_2=n+2}} \ell_2^{(0)}(\Theta_{\mathcal{A}}^{(g_1,n_1)}, \Theta_{\mathcal{A}}^{(g_2,n_2)}) + \Delta_{\mathrm{ns}}\,\Theta_{\mathcal{A}}^{(g-1,n+2)} + \sum_{k\geq 3}\frac{1}{k!}\ell_k^{(\mathrm{tree})}(\Theta_{\mathcal{A}}, \ldots, \Theta_{\mathcal{A}})^{(g,n)} = 0.$$

The four terms have direct geometric meaning. *Internal differential*: the action of d on the bar labels at fixed (g, n). *Separating clutching*: the sum over splittings $g_1 + g_2 = g$, $n_1 + n_2 = n + 2$, identifying the separating node as a binary bracket $\ell_2^{(0)}$ applied to lower-genus and lower-degree MC data. *Non-separating BV operator*: the contraction Δ_{ns} lowers degree by 2 and raises genus by 1, applied to $\Theta^{(g-1,n+2)}$. *Planted-forest corrections*: the higher brackets $\ell_k^{(\mathrm{tree})}$ for $k \geq 3$ encode the genuinely logarithmic corrections from Mok's planted-forest strata.

This recursion determines $\Theta_{\mathcal{A}}^{(g,n)}$ inductively from lower (g', n') in lexicographic order: $g' < g$, or $g' = g$ and $n' < n$. The base cases are $\Theta_{\mathcal{A}}^{(0,3)}$ (the chiral bracket) and $\Theta_{\mathcal{A}}^{(0,4)}$ (the Jacobiator data from weight 1). The recursion terminates at each (g, n): only finitely many stable graphs contribute.

The clutching identity. On each rigid boundary stratum ρ of the log-FM compactification, the restriction of $\Theta_{\mathcal{A}}$ factors as a tensor product over vertices of the dual graph:

$$\Theta_{\mathcal{A}}\big|_{\rho} \;=\; (\kappa_\rho)_*\left(\bigotimes_{v\in V(S_\rho)} \Theta_{\mathcal{A},v}\right).$$

Here S_ρ is the stable dual graph of the stratum ρ, $\Theta_{\mathcal{A},v}$ is the MC element restricted to the irreducible component at vertex v, and κ_ρ is the clutching morphism: the amplitude on the degenerate curve equals the product of amplitudes on the irreducible components, sewn by propagators along the nodes. It is the MC-level incarnation of the sewing axiom for conformal field theory, and the starting point for the clutching law of the quartic shadow (Section 6.3).

Every invariant in this preface is a projection of $\Theta_{\mathcal{A}}$: the curvature κ, the spectral discriminant $\Delta_{\mathcal{A}}$, the shadow metric Q_L, the sewing amplitudes, the $\hat{A}$-genus, the Fredholm determinant, and the quartic contact invariant Q^{contact}. The five main theorems are five projections of $\Theta_{\mathcal{A}}$. The principal projections:

Projection of $\Theta_{\mathcal{A}}$	**Invariant**	**Section**
Scalar ($r = 2$, genus tower)	$\kappa(\mathcal{A})$, F_g, $\hat{A}$-genus	§2
Binary genus-0 residue	classical r-matrix, Yangian seed	§7.7
Binary Verdier dual	complementarity $Q_g \oplus Q^!_g$	§2
Bar-cobar counit	inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$	§1
Chiral Hochschild	$\mathrm{ChirHoch}^*(\mathcal{A})$, deformation ring	§9
Spectral discriminant	$\Delta_{\mathcal{A}}(x)$, genus-1 Hessian	§5
Cubic shadow ($r = 3$)	gauge-trivial obstruction	§6
Quartic resonance ($r = 4$)	$\mathfrak{Q}^{\text{contact}}$, clutching law	§6
MC tautological descent	shadow CohFT, $R^*(\overline{\mathcal{M}}_{g,n})$	§5
Primitive kernel	shell equations, branch BV	§5
Dirichlet–sewing lift	$S_{\mathcal{A}}(u)$, Euler–Koszul class	§8
Swiss-cheese colour decomposition	α_T, six projections (Vol. II)	§10

These five theorems are five projections of a single object: the universal Maurer–Cartan element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d^{(0)}_{\mathcal{A}}$ in the modular convolution algebra $\mathfrak{g}^{\text{mod}}_{\mathcal{A}}$. The existence of $\Theta_{\mathcal{A}}$ is encoded by the bar-cobar adjunction: Theorem A constructs the arena in which $\Theta_{\mathcal{A}}$ lives. Bar-cobar inversion proves that $\Theta_{\mathcal{A}}$ is a complete invariant: Theorem B recovers $\mathcal{A}$ from $\Theta_{\mathcal{A}}$. Verdier duality decomposes $\Theta_{\mathcal{A}}$ into complementary halves: Theorem C splits the genus tower into Lagrangian summands. The leading coefficient of $\Theta_{\mathcal{A}}$ is the modular characteristic: Theorem D extracts the scalar $\kappa(\mathcal{A})$ with $\hat{A}$-genus generating function. The coefficient ring of $\Theta_{\mathcal{A}}$ is the chiral Hochschild complex: Theorem H identifies the deformation ring.

The finite-order projections of $\Theta_{\mathcal{A}}$ form an obstruction tower $\Theta^{\leq 2}_{\mathcal{A}} \to \Theta^{\leq 3}_{\mathcal{A}} \to \Theta^{\leq 4}_{\mathcal{A}} \to \cdots$ whose generating function is algebraic of degree 2: on each primary line L, the weighted shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ for a quadratic polynomial Q_L determined by three seed invariants (κ, α, S_4). The inverse limit $\Theta_{\mathcal{A}} = \varprojlim_r \Theta^{\leq r}_{\mathcal{A}}$ is proved to exist and to be automatically Maurer–Cartan because $D^2_{\mathcal{A}} = 0$.

In three dimensions, $\Theta_{\mathcal{A}}$ extends to an open/closed Maurer–Cartan element $\Theta^{\text{oc}}_{\mathcal{A}} = \Theta_{\mathcal{A}} + \sum_j \mu^{M_j}$ that packages the complete holomorphic-topological quantum field theory on $\mathbb{C}_z \times \mathbb{R}_t$: the bar complex $B(\mathcal{A})$, coassociative over $(\mathrm{ChirAss})^!$, supplies the holomorphic factorization data; the derived center pair $(\mathcal{Z}^{\text{der}}_{\text{ch}}(\mathcal{A}), \mathcal{A})$ carries the $\mathrm{SC}^{\text{ch,top}}$-algebra structure encoding both closed and open colours; and $\mathcal{Z}^{\text{der}}_{\text{ch}}(\mathcal{A})$ is the universal closed sector (Volume II).

The bar complex thus organizes five mathematical structures into projections of one object: the genus expansion $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (uniform-weight) is a GUE matrix model at $N^2 = \kappa$ (Proposition **??**); the ordered bar $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ produces the quantum group $U_q(\mathfrak{g})$ via FRT reconstruction (Proposition **??** for $\mathfrak{sl}_2$); the R-matrix from the quantum group yields knot invariants (Jones, HOMFLYPT); the shadow metric Q_L defines a Bridgeland stability condition with universal wall phase $\pi/4$; and the Stokes constants $S_n/S_1 = (-1)^{n+1}n$ (Proposition **??**) encode the resurgent structure.

Corollary (MC replaces the sum over topologies). On the Koszul locus, the tree-level complex carries a contracting homotopy (from the PBW degeneration), so the genus-g component $\Theta^{(g)}$ is determined (up to the contractible ambiguity in the choice of strong deformation retract) by induction from $\Theta^{(0)}$. No free parameters appear at any genus; the contracting homotopy is unique up to homotopy, and distinct choices yield gauge-equivalent MC elements. No choice of bulk topologies is required. The genus expansion is computed algebraically from the genus-0 chiral bracket by the clutching recursion of Section 4.6, not approximated perturbatively. The Maloney–Witten problem (negative Fourier coefficients in the Poincaré series) does not arise: the MC equation determines each F_g uniquely by algebraic induction, without summing over bulk topologies at fixed genus. The non-perturbative sum over distinct three-manifold topologies is outside the scope of the MC framework.

5. The modular tangent complex and Chern–Weil theory

The MC element $\Theta_{\mathcal{A}}$ exists unconditionally and satisfies $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$. Its deformation theory is controlled by the modular tangent complex, obtained by twisting the convolution algebra by $\Theta_{\mathcal{A}}$ itself.

5.1. The L_∞-twisted differential

The MC element $\Theta_{\mathcal{A}}$ twists the convolution differential. For $x \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$:

$$d_{\Theta_{\mathcal{A}}}(x) \;:=\; \sum_{n\geq 0}\sum_{g\geq 0} \frac{\hbar^g}{n!}\, \ell^{(g)}_{n+1}(\Theta_{\mathcal{A}}^{\otimes n}, x).$$

Expanding:

$$d_{\Theta_{\mathcal{A}}}(x) \;=\; d(x) \;+\; [\Theta_{\mathcal{A}}, x] \;+\; \tfrac{1}{2!}\ell_3^{(0)}(\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}, x) \;+\; \hbar\, \ell_1^{(1)}(x) \;+\; \hbar\, \ell_2^{(1)}(\Theta_{\mathcal{A}}, x) \;+\; \cdots$$

In the strict chart (the dg Lie model $\mathrm{Conv}_{\mathrm{str}}$), this collapses to:

$$d_{\Theta_{\mathcal{A}}}(x) \;=\; d(x) + [\Theta_{\mathcal{A}}, x].$$

The identity $d^2_{\Theta_{\mathcal{A}}} = 0$ is equivalent to the MC equation for $\Theta_{\mathcal{A}}$.

5.2. The modular tangent complex

The *modular tangent complex at* $\Theta_{\mathcal{A}}$ is the twisted cochain complex:

$$T^{\mathrm{mod}}_{\Theta_{\mathcal{A}}} \;:=\; (\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}},\; d_{\Theta_{\mathcal{A}}}).$$

Its cohomology governs deformations of $\Theta_{\mathcal{A}}$:

$$\begin{aligned}
H^0(T^{\mathrm{mod}}_{\Theta_{\mathcal{A}}}) &= \text{infinitesimal automorphisms of } \Theta_{\mathcal{A}} && \text{(stabiliser)},\\
H^1(T^{\mathrm{mod}}_{\Theta_{\mathcal{A}}}) &= \text{first-order deformations of } \Theta_{\mathcal{A}} && \text{(tangent space)},\\
H^2(T^{\mathrm{mod}}_{\Theta_{\mathcal{A}}}) &= \text{obstructions to extending deformations} && \text{(obstruction space)}.
\end{aligned}$$

When $H^2 = 0$, the MC moduli is smooth at $\Theta_{\mathcal{A}}$. When $H^1 = 0$, the MC element is rigid: every nearby element is gauge-equivalent.

For a one-parameter family $\mathcal{A}_t$ varying continuously in level $k(t)$ or central charge $c(t)$, the tangent complex $T^{\mathrm{mod}}_{\Theta_{\mathcal{A}_t}}$ forms a local system over the parameter space, and the obstruction class traces a path in H^2. The vanishing locus of this obstruction path is the moduli of liftable deformations.

5.3. Three characteristic projections

Three successive projections of $\Theta_{\mathcal{A}}$ capture progressively finer structure: $\kappa(\mathcal{A})$ is a *number* (the leading scalar curvature, analogous to the first Chern class); $\Delta_{\mathcal{A}}(x)$ is a *polynomial* (encoding a finite spectrum of branch points, analogous to the Chern character); $\mathfrak{R}^{\mathrm{mod}}_4(\mathcal{A})$ is a *cohomology class* (the first invariant that sees nonlinear OPE interaction beyond the spectral data). Together, they seed the algebraicity theorem.

The scalar curvature $\kappa(\mathcal{A})$: the degree-two, genus-zero trace:

$$\kappa(\mathcal{A}) \;=\; \mathrm{tr}(\Theta_{\mathcal{A}})_{2,0} \;=\; \mathrm{tr}(\pi \circ m_2 \circ \iota^{\otimes 2} \circ \Delta_{\mathrm{sep}}) \;\in\; \Bbbk.$$

This is the first Chern class of the modular bar bundle: $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$, where λ_g is the Hodge class on $\overline{\mathcal{M}}_g$. The Mumford relation $\lambda_g^{g+1} = 0$ confines genus-g scalar obstructions to a space of dimension at most g.

Explicit values across the standard landscape:

Algebra $\mathcal{A}$	$\kappa(\mathcal{A})$	$\kappa(\mathcal{A}^!)$	**Sum rule**
Heisenberg $\mathcal{H}_k$	k	$-k$	0
Affine $\widehat{\mathfrak{g}}_k$	$\dfrac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}$	$-\dfrac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}$	0
Virasoro Vir_c	$c/2$	$(26-c)/2$	13
$\beta\gamma$ system	1	-1	0
Lattice V_Λ	$\mathrm{rank}(\Lambda)$	$-\mathrm{rank}(\Lambda)$	0
$\mathcal{W}_N(\widehat{\mathfrak{sl}}_N)$	$c\cdot(H_N-1)$	$c'\cdot(H_N-1)$	$K_N\cdot(H_N-1)$
Free fermion ψ	$1/4$	$-1/4$	0

The complementarity sum rule $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$ holds for Kac–Moody algebras, free fields, and lattice vertex algebras. For $\mathcal{W}$-algebras the sum is a nonzero constant determined by the anomaly ratio: $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$, and $\kappa(\mathcal{W}_{N,c}) + \kappa(\mathcal{W}_{N,c'}) = K_N\cdot(H_N-1)$ where $K_N = c + c'$ is the family constant. Self-duality occurs at $c = 13$ for Virasoro and $c = 50$ for $\mathcal{W}_3$.

The spectral discriminant $\Delta_{\mathcal{A}}(x)$ and the branch transport operator.

The second characteristic invariant is the characteristic polynomial of a finite-rank operator. The Koszul dual Hilbert series

$$h_{\mathcal{A}^!}(q) \;=\; \sum_{n\geq 0} \dim H^n(\bar{B}(\mathcal{A}))\, q^n,$$

records bar cohomology dimensions. For every standard family, $h_{\mathcal{A}^!}(q)$ is a *rational* function: it satisfies a holonomic recurrence relation of finite order d, and the transfer matrix of this recurrence is a $d \times d$ matrix T_{rec}.

Define the *spectral branch object* to be a finite-rank perfect complex $V_{\mathcal{A}}^{\mathrm{br}}$ together with a *branch transport operator* $T_{\mathrm{br},\mathcal{A}} \colon V_{\mathcal{A}}^{\mathrm{br}} \to V_{\mathcal{A}}^{\mathrm{br}}$ such that

$$\Delta_{\mathcal{A}}(x) \;:=\; \operatorname{sdet}\bigl(1 - x\, T_{\mathrm{br},\mathcal{A}}\bigr).$$

The rank of $V_{\mathcal{A}}^{\mathrm{br}}$ equals the degree of algebraicity of $h_{\mathcal{A}^!}(q)$: the order of the minimal recurrence. The eigenvalues of $T_{\mathrm{br},\mathcal{A}}$ are the reciprocal branch points of the analytic continuation of $h_{\mathcal{A}^!}$.

Three structural properties:

(i) $\Delta_{\mathcal{A}^!}(x) = \Delta_{\mathcal{A}}(x)$: the spectral discriminant is *self-dual*, unlike κ which is anti-symmetric for KM/free fields ($\kappa + \kappa' = 0$).

(ii) Exact factorization functors preserving quadratic OPE data preserve $T_{\mathrm{br},\mathcal{A}}$.

(iii) The spectral radius $\rho(T_{\mathrm{br},\mathcal{A}}) = \max_i |\lambda_i|$ equals $\dim \mathfrak{g}$ for affine Kac–Moody at generic level.

Example: $\mathfrak{sl}_3$. The Koszul dual Hilbert series of $\widehat{\mathfrak{sl}}_3$ satisfies a degree-3 recurrence. The transfer matrix is 3×3 with eigenvalues 8, $(3 + \sqrt{13})/2$, $(3 - \sqrt{13})/2$. The spectral discriminant is

$$\Delta_{\widehat{\mathfrak{sl}}_3}(x) \;=\; (1 - 8x)(1 - 3x - x^2).$$

The dominant eigenvalue $8 = \dim(\mathfrak{sl}_3)$ controls the exponential growth of bar cohomology dimensions. The remaining eigenvalues $(3 \pm \sqrt{13})/2$ are the roots of the DS discriminant factor, shared with the $\mathcal{W}_3$ generating function.

Example: Heisenberg. The Hilbert series $h_{\mathcal{H}_k^!}(q) = 1 + q$ is a polynomial (the recurrence has order 0 after the first term), so $\Delta_{\mathcal{H}_k}(x) = 1$: there is no non-scalar structure. This is the Gaussian archetype: κ alone determines everything.

Example: $\beta\gamma$. The Hilbert series is $\sqrt{(1{+}q)/(1{-}3q)}$; the discriminant is $\Delta_{\beta\gamma}(x) = (1 - 3x)(1 + x)$. The two roots at $x = 1/3$ and $x = -1$ coincide with $\widehat{\mathfrak{sl}}_2$ and the Virasoro algebra.

In the Chern–Weil dictionary, $\Delta_{\mathcal{A}}$ plays the role of the Chern character: the logarithmic derivative

$$\mathrm{ch}_r(\mathcal{A}) \;=\; \frac{1}{r!}\,\frac{d^r}{dt^r}\Big|_{t=0}\bigl[-\log \Delta_{\mathcal{A}}(t)\bigr] \;=\; \frac{1}{r}\,\mathrm{tr}(T^r_{\mathrm{br},\mathcal{A}})$$

recovers the power-sum symmetric functions of the eigenvalues.

The quartic shadow $\mathfrak{R}^{\mathrm{mod}}_{4,g,n}(\mathcal{A})$:

$$\mathfrak{R}^{\mathrm{mod}}_{4,g,n}(\mathcal{A}) \;=\; \mathrm{pr}_{4,g,n}(\Theta_{\mathcal{A}}) \;\in\; H^*(\overline{\mathcal{M}}_{g,n}) \otimes (\bar{\mathcal{A}}^\vee)^{\otimes n}_{\mathrm{cyc}}.$$

This is the first *nonlinear* characteristic class: the first invariant that cannot be recovered from κ and $\Delta_{\mathcal{A}}$. Its genus-zero, four-point specialisation is the quartic contact invariant $Q^{\mathrm{contact}}(\mathcal{A})$. Explicit:

$$Q^{\mathrm{contact}}_{\mathrm{Vir}} = \frac{10}{c(5c+22)}, \qquad Q^{\mathrm{contact}}_{\mathcal{H}} = 0, \qquad Q^{\mathrm{contact}}_{\beta\gamma} \neq 0.$$

For Heisenberg, the quartic vanishes (no cubic OPE pole). For $\beta\gamma$, the quartic is the first gauge-nontrivial shadow (the cubic is gauge-removable by Theorem **??**). For Virasoro, both cubic and quartic are nonvanishing.

5.4. The Chern–Weil dictionary

The analogy between $\Theta_{\mathcal{A}}$ and a connection 1-form is exact. The modular bar bundle $\mathrm{B}^{\log}_{\mathrm{mod}}(\mathcal{A}) \to \overline{\mathcal{M}}_{g,n}$ is a principal bundle for the gauge group of the convolution algebra; $\Theta_{\mathcal{A}}$ is its flat connection (flatness automatic from the bar-intrinsic construction); and the shadow tower is its Chern–Weil theory. Holonomy becomes factorisation homology, gauge equivalence becomes MC gauge, and the Weil homomorphism becomes the shadow algebra $\mathcal{A}^{\mathrm{sh}} = H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}})$.

Classical Chern–Weil	**Modular factorization**
Principal G-bundle $P \to M$	Modular bar bundle $\mathrm{B}^{\log}_{\mathrm{mod}}(\mathcal{A}) \to \overline{\mathcal{M}}_{g,n}$
Connection 1-form $\omega \in \Omega^1(P, \mathfrak{g})$	MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$
Curvature $\Omega = d\omega + \frac{1}{2}[\omega, \omega]$	MC equation $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$
Flatness $\Omega = 0$	Automatic (bar-intrinsic)
First Chern class $c_1 \in H^2(M; \mathbb{Z})$	Scalar curvature $\kappa(\mathcal{A}) = \mathrm{tr}(\Theta_{\mathcal{A}})_{2,0}$
Chern character $\mathrm{ch} = \mathrm{tr}\exp(\Omega/2\pi i)$	Spectral determinant $\Delta_{\mathcal{A}}(t) = \mathrm{sdet}(1 - tH_{d_\Theta})$
Higher Chern classes c_r	Shadows $\mathfrak{R}^{\mathrm{mod}}_{4,g,n}$, Sh_r for $r \geq 5$
Weil homomorphism $W\colon \mathrm{Sym}^*(\mathfrak{g}^*)^G \to H^*(M)$	Shadow algebra $\mathcal{A}^{\mathrm{sh}} = H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}})$
ad-invariant polynomial	Cyclic cocycle in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$
Cyclic pairing $\mathrm{tr}(XY)$	Verdier duality on $\mathrm{Ran}(X)$
Group multiplication $G \times G \to G$	Operadic composition in $\mathcal{P}^{\mathrm{ch}}$
Coproduct on $\mathcal{O}(G)$	Operadic cocomposition / clutching / degeneration
Lagrangian splitting $T^*G = \mathfrak{g} \oplus \mathfrak{g}^*$	Complementarity $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}(\mathcal{A}))$
Holonomy $\mathrm{Hol}(\gamma) \in G$	Factorisation homology $\int_\Sigma \mathcal{A}$
Gauge equivalence $\omega \sim g^{-1}\omega g + g^{-1}dg$	MC gauge in $\mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$
Transgression $\mathrm{CS}(\omega) \in \Omega^3$	Shadow obstruction tower (Section 6)
Flat sections $\nabla s = 0$	Twisted cochains $d_{\Theta_{\mathcal{A}}}(x) = 0$

5.5. The genus spectral sequence

The genus filtration $F^g L_{\mathrm{mod}} := \bigoplus_{g' \geq g} (\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})_{g',*,*}$ yields a convergent spectral sequence. Write $D_0 = D_{\mathrm{int}} + D_{\mathrm{sep}}$ for the genus-preserving part of the differential and $D_1 = D_{\mathrm{nsep}}$ for the genus-raising part. These satisfy $D_0^2 = 0$, $D_0 D_1 + D_1 D_0 = 0$, $D_1^2 = 0$ (a bicomplex structure).

The spectral sequence:

(i) $E_0^{p,q} = \mathrm{gr}^p_F L^{p+q}_{\mathrm{mod}}$ with differential $d_0 = [D_0, -]$.

(ii) $E_1^{p,*} = H^*(\mathrm{gr}^p_F L_{\mathrm{mod}}, d_0)$: the associated graded cohomology, genus by genus. $p = 0$: tree-level (unlooped homotopy transfer). $p = 1$: one-loop (BV trace and deformations). $p = 2$: genus-two shells (loops, clutching, planted forests).

(iii) $d_1 \colon E_1^{0,*} \to E_1^{1,*}$ is the genus-one obstruction Ob_1: the obstruction to lifting a tree-level MC element to one-loop. More generally, $d_r \colon E_r^{p,*} \to E_r^{p+r,*-r+1}$ encodes Ob_r.

(iv) $E_\infty^{p,q} \cong \mathrm{gr}^p_F H^{p+q}(L_{\mathrm{mod}})$: the associated graded of the full modular cohomology.

Convergence follows from completeness of the depth filtration and pronilpotence of the weight filtration: the $\varprojlim^1$ term vanishes by Mittag-Leffler (finite-dimensionality in each conformal weight).

This spectral sequence is *distinct* from the PBW spectral sequence (which decomposes by conformal weight within a fixed genus). PBW computes *what* bar cohomology is; the genus spectral sequence determines *whether* the MC element extends across genera. For Heisenberg, it degenerates at E_1. For Virasoro, $d_1 \neq 0$ and nontrivial differentials persist at every page.

5.6. Homotopy invariance and functoriality

The shadow algebra $\mathcal{A}^{\mathrm{sh}} = H_\bullet(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$ is a homotopy invariant of the chiral algebra. For any ∞_α-quasi-isomorphism $f \colon \mathcal{A} \xrightarrow{\sim} \mathcal{A}'$, the induced map on MC ∞-groupoids

$$f_* \colon \mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}) \xrightarrow{\sim} \mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}'})$$

is a weak homotopy equivalence (a quasi-isomorphism on tangent complexes at corresponding MC elements). In particular:

- The scalar $\kappa(\mathcal{A})$ is a quasi-isomorphism invariant.
- The spectral determinant $\Delta_{\mathcal{A}}(t)$ is a quasi-isomorphism invariant.
- The full shadow obstruction tower $(\kappa, \Delta, \mathfrak{C}, \mathfrak{Q}, \mathrm{Sh}_5, \mathrm{Sh}_6, \dots)$ consists of quasi-isomorphism invariants.
- The MC element $\Theta_{\mathcal{A}}$ is a quasi-isomorphism invariant up to gauge: $f_*(\Theta_{\mathcal{A}})$ and $\Theta_{\mathcal{A}'}$ are gauge-equivalent in $\mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}'})$.

The assignment $\mathcal{A} \mapsto (\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, \Theta_{\mathcal{A}})$ is a functor from chiral algebras (with ∞_α-quasi-isos inverted) to MC moduli problems: modular Koszul duality is a derived invariant.

The convolution L_∞ algebra $\mathrm{Conv}_\infty(C, \mathcal{P})$ is functorial in each slot separately, but *not* simultaneously as a bifunctor (§3). Functoriality of $\mathcal{A} \mapsto \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ proceeds through the source slot, with the target operad held fixed.

5.7. The shadow CohFT and topological recursion

Projected to $\overline{\mathcal{M}}_{g,n}$, the MC equation produces tautological classes and tautological relations. The MC element $\Theta_{\mathcal{A}}$ determines shadow tautological classes $\tau_{g,n}(\mathcal{A}) \in R^*(\overline{\mathcal{M}}_{g,n+1})$ that assemble into a cohomological field theory (Theorem ??): the factorisation axiom holds by the clutching law for $\Theta_{\mathcal{A}}$, the Σ_n-equivariance by the symmetry of stable-graph summation, and the flat identity conditionally (when the vacuum lies in V).

Projecting the MC equation $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ to the (g,n)-component yields a tautological relation in $R^*(\overline{\mathcal{M}}_{g,n})$ at each (g,n) (Theorem ??):

$$\sum_{\text{sep/nsep}} \xi_{\text{sep},*}\big(\tau_{g_1,|S|}(\mathcal{A}) \cdot \tau_{g_2,|S^c|}(\mathcal{A})\big) + \delta_{\text{pf}}^{(g,n)} = 0.$$

WDVV and Mumford from MC. At genus 0, degree 3: the MC tautological relation on $\overline{\mathcal{M}}_{0,4}$ reduces to the WDVV equations (Proposition ??). At degree 2, genus $g \geq 2$: substituting $\tau_{g,2}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \pi^*\lambda_g$ recovers the Mumford λ-class clutching relation on $\overline{\mathcal{M}}_g$ (Proposition ??).

The complementarity propagator $P_{\mathcal{A}} \colon H^*(\mathcal{A}) \otimes H^*(\mathcal{A}^!) \to \mathbb{C}$ is the Givental R-matrix of the shadow CohFT; when it carries a flat unit (all standard families with vacuum in V; Theorem ??), Teleman reconstruction recovers $\{\tau_{g,n}\}$ at all (g,n) from genus-0 data and $P_{\mathcal{A}}$ (Theorem ??). Eynard–Orantin topological recursion is the MC equation applied to the shadow obstruction tower (Corollary ??).

MC datum	Tautological output
$\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$	shadow CohFT $\Omega_{g,n}^{\mathcal{A}}$
$D^2 = 0$ on $\mathrm{FM}^{\log}$	CohFT axioms
$D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at (g,n)	tautological relation
$[\Theta, \Theta]\vert_{\text{sep}}$	WDVV / associativity
$\hbar\Delta(\Theta)\vert_{\text{nsep}}$	Mumford λ-class relations
$P_{\mathcal{A}}$	Givental R-matrix
MC recursion	EO topological recursion

The Pixton shadow bridge. For class-M algebras (infinite shadow obstruction tower), $\Theta_{\mathcal{A}}$ generates tautological classes at every genus. At genus 2–3, the relations $\tau_{g,n}(\mathrm{Vir}_c)$ for generic c produce elements of the Pixton ideal in $R^*(\overline{\mathcal{M}}_{g,n})$: Pixton's tautological ring relations arise as shadows of the Virasoro MC equation. Theorem ??: on the semisimple locus, MC-descended relations together with Mumford relations generate the Pixton ideal. Theorem ?? (Pixton ideal generation on the semisimple locus); the non-semisimple frontier remains open.

The Böcherer bridge. At genus 2, the shadow tautological class $\tau_{2,0}(\mathrm{Vir}_c)$ for lattice vertex algebras computes a central L-value via the Böcherer conjecture (now a theorem of Furusawa–Morimoto). For the Leech lattice: $\chi_{12} = \mathrm{SK}(f_{22})$ (the Saito–Kurokawa lift of the unique weight-22 cuspidal eigenform), and the MC-derived second Fourier coefficient $c_2 \approx -1.918 \times 10^{-6} \neq 0$ is the first nonzero central L-value produced by the shadow obstruction tower.

5.8. The primitive kernel and cofree reduction

The MC element $\Theta_{\mathcal{A}}$ lives in the cofree conilpotent coalgebra underlying the modular bar construction. By Milnor–Moore for modular operads, $\Theta_{\mathcal{A}}$ is determined by its restriction to primitive corollas and rigid cutting strata, the *primitive logarithmic modular kernel*

$$\mathfrak{K}_{\mathcal{A}} = \sum_{g\geq 0,\, n\geq 2} K_{g,n}^{\mathcal{A}} + \sum_{\rho\in\mathrm{Rig}} K_{\rho}^{\mathcal{A}},$$

where $K_{g,n}^{\mathcal{A}}$ are genus-g corolla coefficients and $K_{\rho}^{\mathcal{A}}$ are rigid boundary cutting terms. The fundamental equivalence (Theorem **??**):

$$(Q_{\mathcal{A}}^{\mathrm{mod}})^2 = 0 \qquad\Longleftrightarrow\qquad d\,\mathfrak{K}_{\mathcal{A}} + \mathfrak{K}_{\mathcal{A}} \star \mathfrak{K}_{\mathcal{A}} + \hbar\,\Delta_{\mathrm{ns}}(\mathfrak{K}_{\mathcal{A}}) = 0,$$

where $\star$ is the primitive pre-Lie product from log-FM residues, replaces the genus-by-genus obstruction programme by a single equation on primitive data.

Shell equations. Projecting the primitive master equation to genus 2 and genus 3 shells gives explicit constraints:

$$\begin{aligned}
R_2(\mathfrak{K}_{\mathcal{A}}) &= \Delta_{\mathrm{ns}}(K_1) + \tfrac{1}{2}[K_1, K_1] + \textstyle\sum_{\rho\in\mathrm{Rig}_2} R_{\rho}^{\mathcal{A}}(K_{0,2}^{\mathcal{A}}),\\
R_3(\mathfrak{K}_{\mathcal{A}}) &= \Delta_{\mathrm{ns}}(K_2) + [K_1, K_2] + \tfrac{1}{3!}\ell_3(K_1, K_1, K_1) + \textstyle\sum_{\rho} R_{\rho}^{\mathcal{A}}(\ldots).
\end{aligned}$$

For the Heisenberg algebra, only the BV term Δ_{ns} survives. For affine Kac–Moody, the bracket $[K_1, K_1]$ contributes the cubic shadow. For $\beta\gamma$, cubic gauge triviality kills the bracket term and the rigid stratum contributes the quartic contact invariant. For Virasoro and $\mathcal{W}_N$, all three terms are active at every genus.

The branch BV action. The finite-rank branch-active quotient $V_{\mathcal{A}}^{\mathrm{br}} = V_{\mathcal{A}}^{+} \oplus V_{\mathcal{A}}^{-}$, a quotient of the modular tangent complex, spanned by spectral generators, carries a BV master action

$$S_{\mathcal{A}}^{\mathrm{br}}(x,\xi) = \tfrac{1}{2}\Omega_{\mathcal{A}}(K_{\mathcal{A}}^{\mathrm{br}}x, x) + \sum_{m+n\geq 3} \frac{1}{m!\,n!}\mu_{m,n}^{\mathcal{A}}(x^{\otimes m}; \xi^{\otimes n}),$$

satisfying $\hbar\,\Delta_{\mathcal{A}}^{\mathrm{br}} S_{\mathcal{A}}^{\mathrm{br}} + \tfrac{1}{2}\{S_{\mathcal{A}}^{\mathrm{br}}, S_{\mathcal{A}}^{\mathrm{br}}\}_{\mathrm{BV}} = 0$, where $\{-,-\}_{\mathrm{BV}}$ is the odd Poisson bracket induced by the invariant pairing on $\mathcal{A}$. The *metaplectic half-density* $\delta_{\mathcal{A}} \in 1 + x\,\Bbbk[\![x]\!]$ satisfies $\delta_{\mathcal{A}}^2 = \Delta_{\mathcal{A}} = \det(1 - x\,T_{\mathcal{A}}^{\mathrm{br,red}})$: the spectral discriminant is the square of a canonical half-density.

5.9. Entanglement from the modular characteristic

The bar coalgebra carries a natural bipartite structure: its coproduct splits a configuration of points into two halves, one on each side of a spatial cut. This splitting connects the modular characteristic directly to entanglement.

For a spatial bipartition of an interval of length L with UV cutoff ε, the scalar projection of $\Theta_{\mathcal{A}}$ restricted to the bipartite cut yields the Ryu–Takayanagi formula:

$$S_{\mathrm{EE}}(\mathcal{A}) = \frac{c(\mathcal{A})}{3}\log\frac{L}{\varepsilon},$$

where $c(\mathcal{A})$ is the central charge. (For the Virasoro algebra, $c = 2\kappa$; the relationship between c and κ is family-dependent.) Complementarity gives the entanglement sum rule:

$$S_{\mathrm{EE}}(\mathcal{A}) + S_{\mathrm{EE}}(\mathcal{A}^!) \;=\; \frac{c(\mathcal{A}) + c(\mathcal{A}^!)}{3} \cdot \log \frac{L}{\varepsilon}.$$

For the Virasoro family $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$, this gives $c + c' = 26$ and the universal sum $26/3 \cdot \log(L/\varepsilon)$.

The four shadow-depth classes (G/L/C/M) classify entanglement complexity: class G algebras (Gaussian, $r_{\max} = 2$) exhibit area-law entanglement only; class L ($r_{\max} = 3$) adds subleading logarithmic corrections from the cubic shadow; class C ($r_{\max} = 4$) adds a contact correction from the quartic invariant; and class M ($r_{\max} = \infty$) carries an infinite tower of entanglement corrections, one for each shadow.

The entanglement programme (G11–G16). Six theorem targets extend the scalar result. *G11*: the Ryu–Takayanagi formula above, proved from the scalar projection of $\Theta_{\mathcal{A}}$. *G12* (Koszulness = exact quantum error correction): the Knill–Laflamme condition for quantum error correction is the Lagrangian isotropy condition from the complementarity pairing; shadow depth $r_{\max}$ equals the number of redundancy channels; the twelve Koszulness characterizations (K1–K12; eight genuinely independent bidirectional equivalences, one listed Barr–Beck–Lurie consequence, one one-way chiral Hochschild consequence, one conditional Lagrangian criterion, one one-directional $\mathcal{D}$-module purity) translate into a twelve-fold code dictionary; failure of Koszulness is failure of the code. *G13*: the replica trick from the MC equation (the n-sheeted cover computation is the n-fold bar coproduct). *G14*: modular flow from the tangent complex (the Tomita modular operator is the L_∞-twisted differential $d_{\Theta_{\mathcal{A}}}$, restricted to the bipartite subalgebra). *G15*: the quantum extremal surface as the shadow connection Ward identity: the QES equation is the stationarity condition for the shadow metric $Q_L(t)$ restricted to the entanglement cut. *G16*: the Page curve from the shadow connection at $c = 13$ (the self-dual point): the Page transition occurs when the shadow growth rate $\rho(\mathcal{A})$ crosses the entanglement threshold, and the scrambling time is controlled by the spectral discriminant $\Delta_{\mathcal{A}}$.

6. The shadow obstruction tower

6.1. The extension tower

Can $\Theta_{\mathcal{A}}$ be approximated by finitely many degrees, and what controls the passage from one finite approximation to the next? The total weight filtration answers both.

Define the *total weight* $w(g, r, d) := 2g - 2 + r + d$, where g is loop genus, r is degree, and d is log boundary depth. The total weight filtration $F^N \mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}} := \bigoplus_{w \geq N} (\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}})_{g,r,d}$ is an exhaustive, separated, complete filtration. Here $\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}}$ denotes the *ambient* (planted-forest-extended) modular convolution algebra, the extension of $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ by the planted-forest coefficient algebra $\mathcal{G}_{\mathrm{pf}}$ that incorporates log boundary strata. The *modular extension tower*:

$$\mathcal{E}_{\mathcal{A}}(N) := \mathrm{MC}\big(\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}} / F^{N+1}\big), \qquad N \geq 1.$$

Restriction maps $\mathcal{E}_{\mathcal{A}}(N+1) \to \mathcal{E}_{\mathcal{A}}(N)$ truncate the top weight. A *full lift* is a compatible point of the inverse system:

$$\Theta_{\mathcal{A}} \;\in\; \varprojlim_{N} \mathcal{E}_{\mathcal{A}}(N).$$

The bar-intrinsic construction provides such a point unconditionally. The shadow obstruction tower is the study of the extension tower $\mathcal{E}_{\mathcal{A}}(1) \leftarrow \mathcal{E}_{\mathcal{A}}(2) \leftarrow \mathcal{E}_{\mathcal{A}}(3) \leftarrow \cdots$ and the obstructions that arise at each stage.

6.2. Obstruction classes and the all-degree master equation

Define the truncated MC elements $\Theta_{\mathcal{A}}^{\leq r}$ by projecting $\Theta_{\mathcal{A}}$ to total weight $\leq r$. The *shadow obstruction tower* is the sequence

$$\Theta_{\mathcal{A}}^{\leq 2}, \quad \Theta_{\mathcal{A}}^{\leq 3}, \quad \Theta_{\mathcal{A}}^{\leq 4}, \quad \Theta_{\mathcal{A}}^{\leq 5}, \quad \ldots$$

At each truncation, the MC equation fails by a remainder:

$$D\Theta_{\mathcal{A}}^{\leq r} + \tfrac{1}{2}[\Theta_{\mathcal{A}}^{\leq r}, \Theta_{\mathcal{A}}^{\leq r}] \;=\; -o_{r+1}(\mathcal{A}),$$

where the *obstruction class* $o_{r+1}(\mathcal{A})$ lives in the $(r+1)$-graded piece:

$$o_{r+1}(\mathcal{A}) \;\in\; Z^2\!\left(\frac{F^{r+1}\mathfrak{g}_{\mathcal{A}}^{\mathrm{amb}}}{F^{r+2}\mathfrak{g}_{\mathcal{A}}^{\mathrm{amb}}},\; d_{\leq r}\right).$$

The truncation extends to $\Theta^{\leq r+1}$ if and only if $[o_{r+1}(\mathcal{A})] = 0$ in $H^2(F^{r+1}/F^{r+2}, d_{\leq r})$. The bar-intrinsic construction guarantees that extension always exists, so the obstruction class is always exact; the shadow obstruction tower records the gauge choices at each stage.

The all-degree master equation at degree r is:

$$\nabla_H(\mathrm{Sh}_r(\mathcal{A})) + o^{(r)}(\mathcal{A}) = 0, \qquad r \geq 3,$$

where ∇_H is the Hodge covariant derivative (the linearised MC operator twisted by the previous stages) and $\mathrm{Sh}_r(\mathcal{A})$ is the degree-r shadow (the named invariant extracted from $\Theta^{\leq r}$). This equation determines Sh_r uniquely modulo gauge if $H^1(F^r/F^{r+1}, d_{\leq r-1}) = 0$, and up to the image of a gauge map otherwise.

6.3. Cubic gauge triviality and the canonical quartic

> **Cubic gauge triviality.** When $H^1(F^3\mathfrak{g}/F^4\mathfrak{g},\, d_2) = 0$, the cubic MC term $\Theta_{\mathcal{A}}^{[3]}$ is gauge-trivial: there exists a gauge transformation $\exp(\xi_3)$ with $\xi_3 \in F^3$ such that $\exp(\xi_3) \cdot \Theta_{\mathcal{A}}^{\leq 3}$ has vanishing weight-3 component.

After this gauge transformation, the quartic obstruction class $[\mathfrak{Q}(\mathcal{A})] \in H^2(F^4\mathfrak{g}/F^5\mathfrak{g},\, d_2)$ is *canonical*: independent of homotopy retraction data, cubic gauge, and MC gauge orbit representative. This is the first characteristic class not polynomial in κ.

The mechanism is standard Postnikov: $o_3(\mathcal{A}) \in H^2(F^3/F^4, d_2)$ is exact (since $\Theta_{\mathcal{A}}$ exists), and the cochain $\Theta^{[3]}$ lifting it is unique up to coboundary when $H^1(F^3/F^4, d_2) = 0$. After removal, the quartic error is a well-defined cohomology class: trivial H^1 at level n makes the H^2 class at level $n+1$ canonical.

The clutching law for the quartic class follows from Mok's log FM degeneration formula. For a degeneration $C \rightsquigarrow C_0 = C_1 \cup_p C_2$:

$$\mathfrak{Q}(\mathcal{A})\big|_{C_0} \;=\; \sum_{a+b=4} \mathfrak{Q}^{(a)}(\mathcal{A})\big|_{C_1} \cdot \mathfrak{Q}^{(b)}(\mathcal{A})\big|_{C_2} \;+\; \text{planted-forest correction.}$$

The planted-forest correction is genuinely logarithmic, with no counterpart in the classical Deligne–Mumford setting.

The quartic contact invariant:

$$Q^{\mathrm{contact}}_{\mathrm{Vir}} \;=\; \frac{10}{c(5c+22)} \;\in\; H^2(F^4/F^5, d_2).$$

The denominator $c(5c+22)$ has physical meaning: $c = 0$ is the logarithmic threshold; $c = -22/5$ is the minimal model accumulation point $\mathcal{M}(2,5)$. The numerator $10 = \binom{5}{2}$ counts the degree-four Catalan channels.

6.4. The Hamilton–Jacobi generating function

For a multi-generator algebra ($\mathcal{W}_3$ with generators T (weight 2) and W (weight 3)), the deformation space is two-dimensional. The Riccati ODE lifts to a Hamilton–Jacobi PDE. Write $U = \sum_{s\geq 3} \mathrm{Sh}_s(x_T, x_W)$. The master equation becomes

$$2\,E(U) + \tfrac{1}{2}\|\nabla U\|^2_H \;=\; R(x_T, x_W),$$

with Euler operator $E = x_T\,\partial_T + x_W\,\partial_W$, kinetic energy $\|\nabla U\|^2_H = (2/c)(\partial_T U)^2 + (3/c)(\partial_W U)^2$, and R a polynomial of degree ≤ 4 encoding the three inputs $(\kappa, \mathfrak{C}, \mathfrak{Q})$. The gradient $dU\colon X \to T^*X$ embeds the deformation space as a Lagrangian submanifold of T^*X ($X = \mathbb{C}\cdot x_T \oplus \mathbb{C}\cdot x_W$, kinetic form positive-definite for $c > 0$), by the Hamilton–Jacobi theorem applied to the stationary equation with Euler operator E as degree-filtration weight. Whether the genus expansion $\Theta_{\mathcal{A}} = \sum_{g\geq 0} \hbar^g \Theta^{(g)}_{\mathcal{A}}$ admits a quantisation in which U is the leading semiclassical term is an open problem: it would require constructing a quantum Hamiltonian whose WKB expansion has leading term U and whose quantum corrections reproduce the genus-g shadow amplitudes.

On the T-line ($x_W = 0$): conformal weight forces the transverse momentum $p_W = (\partial_W U)|_{x_W=0}$ to vanish, and the Hamilton–Jacobi PDE collapses to the Riccati ODE. On the W-line ($x_T = 0$): the cubic shadow $\mathrm{Sh}_3 = 2x_T^3 + 3x_T x_W^2$ has nonzero normal derivative $(\partial_T \mathrm{Sh}_3)|_{x_T=0} = 3x_W^2$, so the transverse kinetic energy is nonzero. The first inter-channel coupling correction enters at degree 6:

$$\delta_6^{W\text{-line}} \;=\; \frac{576(5c+38)}{c^3(5c+22)^3}.$$

The T-line is autonomous; the W-line is not. The asymmetry is intrinsic: it traces to the conformal-weight asymmetry between T (weight 2) and W (weight 3).

6.5. The four depth classes

The *shadow depth* $r_{\max}(\mathcal{A})$ is the smallest integer r such that $o_{r+1}(\mathcal{A}) = 0$ for all subsequent levels, or $r_{\max} = \infty$ if the tower never terminates. Equivalently, $r_{\max}$ is the largest degree at which the shadow obstruction tower carries a nontrivial invariant.

The standard landscape partitions into exactly four shadow-depth classes; no other values occur on any one-dimensional primary slice. The single-line dichotomy (Theorem ??): the shadow metric $Q_L(t)$ is quadratic in (κ, α, S_4), and its factorisation type (perfect square or irreducible) determines whether the tower terminates.

Class	Symbol	$r_{\max}$	$\ell_3^{(0)}$, $\ell_4^{(0)}$	Potential type	Examples
Gaussian	G	2	0, 0	quadratic	Heisenberg $\mathcal{H}_k$, free fermion ψ
Lie/tree	L	3	$\neq 0$, 0	cubic	Affine $\widehat{\mathfrak{sl}}_N$, current algebras
Contact	C	4	0, $\neq 0$	quartic	$\beta\gamma$, bc ghosts
Mixed	M	∞	$\neq 0$, $\neq 0$	non-polynomial	Virasoro, $\mathcal{W}_N$, $\mathcal{W}_\infty$

The depth class is determined by the OPE structure:

- **Gaussian** ($r_{\max} = 2$): the transferred brackets $\ell_k^{(0)}$ vanish for $k \geq 3$ on bar cohomology, so the tower terminates at κ. Two mechanisms: Heisenberg has no simple pole ($S_3 = 0$, no Lie bracket); the free fermion has a simple pole but fermionic antisymmetry forces $S_3 = 0$ (parity on three-point configuration space). Sewing is by Fredholm determinant (Heisenberg) or Pfaffian (fermion). The potential is quadratic.
- **Lie/tree** ($r_{\max} = 3$): the OPE has a simple pole generating a Lie bracket, but the transferred quaternary bracket $\ell_4^{(0)}$ vanishes on bar cohomology. The ternary bracket is the last nontrivial shadow. The potential is cubic.
- **Contact** ($r_{\max} = 4$): the simple-pole bracket vanishes on bar cohomology (as in Heisenberg), but the quaternary bracket survives. The first nontrivial shadow is $\mathfrak{Q}(\mathcal{A})$, and the tower terminates there. The potential is quartic.
- **Mixed** ($r_{\max} = \infty$): both cubic and quartic shadows are nontrivial, and their interaction (the Poisson bracket $\{\mathfrak{C}, \mathfrak{Q}\}_H$) generates a nonvanishing quintic obstruction, which in turn generates a sextic, and so on. The tower never terminates. The potential is transcendental.

Shadow depth is not Koszulness. All four depth classes contain chirally Koszul algebras. Heisenberg (depth 2), affine at generic level (depth 3), $\beta\gamma$ (depth 4), and Virasoro (depth ∞) are all chirally Koszul. Shadow depth classifies the *complexity* of Koszul algebras, not Koszulness itself.

The *operadic complexity theorem* (Theorem **??**): the shadow depth equals the A_∞-depth equals the L_∞-formality level:

$$r_{\max}(\mathcal{A}) \;=\; A_\infty\text{-depth}(\mathcal{A}) \;=\; L_\infty\text{-formality level}(\mathcal{A}).$$

Here A_∞-depth is the largest r with $m_r \neq 0$ in the minimal A_∞ model; L_∞-formality level is the largest r with a nontrivial r-ary bracket on bar cohomology (not removable by L_∞-isomorphism). The shadow obstruction tower equals the L_∞ formality obstruction tower at all degrees (Theorem **??**).

The algebraic depth gap. In classical homotopy theory, the depth of the Massey product tower is an arbitrary nonnegative integer: a Postnikov system can terminate at any stage. The shadow obstruction tower is more constrained. The algebraicity of the shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ forces the depth into a four-element set; the intermediate values $d_{\text{alg}} = 3, 4, 5, \ldots$ are excluded by the degree-2 polynomial Q_L. This gap is the shadow-tower analogue of the fact that a quadratic form is either degenerate or nondegenerate, with no intermediate rank condition.

The four shadow-depth classes **G**, **L**, **C**, **M** correspond to algebraic depths $d_{\text{alg}} = 0, 1, 2, \infty$ respectively, and no other values occur.

Proposition 0.0.3 (Algebraic depth gap). *Let $\mathcal{A}$ be a chirally Koszul algebra in the standard landscape with $\kappa(\mathcal{A}) \neq 0$. Then*

$$d_{\mathrm{alg}}(\mathcal{A}) \;\in\; \{0,\,1,\,2,\,\infty\}.$$

No finite value $d_{\mathrm{alg}} \geq 3$ is realized.

The proof uses the Riccati algebraicity of the shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ ((0.0.3)): when the discriminant $\Delta = 8\kappa(\mathcal{A}) \cdot S_4 = 0$, the polynomial Q_L is a perfect square, forcing H to be a cubic polynomial and $d_{\mathrm{alg}} \leq 2$. When $\Delta \neq 0$, the square root $\sqrt{Q_L}$ is irrational and the Taylor coefficients S_r are generically nonzero for all $r \geq 2$: the tower never terminates, so $d_{\mathrm{alg}} = \infty$. The gap at $d_{\mathrm{alg}} = 3$ (and all finite values ≥ 3) is a consequence of the quadratic nature of Q_L: a polynomial of degree 2 is either a perfect square or irreducible, with no intermediate possibility. The depth gap is therefore not a classification *theorem* about specific families but a structural consequence of the MC equation itself: any chiral algebra whose shadow generating function satisfies the Riccati relation inherits the gap.

6.6. The genus–degree bigrading

The MC equation decomposes along two orthogonal axes (Remark **??**). The *genus axis* extends $\Theta_{\mathcal{A}}$ from genus 0 to 1 to 2, controlled by the genus spectral sequence (Construction **??**). The *degree axis* extends $\Theta_{\mathcal{A}}^{\leq 2}$ to $\Theta_{\mathcal{A}}^{\leq 3}$ to $\Theta_{\mathcal{A}}^{\leq 4}$, controlled by cyclic obstruction classes. At fixed degree 2 the genus tower is governed by κ alone; at fixed genus 0 each new shadow coefficient introduces independent data. The axes couple through the *visibility formula* (Corollary **??**): $g_{\min}(S_r) = \lfloor r/2 \rfloor + 1$, so S_r first contributes at genus $\lfloor r/2 \rfloor + 1$ and two new shadow coefficients enter at each genus.

6.7. Genus-two shells

Genus 2 is the first level where all three boundary strata interact. The genus-two MC component decomposes as:

$$\Theta_{\mathcal{A}}^{(2)} \;=\; \underbrace{\Theta^{(2)}_{\mathrm{loop}\circ\mathrm{loop}}}_{\text{iterated BV: two self-loops}} \;+\; \underbrace{\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}}}_{\text{clutching}\,\times\,\text{BV: one separating + one self-loop}} \;+\; \underbrace{\Theta^{(2)}_{\mathrm{pf}}}_{\text{planted-forest: Mok's log-FM strata}}.$$

The iterated-BV term $\Theta^{(2)}_{\mathrm{loop}^2}$ involves the graph with one vertex of genus 0 and two self-loops (the "figure-eight"). Its amplitude is the iterated trace:

$$\Theta^{(2)}_{\mathrm{loop}^2} \;=\; \tfrac{1}{2}\,\mathrm{tr}_1\,\mathrm{tr}_2\big(P_{\mathcal{A}}^{\otimes 2} \circ m_{0,4}^{(h)} \circ \Delta^{\otimes 2}\big).$$

The separating-times-loop term $\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}}$ is the graph with two vertices connected by a separating edge, one vertex carrying a self-loop. Its amplitude is:

$$\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}} \;=\; \mathrm{tr}_{\mathrm{ns}}\big(P_{\mathcal{A}} \circ \ell_2^{(0)}(\Theta^{(1,\bullet)}, \Theta^{(1,\bullet)})\big).$$

The planted-forest term is the genuinely logarithmic contribution, present only when the algebra has nontrivial cubic or quartic shadows. It arises from depth-2 strata of $\mathrm{FM}_n(X \mid D)$ that have no classical counterpart.

The family-by-family structure:

	$\Theta^{(2)}_{\mathrm{loop}^2}$	$\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}}$	$\Theta^{(2)}_{\mathrm{pf}}$	**Depth class**
Heisenberg $\mathcal{H}_k$	✓	--	--	G
Affine $\widehat{\mathfrak{g}}_k$	✓	✓	--	L
$\beta\gamma$ **system**	✓	--	--	C
Virasoro Vir_c	✓	✓	✓	M
$\mathcal{W}_N$	✓	✓	✓	M

For Heisenberg: only the iterated BV contributes (two independent loop contractions on a free field). No separating clutching (the Lie bracket on bar cohomology is central) and no planted forests (no higher interactions). For affine: the separating-times-loop term appears (the Lie bracket creates a nontrivial genus-one correction after one clutching), but planted forests are absent (the cubic shadow terminates the tower). For Virasoro and $\mathcal{W}_N$: all three terms contribute, and the planted-forest term carries the quartic resonance data. For $\beta\gamma$: both the separating and planted-forest terms vanish: the separating term because the Lie bracket is trivial on bar cohomology, the planted-forest term because the quartic contact invariant $\mu_{\beta\gamma} = 0$ (rank-one abelian rigidity). Only the iterated BV shell contributes, as for Heisenberg, but via contact termination rather than Gaussian collapse.

6.8. The genus loop operator

The *genus loop operator* $\Lambda_P : \mathrm{Sym}^r(\bar{\mathcal{A}}^\vee)_{\mathrm{cyc}} \to \mathrm{Sym}^{r-2}(\bar{\mathcal{A}}^\vee)_{\mathrm{cyc}}$ contracts two legs of a degree-r cyclic cochain with the propagator:

$$\Lambda_P(\phi)(a_1, \ldots, a_{r-2}) = \sum_i \phi(a_1, \ldots, a_{r-2}, e_i, e^i),$$

where $\{e_i\}, \{e^i\}$ is a dual basis for the cyclic pairing. This raises genus by 1 and lowers degree by 2: the shadow-level incarnation of the BV operator Δ_{ns}.

Explicit Virasoro computation. On the Virasoro primary line, the conformal-weight-2 bar cohomology is one-dimensional, spanned by the class x dual to the stress tensor T. The Killing form is $H = \frac{c}{2}x^2$ and the propagator is $P = \frac{2}{c}$. The quartic contact invariant is $Q^{\mathrm{contact}}_{\mathrm{Vir}} \cdot x^4$ with $Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c+22)]$. The genus loop operator applied to this quartic:

$$\Lambda_P\left(\frac{10}{c(5c+22)} \cdot x^4\right) = \binom{4}{2} \cdot \frac{2}{c} \cdot \frac{10}{c(5c+22)} \cdot x^2 = \frac{120}{c^2(5c+22)}\, x^2.$$

Here $\binom{4}{2} = 6$ counts leg selections and $2/c$ is the propagator. The result is the *genus-one Hessian correction* $\delta^{(1)}_{H,\mathrm{Vir}} = 120/[c^2(5c+22)] \cdot x^2$.

The *loop ratio*:

$$\rho^{(1)}_{\mathrm{Vir}} = \frac{\Lambda_P(Q \cdot x^4)}{\kappa \cdot x^2} = \frac{120/[c^2(5c+22)]}{c/2} = \frac{240}{c^3(5c+22)}.$$

This ratio is not determined by $\kappa = c/2$: it detects the quartic structure of the Virasoro algebra through its genus-one projection. The asymptotic behaviour $\rho^{(1)} \sim 48/c^4$ for $c \to \infty$ reflects the weakening of quartic corrections in the semiclassical limit.

6.9. Quintic forcing and infinite towers

The quintic obstruction determines whether the tower terminates at degree 4.

Virasoro. The cubic shadow $\mathfrak{C}_{\mathrm{Vir}}$ and the quartic contact invariant $Q^{\mathrm{contact}}_{\mathrm{Vir}}$ are both nonzero. Their Poisson bracket under the Hodge pairing gives the quintic obstruction:

$$o^{(5)}_{\mathrm{Vir}} = \{\mathfrak{C}_{\mathrm{Vir}}, Q^{\mathrm{contact}}_{\mathrm{Vir}}\}_H = \frac{480}{c^2(5c+22)}\, x^5 \neq 0.$$

Since $o^{(5)} \neq 0$ for all $c \neq 0, -22/5$, the Virasoro shadow tower does not terminate at any finite degree. The tower is infinite: $r_{\max}(\mathrm{Vir}_c) = \infty$. The same holds for every $\mathcal{W}_N$ algebra ($N \geq 2$): the cubic and quartic shadows coexist and their bracket forces an infinite sequence of higher obstructions.

Heisenberg. $\mathfrak{C}_{\mathcal{H}} = 0$ and $Q^{\mathrm{contact}}_{\mathcal{H}} = 0$. The quintic obstruction $o^{(5)} = \{0, 0\}_H = 0$. Every higher obstruction also vanishes. The tower terminates at $r_{\max} = 2$.

Affine. $\mathfrak{C}_{\widehat{\mathfrak{g}}} \neq 0$ but $Q^{\mathrm{contact}}_{\widehat{\mathfrak{g}}} = 0$ (the quaternary bracket vanishes on bar cohomology by the PBW theorem for Kac–Moody algebras). The quintic obstruction $o^{(5)} = \{\mathfrak{C}, 0\}_H = 0$. The tower terminates at $r_{\max} = 3$.

$\beta\gamma$**.** $\mathfrak{C}_{\beta\gamma} = 0$ (the Lie bracket is trivial on bar cohomology) but $Q^{\mathrm{contact}}_{\beta\gamma} \neq 0$. The quintic obstruction $o^{(5)} = \{0, Q\}_H = 0$. The tower terminates at $r_{\max} = 4$.

The pattern:

	$\mathfrak{C}$	$\mathfrak{Q}$	$o^{(5)} = \{\mathfrak{C}, \mathfrak{Q}\}_H$	$r_{\max}$
Heisenberg	0	0	0	2
Affine	$\neq 0$	0	0	3
$\beta\gamma$	0	$\neq 0$	0	4
Virasoro	$\neq 0$	$\neq 0$	$\neq 0$	∞

Finite termination requires $\mathfrak{C} = 0$ or $\mathfrak{Q} = 0$. When both are nonzero, their bracket generates the quintic, the quintic and quartic generate the sextic, and the induction continues without bound: a transcendental potential on the MC moduli.

The shadow growth rate. For class M algebras (infinite shadow depth), the Taylor coefficients S_r of $\sqrt{Q_L}$ grow as $S_r \sim A \cdot \rho(\mathcal{A})^r \cdot r^{-5/2}\cos(r\theta + \varphi)$, where the *shadow growth rate* $\rho(\mathcal{A}) = \sqrt{9\alpha^2 + 2\Delta}\,/\,(2|\kappa|)$ is a continuous invariant on the modular deformation space. Self-dual Virasoro ($c = 13$) has $\rho \approx 0.467$. The critical cubic $5c^3 + 22c^2 - 180c - 872 = 0$ has root $c^* \approx 6.125$: the tower converges for $c > c^*$ and diverges for $c < c^*$.

The shadow partition function. The double sum $Z^{\mathrm{sh}}_{\mathcal{A}} = \sum_{g,r} \hbar^{2g-2}\, Z^{(r)}_g$ converges absolutely. Genus decay is at rate $1/(2\pi)^{2g}$ (Faber–Pandharipande λ_g-integral); degree decay is at rate $\rho^r \cdot r^{-5/2}$ (Darboux analysis of the shadow metric). By contrast, the string partition function diverges as $(2g-2)!$. For Virasoro at $c = 26$: $\rho \approx 0.234$, and Z^{sh} has radius of absolute convergence in both $\hbar$ and degree.

6.10. Independent sum factorisation

For a direct sum $L = L_1 \oplus L_2$ of Lie conformal algebras with vanishing mixed OPE ($[L_{1\lambda} L_2] = 0$), all shadow invariants separate:

$$\begin{aligned}
\kappa(L_1 \oplus L_2) &= \kappa(L_1) + \kappa(L_2) && \text{(additive)},\\
T^{\mathrm{br}}(L_1 \oplus L_2) &= T^{\mathrm{br}}(L_1) \oplus T^{\mathrm{br}}(L_2) && \text{(direct sum)},\\
\Delta_{L_1 \oplus L_2}(t) &= \Delta_{L_1}(t) \cdot \Delta_{L_2}(t) && \text{(multiplicative)},\\
R_4^{\mathrm{mod}}(L_1 \oplus L_2) &= R_4^{\mathrm{mod}}(L_1) + R_4^{\mathrm{mod}}(L_2) && \text{(additive)}.
\end{aligned}$$

More generally, the full MC element factorises:

$$\Theta_{L_1 \oplus L_2} \;=\; \Theta_{L_1} \hat{\otimes} 1 + 1 \hat{\otimes} \Theta_{L_2} \;\in\; \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[L_1] \,\hat{\otimes}\, \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[L_2],$$

where $\hat{\otimes}$ is the completed tensor product with respect to the depth filtration; at the level of partition functions, $Z_{L_1 \oplus L_2} = Z_{L_1} \cdot Z_{L_2}$.

The factorisation principle reduces computation for composite algebras to their primitive summands. Every algebra in the standard landscape is either primitive (Heisenberg, Virasoro, $\beta\gamma$) or built from primitive pieces by direct sum plus extension (affine = current algebra over simple Lie, lattice = exponential extension of Heisenberg, $\mathcal{W}_N$ = quantum Drinfeld–Sokolov reduction of affine). The full shadow obstruction tower of any composite algebra is determined by its primitive constituents. Drinfeld–Sokolov reduction is the *shadow-depth escalator* (Principle **??**): it sends current-algebra systems (class **L**, $r_{\max} = 3$, finite towers) to $\mathcal{W}$-algebra systems (class **M**, $r_{\max} = \infty$, infinite towers). The nonlinearity of the Dirac bracket on the Slodowy slice creates the quartic and all higher shadows; no finite truncation survives the reduction.

6.11. The representation-theoretic perspective

The categorical logarithm $\eta_{ij} = d\log(z_i - z_j)$ of Section 1 is the integral kernel of a tautology: $D^2_{\mathcal{A}} = 0$ on the modular bar complex. From this identity, three structure theorems emerge. *Algebraicity*: on each primary slice L, the shadow generating function satisfies $H(t)^2 = t^4 Q_L(t)$, where Q_L is a quadratic polynomial in three invariants κ, α, S_4 (Theorem **??**). *Formality*: the genus-0 shadow obstruction tower is the L_∞ formality obstruction tower (Theorem **??**); the four depth classes G/L/C/M are the corresponding formality types, while positive-genus corrections form a separate quantum layer. *Complementarity*: for a Koszul pair $(\mathcal{A}, \mathcal{A}^!)$, the summands $\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!)$ are Lagrangian (Theorem C).

The discriminant $\Delta = 8\kappa S_4$ governs the single-line dichotomy: on each primary deformation line, $\Delta = 0$ forces tower termination (L_∞-formality) and $\Delta \neq 0$ forces an infinite tower (intrinsic non-formality). The contact class C ($r_{\max} = 4$, $\beta\gamma$) escapes via stratum separation. No invariant of the classical bar complex of a graded algebra sees Δ: the discriminant requires curve geometry. The Arnold relation holds exactly at genus 0; at genus $g \geq 1$ it acquires a period-matrix correction, and Δ governs whether the resulting tower terminates.

Classical Koszul duality (Priddy, Beilinson–Ginzburg–Soergel) embeds into the genus 0, degree 2, $\Delta = 0$ stratum via formal-disk restriction (the formal, class-G case); the E_1/E_∞ distinction and Fulton–MacPherson geometry are already absent over a bare point. The R-matrix of quantum groups arises as the genus-0 shadow $\mathrm{Sh}_{0,2}(\Theta_{\widehat{\mathfrak{g}}_k})$ (§7.7). At the critical level $k = -h^\vee$, the scalar class vanishes ($\kappa = 0$)

and bar cohomology recovers the Feigin–Frenkel center $\mathrm{Fun}(\mathrm{Op}_{^L\mathfrak{g}})$; the Langlands programme lives on this orthogonal axis, not in the MC element but in the cohomology that emerges on the uncurved scalar locus.

7. The standard landscape

The machine of Sections 1–6 accepts any chiral algebra as input. The seven families below exhaust the standard Lie-theoretic landscape: each exhibits a different bar-complex structure, shadow archetype, genus-two shell profile, and completion behaviour. No two families share all four invariants; taken together, they realise every depth class (G/L/C/M), every curvature sign, and every pattern of genus-two shell activation.

For each family we display the complete OPE, bar complex with explicit differential, Koszul dual, shadow obstruction tower with depth classification, modular characteristic with complementarity, genus-two shell profile, and any structural feature unique to that family.

7.1. Heisenberg

The Gaussian archetype: shadow depth 2, the tower terminates at degree 2, and every higher-genus invariant is determined by κ alone.

A single bosonic field α of conformal weight 1. The OPE is

$$\alpha(z)\,\alpha(w) \;\sim\; \frac{k}{(z-w)^2}.$$

No simple pole: the singularity is purely of order 2. The mode algebra is the Heisenberg Lie algebra $[\alpha_m, \alpha_n] = km\,\delta_{m+n,0}$, and the vertex algebra $\mathcal{H}_k$ is the vacuum module at level $k \neq 0$. Central charge $c = 1$.

Bar complex. The augmentation ideal $\bar{\mathcal{H}}_k = \mathcal{H}_k/\mathbb{C}\cdot\mathbf{1}$ is spanned by normally ordered monomials in α and its derivatives. The bar construction is the tensor coalgebra on the desuspension:

$$\bar{B}(\mathcal{H}_k) \;=\; T^c\big(s^{-1}\bar{\mathcal{H}}_k\big), \qquad d_{\bar{B}}\big(s^{-1}a_1|\cdots|s^{-1}a_n\big) \;=\; \sum_{i<j} \pm\, s^{-1}a_1|\cdots|s^{-1}\,\mathrm{Res}_{z_i=z_j}\big(a_i(z_i)\,a_j(z_j)\,\eta_{ij}\big)|\cdots|s^{-1}a_n.$$

Because the only singularity is a double pole (producing a scalar), $d_{\bar{B}}$ acts by contraction: it pairs two tensor factors and replaces them by a scalar. Explicitly:

$$\begin{aligned} d_{\bar{B}}(s^{-1}\alpha) &= 0,\\ d_{\bar{B}}(s^{-1}\alpha|s^{-1}\alpha) &= k\cdot s^{-1}\mathbf{1} = 0 \quad \text{(in the reduced complex)},\\ d_{\bar{B}}(s^{-1}\partial^m\alpha|s^{-1}\partial^n\alpha) &= k\binom{m+n+1}{m}\cdot s^{-1}\mathbf{1} = 0 \quad (m+n\geq 1). \end{aligned}$$

Bar cohomology concentrates:

$$H^1\big(\bar{B}(\mathcal{H}_k)\big) = \mathbb{C}\cdot\alpha^*, \qquad H^n\big(\bar{B}(\mathcal{H}_k)\big) = 0 \;\text{ for }\; n\geq 2.$$

PBW concentration: bar cohomology is one-dimensional in degree 1.

Koszul dual. The bar cohomology $\mathcal{A}^{\text{i}} = H^*(\bar{B}(\mathcal{H}_k))$ is a one-dimensional graded coalgebra concentrated in degree 1. Taking graded linear duals gives the Koszul dual algebra:

$$\mathcal{H}_k^! \;:=\; (\mathcal{A}^{\text{i}})^\vee \;=\; \big(\mathrm{Sym}^{\mathrm{ch}}(V^*),\; m_0 = -k\omega\big),$$

where $V^* = \mathbb{C} \cdot \alpha^*$ is the dual generator space and $m_0 = -k\omega$ is the curvature: a curving element, not a differential. The curved A_∞-relation reads $m_1^2(a) = [m_0, a]$ (commutator, with sign). The Koszul dual of Heisenberg is *not* Heisenberg: $\mathcal{H}_k^!$ is a *curved* symmetric algebra, while $\mathcal{H}_k$ is a *flat* vertex algebra. Chiral Koszul duality is not an involution but a Verdier-type reflection.

Shadow obstruction tower. The OPE has no simple pole: $\alpha_{(0)}\alpha = 0$. Every transferred operation beyond degree 2 vanishes, so the only stable graphs contributing to (0.0.7) have all vertices of valence 2; stability forces $g(v) \geq 1$ at each bivalent vertex. Since the only nonvanishing OPE coefficient is the scalar k and the desuspended generator $s^{-1}\eta$ has odd degree, the parity argument (Theorem **??**) forces the graph sum to collapse:

$$\Theta_{\mathcal{H}_k} \;=\; k \cdot \eta \otimes \Lambda,$$

where $\eta = d\log(z_1 - z_2)$ is the propagator on $\mathrm{FM}_2(X)$ and $\Lambda = \lambda_1 \in H^2(\overline{\mathcal{M}}_{1,1})$ is the generator of $H^*(\overline{\mathcal{M}}_{1,1})$. The five-component differential collapses completely:

- $d_{\mathrm{Ch}_\infty} = 0$: no higher transferred operations,
- $d_{\mathrm{coll}} = 0$: collisions produce scalars only,
- $d_{\mathrm{pf}} = 0$: no planted-forest contributions,
- only $d_{\check{C}} + d_{\mathrm{loop}}$ survive.

Cubic shadow: $\mathfrak{C}(\mathcal{H}_k) = 0$. Quartic contact shadow: $Q^{\mathrm{contact}}(\mathcal{H}_k) = 0$. All higher obstructions vanish. Shadow depth 2. Gaussian archetype.

Modular characteristic and complementarity.

$$\kappa(\mathcal{H}_k) = k, \qquad \kappa(\mathcal{H}_k^!) = -k, \qquad \kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = 0.$$

A single scalar $\kappa(\mathcal{H}_k) = k$ determines every projection: the generating function is $\sum_{g\geq 1} F_g\, x^{2g} = k\big(\frac{x/2}{\sin(x/2)} - 1\big) = k\big(\hat{A}(ix) - 1\big)$, giving $F_1 = k/24$, $F_2 = 7k/5760$, $F_3 = 31k/967680$, …(the Taylor coefficients of the $\hat{A}$-genus). At genus g, $Q_g(\mathcal{H}_k)$ is a polynomial of degree $2g$ in k, computed entirely from iterated loop contractions of the single propagator η.

Genus-two shell profile. Only the loop∘loop shell contributes:

$$\Theta^{(2)}_{\mathrm{loop}\circ\mathrm{loop}} \neq 0, \qquad \Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}} = 0, \qquad \Theta^{(2)}_{\mathrm{pf}} = 0.$$

Every genus correction reduces to iterated loop contractions of a single propagator: the Gaussian measure of chiral algebra theory.

7.2. Affine Kac–Moody

The Lie/tree archetype: shadow depth 3, the cubic shadow is the Lie bracket, and the Jacobi identity kills all higher obstructions. The first family with a nonabelian bar differential.

Let $\mathfrak{g}$ be a simple Lie algebra of rank ℓ, dimension $d = \dim \mathfrak{g}$, dual Coxeter number $h^\vee$, with structure constants $f^{ab}{}_c$ and normalised Killing form κ^{ab}. The affine vertex algebra $\widehat{\mathfrak{g}}_k$ has generators J^a, $a = 1, \ldots, d$, of conformal weight 1, with OPE

$$J^a(z)\, J^b(w) \;\sim\; \frac{f^{ab}{}_c\, J^c(w)}{z - w} \;+\; \frac{k\,\kappa^{ab}}{(z-w)^2}.$$

Both pole orders are present. The simple pole carries the Lie bracket; the double pole carries the invariant inner product at level k. The mode algebra is the affine Lie algebra $\widehat{\mathfrak{g}}$:

$$[J^a_m, J^b_n] \;=\; f^{ab}{}_c\, J^c_{m+n} \;+\; km\,\kappa^{ab}\,\delta_{m+n,0}.$$

Bar complex. The simple-pole residue contributes a bracket component to $d_{\bar{B}}$:

$$d_{\bar{B}}(s^{-1}J^a | s^{-1}J^b) \;=\; f^{ab}{}_c\, s^{-1}J^c \;+\; k\,\kappa^{ab} \cdot (\text{curvature term}).$$

The bracket component is the Chevalley–Eilenberg differential of $L\mathfrak{g}$; the curvature component is the central extension by k. The identity $d_{\bar{B}}^2 = 0$ requires the *Borcherds identity*, not Jacobi: the n-fold residue relations from all pole orders simultaneously, replacing $[d, [d, -]] = 0$.

PBW spectral sequence. Filter $\bar{B}(\widehat{\mathfrak{g}}_k)$ by conformal weight. The spectral sequence $E_1^{p,q}(g)$ at genus g has E_1 page given by the bar cohomology of the associated graded, a free-field computation in the Heisenberg algebra $\mathcal{H}_k^{\otimes d}$. PBW concentration at all genera is proved: $E_\infty^{p,q}(g) = 0$ for $q \neq 0$. The enriched genus-g E_1 page carries the structure of a free resolution of the diagonal $\widehat{\mathfrak{g}}_k$-bimodule, with the Hodge bundle $\mathbb{E}$ supplying the curvature twist at each genus.

Sugawara construction. At non-critical level $k \neq -h^\vee$, the stress tensor is

$$T(z) \;=\; \frac{1}{2(k+h^\vee)} \sum_{a=1}^{d} {:}J^a(z)J^a(z){:}\,, \qquad c \;=\; \frac{k\,\dim\mathfrak{g}}{k+h^\vee}.$$

The Sugawara field is composite, not in the generating set. At $k = -h^\vee$, the construction is *undefined* (c does not exist, not "c diverges"). The Feigin–Frenkel center $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \mathrm{Fun}(\mathrm{Op}_{{}^L\mathfrak{g}}(D))$ is a commutative vertex algebra of ${}^L\mathfrak{g}$-opers on the formal disk. At critical level, the bar complex is flat but noncommutative.

Shadow obstruction tower. The cubic shadow is the Lie bracket itself:

$$\mathfrak{C}(\widehat{\mathfrak{g}}_k) \;=\; \big[f_{abc}\, J^a \otimes J^b \otimes J^c\big] \;\in\; H^1(F^3\mathfrak{g}^{\mathrm{mod}}/F^4\mathfrak{g}^{\mathrm{mod}},\, d_2).$$

The quartic obstruction $o^{(4)}(\widehat{\mathfrak{g}}_k) = 0$: the Jacobi identity on $\mathfrak{g}$ kills the only candidate quartic cocycle via $f^{ab}{}_e f^{ec}{}_d$ + cyclic $= 0$. The quintic obstruction vanishes by the same mechanism. Shadow depth 3. Lie/tree archetype.

Modular characteristic and complementarity.

$$\kappa(\widehat{\mathfrak{g}}_k) \;=\; \frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}, \qquad \kappa(\widehat{\mathfrak{g}}^!_k) \;=\; -\frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}.$$

Complementarity: $\kappa(\widehat{\mathfrak{g}}_k) + \kappa(\widehat{\mathfrak{g}}^!_k) = 0$ for all $k \neq -h^\vee$. The Feigin–Frenkel involution

$$k \longleftrightarrow -k - 2h^\vee$$

is Koszul duality in the modular sense: the level of the dual algebra is the Langlands dual level. The complementarity sum $c(k) + c(-k - 2h^\vee) = 2 \dim \mathfrak{g}$ becomes the complementarity relation evaluated on this family.

Genus-two shell profile. At genus 2, the loop∘loop and sep∘loop shells both contribute:

$$\Theta^{(2)}_{\text{loop}\circ\text{loop}} \neq 0, \qquad \Theta^{(2)}_{\text{sep}\circ\text{loop}} \neq 0, \qquad \Theta^{(2)}_{\text{pf}} = 0.$$

The planted-forest shell vanishes because the quartic shadow is zero and planted forests require degree ≥ 4. Separating degeneration activates through the cubic shadow (the Lie bracket) composed with the loop contraction: the genus-two correction factors through $\mathfrak{g} \otimes \mathfrak{g} \xrightarrow{[-,-]} \mathfrak{g} \xrightarrow{\text{loop}} \mathbb{C}$.

Special feature: Feigin–Frenkel at critical level. At $k = -h^\vee$, the vertex algebra $\widehat{\mathfrak{g}}_{-h^\vee}$ has a large center. The center $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \operatorname{Fun}(\operatorname{Op}_{{}^L\mathfrak{g}}(D))$ is the ring of opers on the formal disk for the Langlands dual. The bar complex at critical level is *flat* ($m_0^{(1)} = 0$): the modular characteristic $\kappa(-h^\vee) = 0$. Critical-level chiral Koszul duality is the geometric Langlands correspondence at the level of the bar construction.

7.3. Virasoro

The mixed archetype: shadow depth infinity, the tower never terminates, and new invariants appear at every degree. The self-coupling $T_{(1)}T = 2T$ drives all nonlinear complexity.

One generator T of conformal weight 2. The OPE is

$$T(z)\,T(w) \;\sim\; \frac{c/2}{(z-w)^4} \;+\; \frac{2\,T(w)}{(z-w)^2} \;+\; \frac{\partial T(w)}{z-w}.$$

Three pole orders: the fourth-order pole carries c; the second-order pole carries $T_{(1)}T = 2T$; the simple pole carries ∂T. The mode algebra:

$$[L_m, L_n] \;=\; (m-n)L_{m+n} \;+\; \frac{c}{12}(m^3 - m)\,\delta_{m+n,0}.$$

Bar complex. The bar differential on the Virasoro vacuum module has three components:

$$d_{\bar{B}}(s^{-1}T|s^{-1}T) = 2\,s^{-1}T + \frac{c}{2}\cdot(\text{curvature}) \quad (\text{from } (z-w)^{-2} \text{ and } (z-w)^{-4} \text{ poles}),$$
$$d_{\bar{B}}(s^{-1}T|s^{-1}\partial T) = s^{-1}\partial T + 3\,s^{-1}T \quad (\text{from } (z-w)^{-1} \text{ and } (z-w)^{-2} \text{ poles}).$$

At higher tensor degree, the bar differential on $s^{-1}T|s^{-1}T|s^{-1}T$ produces terms involving the self-coupling $T_{(1)}T = 2T$ composed with itself, the first instance of a bar differential that is genuinely quadratic in the structure constants. PBW concentration still holds: at every genus g, the spectral sequence collapses to $E_\infty^{p,0}$.

Koszul dual.

$$\mathrm{Vir}^!_c \;=\; \mathrm{Vir}_{26-c}\,.$$

The duality is an involution on the one-parameter Virasoro family. The fixed point is $c = 13$, *not* $c = 26$. Self-duality: $\mathrm{Vir}_c \cong \mathrm{Vir}_c^!$ iff $c = 13$. The central charge 26 plays no special role in Koszul duality; its appearance in string theory is the vanishing of $\kappa(\mathrm{Vir}_{26}) = 13$, $\kappa(\mathrm{Vir}_{26}^!) = \kappa(\mathrm{Vir}_0) = 0$: the *anomaly-cancellation* condition, not the self-duality condition.

The same-family shadow Vir_{26-c} is the depth-zero resonance shadow: the image of the finite-dimensional resonance truncation R_{Vir}, not the final Koszul dual in the completed sense. The genuine $\mathcal{W}_\infty$-type dual requires the full resonance-filtered completion.

Shadow obstruction tower. The cubic shadow is determined by the self-coupling $T_{(1)}T = 2T$:

$$\mathfrak{C}_{\mathrm{Vir}} \neq 0.$$

The quartic contact shadow, extracted from the four-point collision residue on $\overline{C}_4(\mathbb{P}^1)$, is

$$Q_{\mathrm{Vir}}^{\mathrm{contact}} = \frac{10}{c(5c+22)}.$$

The poles at $c = 0$ and $c = -22/5$ reflect the degeneration of the Virasoro vacuum module: at $c = 0$ the identity representation degenerates, and at $c = -22/5$ the $(2, 5)$ minimal model forces the vacuum Verma module to acquire a singular vector at weight 4.

The quintic obstruction is the Poisson bracket in the shadow algebra:

$$o_{\mathrm{Vir}}^{(5)} = \left\{C_{\mathrm{Vir}}, Q_{\mathrm{Vir}}^{\mathrm{contact}}\right\}_H = \frac{480}{c^2(5c+22)}\, x^5 \neq 0.$$

Since $o^{(5)} \neq 0$, no finite truncation captures the Virasoro algebra. Shadow depth ∞. Mixed archetype.

Modular characteristic and complementarity.

$$\kappa(\mathrm{Vir}_c) = \frac{c}{2}, \qquad \kappa(\mathrm{Vir}_{26-c}) = \frac{26-c}{2} = 13 - \frac{c}{2}, \qquad \kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_c^!) = 13.$$

This is the unique standard family where complementarity gives a *nonzero* constant: the scalar invariant $\varkappa := \kappa + \kappa^! = 13$ rather than 0 (distinct from the Trinity ghost-charge conductor $K(\mathrm{Vir}) := c + c^! = 26$; related by $\varkappa = \varrho_{\mathrm{Vir}} \cdot K$, $\varrho_{\mathrm{Vir}} = 1/2$). The constant 13 is half the critical dimension 26, and its nonvanishing reflects that Vir is a one-generator algebra whose bar complex has nontrivial self-coupling: the curvature of the dual cannot cancel the curvature of the algebra when both live in the same one-dimensional family.

Higher-genus corrections. At genus 1, the shadow obstruction tower produces corrections beyond the linear curvature. The genus-1 Hessian correction:

$$\delta H_{\mathrm{Vir}}^{(1)} = \frac{120}{c^2(5c+22)} \cdot x^2.$$

The loop ratio:

$$\rho_{\mathrm{Vir}}^{(1)} = \frac{240}{c^3(5c+22)}.$$

The Hessian correction $\delta H^{(1)}$ is the genus-1 quadratic deviation from the linearised curvature; the loop ratio $\rho^{(1)}$ is the ratio of the quadratic correction to κ^2. Both are determined by the quartic contact shadow and are not determined by $\kappa = c/2$: they are the first nonlinear genus-1 invariants, visible only because the shadow tower has depth ≥ 4.

Genus-two shell profile. All three primitive shells contribute:

$$\Theta^{(2)}_{\mathrm{loop}\circ\mathrm{loop}} \neq 0, \qquad \Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}} \neq 0, \qquad \Theta^{(2)}_{\mathrm{pf}} \neq 0.$$

The loop∘loop shell comes from iterated curvature; the sep∘loop shell from mixed clutching; and the planted-forest shell from the quartic contact channel Q^{contact}. The Virasoro algebra is the simplest family in which the planted-forest shell is nonzero, the first family that sees the log-FM boundary strata of $\overline{\mathcal{M}}_{2,n}$.

The critical dimension.

$$\nabla^{\mathrm{hol}}_c \;=\; d \;-\; \frac{c}{2}\,\omega_g \;-\; \frac{10}{c(5c+22)}\,\delta_4 \;-\; \frac{120}{c^2(5c+22)}\,\delta_H \;-\; \cdots.$$

The connection is flat for every c by the MC equation. The critical dimension $c = 26$ arises from the *dual* curvature vanishing:

$$\kappa(\mathrm{Vir}_{26-c}) \;=\; \frac{26-c}{2} \;=\; 0 \qquad \Longrightarrow \qquad \boxed{c \;=\; 26}.$$

At $c = 26$ the Koszul dual $\mathrm{Vir}^!_{26} = \mathrm{Vir}_0$ is *uncurved*: the dual bar complex is a genuine cochain complex. The quartic contact correction evaluates to $\mathfrak{Q}^{\mathrm{contact}}_{26} = 10/(26\cdot 152) = 5/1976$. The algebra itself retains curvature $\kappa = 13$; it is the depth-zero resonance shadow Vir_{26-c} at $c = 26$ that is trivial. The holographic datum $\mathcal{H}(\mathrm{Vir}_{26})$ has unobstructed dual; this is the string-theoretic content of the critical dimension.

The Polyakov formula ([?Pol87]) for a single scalar on a surface Σ with conformal factor σ ($g_1 = e^{2\sigma} g_0$) is

$$\log\frac{\det'\Delta_{g_1}}{\det'\Delta_{g_0}} \;=\; -\frac{1}{6\pi}\int_\Sigma \big(|\nabla\sigma|^2 + R_{g_0}\,\sigma\big)\,d\mu_{g_0}.$$

This is the degree-2 shadow evaluation at $\kappa = 1$: for general κ, on the proved uniform-weight scalar lane (Definition ??) the minimal shadow satisfies $\Theta^{\min}_{\mathcal{A}} = \kappa\cdot\eta\otimes\Lambda$ evaluates against σ to give $\langle\Theta^{\min},\sigma\rangle = -(\kappa/6\pi)\int(|\nabla\sigma|^2 + R\sigma)\,d\mu$, reproducing the conformal anomaly with coefficient $\kappa/(6\pi)$. The higher-degree corrections $\delta_4, \delta_H, \ldots$ displayed in the shadow connection above are inaccessible to the free-field path integral: they require the shadow obstruction tower. For uniform-weight algebras, the genus expansion is controlled by the $\hat{A}$-genus:

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A})\cdot\lambda_g^{\mathrm{FP}}, \qquad \textstyle\sum_{g\geq 1} F_g\,x^{2g} \;=\; \kappa\big(\hat{A}(ix) - 1\big) \;=\; \kappa\Big(\frac{x/2}{\sin(x/2)} - 1\Big).$$

The Taylor coefficients λ_g^{FP} are the Faber–Pandharipande evaluations of the Hodge class: $\lambda_1^{\mathrm{FP}} = 1/24$, $\lambda_2^{\mathrm{FP}} = 7/5760$, $\lambda_3^{\mathrm{FP}} = 31/967680$ (Theorem ??). The series converges: $|F_g| \sim 2|\kappa|/(2\pi)^{2g}$ (exponential decay), in contrast to the bosonic string where $F_g \sim (2g-2)!$ from the noncompactness of $\mathbf{R}^d$. The critical dimension is the zero of the dual curvature $\kappa(\mathrm{Vir}_{26-c}) = (26-c)/2$.

7.4. $\mathcal{W}_3$

The first multi-generator family: two generators of different conformal weights produce cross-channel mixing in the genus-1 Hessian, a 2×2 spectral discriminant, and the first multi-channel genus-2 amplitude. Infinite shadow depth, with propagator mixing as a new structural feature.

Two generators: the stress tensor T (weight 2) and a primary field W (weight 3). The TT OPE is Virasoro. The TW OPE encodes the primary property:

$$T(z)\,W(w) \;\sim\; \frac{3\,W(w)}{(z-w)^2} \;+\; \frac{\partial W(w)}{z-w}.$$

The WW OPE is the first genuinely nonlinear datum in the standard landscape:

$$\begin{aligned} W(z)\,W(w) \;\sim\; &\frac{c/3}{(z-w)^6} \;+\; \frac{2\,T(w)}{(z-w)^4} \;+\; \frac{\partial T(w)}{(z-w)^3} \\ &+\; \frac{1}{(z-w)^2}\left[\frac{16}{22+5c}\,\Lambda(w) \;+\; \frac{3}{10}\,\partial^2 T(w)\right] \;+\; \cdots, \end{aligned}$$

where the composite field

$$\Lambda \;=\; :TT: \;-\; \tfrac{3}{10}\,\partial^2 T$$

is the unique quasi-primary of weight 4 in the vacuum module. The coefficient $-3/10$ is fixed by requiring $T(z)\Lambda(w)$ to have no third-order pole: quasi-primary projection orthogonal to the descendant $\partial^2 T$. The structure constant $16/(22+5c)$ governs the $WW\Lambda$ coupling.

The $c = -22/5$ degeneration. At $c = -22/5$, the denominator vanishes and Λ becomes a null field. The vacuum Verma module has a singular vector at weight 4: the Kac determinant vanishes there, and $\mathcal{W}_3$ is not finitely generated by T and W alone. Infinitely many new generators are required. This is the first $\mathcal{W}$-algebra where the structure constants have poles in the central charge.

Shadow obstruction tower. The cubic shadow inherits the Virasoro self-coupling $T_{(1)}T = 2T$ and acquires a contribution from $T_{(1)}W = 3W$: $\mathfrak{C}_{\mathcal{W}_3} \neq 0$. The quartic contact channel is the composition

$$W \otimes W \;\xrightarrow{\text{OPE at }(z-w)^{-2}}\; \Lambda \;\xrightarrow{\text{OPE of }:TT:}\; T \otimes T,$$

which factors through the quasi-primary projection and then the Virasoro OPE. This factorisation is represented in the bar complex by a planted forest with two leaves and an internal vertex of weight 4: the first time a planted-forest correction appears in the bar differential.

The quartic contact shadow includes the Virasoro contribution $Q^{\text{contact}}_{\text{Vir}} = 10/[c(5c+22)]$ plus an additional term from the $WW\Lambda$ channel, proportional to $16/(22+5c)$. Both are singular at $c = -22/5$: the quartic shadow detects the vacuum degeneracy. Shadow depth ∞. Mixed modular archetype.

Modular characteristic.

$$\kappa(\mathcal{W}_3) \;=\; \frac{5c}{6}\,.$$

The weight-2 generator contributes $c/2$ and the weight-3 generator contributes $c/3$, giving $\kappa = c/2 + c/3 = 5c/6$. The Koszul dual $\mathcal{W}_3^!$ is a $\mathcal{W}_3$-algebra at the dual central charge $c' = 100 - c$ (self-dual at $c = 50$); complementarity: $\kappa(\mathcal{W}_{3c}) + \kappa(\mathcal{W}_{3100-c}) = 250/3$.

Genus-two shell profile. All three shells are active: $\Theta^{(2)}_{\mathrm{loop}\circ\mathrm{loop}} \neq 0, \Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}} \neq 0, \Theta^{(2)}_{\mathrm{pf}} \neq 0$. The planted-forest shell is sensitive to the $WW\Lambda$ channel: $\mathcal{W}_3$ is the first family that is genuinely planted-forest sensitive, detecting log-FM boundary strata that are invisible to Heisenberg and affine Kac–Moody. The genus-2 free energy decomposes as

$$F_2(\mathcal{W}_3) = \underbrace{\frac{7c}{6912}}_{\kappa\cdot\lambda_2^{\mathrm{FP}}} + \underbrace{\frac{c+204}{16c}}_{\delta F_2},$$

where $\delta F_2 = (c{+}204)/(16c)$ is the cross-channel correction from mixed TW-propagator graphs, verified by graph-by-graph recomputation over all seven genus-2 stable graphs (Theorem ??(vi), Computation ??). The scalar part $\kappa \cdot \lambda_2^{\mathrm{FP}}$ is unconditional; the cross-channel correction $\delta F_2 = (c{+}204)/(16c) \neq 0$ resolves Open Problem ?? negatively: the scalar formula fails for multi-weight algebras at genus ≥ 2.

The cross-channel correction accelerates with genus: $\delta F_g^{\mathrm{cross}}$ dominates the scalar $\kappa \cdot \lambda_g^{\mathrm{FP}}$ by a factor of 24 at genus 4 (Proposition ??). No classical matrix model spectral curve reproduces this tower (Remark ??). The gravitational cross-channel correction admits a universal closed form for all principal $\mathcal{W}_N$ (Proposition ??): $\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) = (N{-}2)(N{+}3)/96 + (N{-}2)(3N^3{+}14N^2{+}22N{+}33)/(24c)$, exact for $\mathcal{W}_3$ and a lower bound for $N \geq 4$.

The $\mathcal{W}_3$ holographic datum. The deformation space of the modular MC problem is two-dimensional: deformations along $\partial_c T$ and along $\partial_c W$. At $c = 0$: $\kappa = 0$ and the bar complex is uncurved. The dual curvature vanishes at $c = 100$ ($K = c + c' = 100$). The per-channel Hessian eigenvalues $\kappa_T = c/2$ and $\kappa_W = c/3$ govern the spectral discriminant; at $c = 50$ (the FF fixed point, $\mathcal{W}_3$ self-dual central charge) these give $\kappa_T = 25$ and $\kappa_W = 50/3$.

The quartic contact shadow is controlled by the quasi-primary

$$\Lambda \;=\; :TT: - \tfrac{3}{10}\,\partial^2 T, \qquad [\ell_4^{(0)}] \;=\; \frac{16}{22+5c}\,\Lambda.$$

At $c = -22/5$ (the $(2,5)$ minimal model) the quartic diverges; this is a resonance point of the $\mathcal{W}_3$ shadow obstruction tower, shared with Virasoro.

The genus-1 tangent map is a 2×2 matrix

$$H^{(1)}_{\mathcal{W}_3} \;=\; \begin{pmatrix} \kappa_T & 0 \\ 0 & \kappa_W \end{pmatrix}, \qquad \kappa_T = \frac{c}{2}, \quad \kappa_W = \frac{c}{3},$$

in the basis $(\partial_c T,\ \partial_c W)$; these are the per-channel eigenvalues from $T_{(3)}T = c/2$ and $W_{(5)}W = c/3$, *not* the total modular characteristic $\kappa = 5c/6$. The spectral discriminant is

$$\Delta_{\mathcal{W}_3}(x) \;=\; \big(1 - \kappa_T\, x\big)\big(1 - \kappa_W\, x\big) \;=\; \Big(1 - \frac{c}{2}\,x\Big)\Big(1 - \frac{c}{3}\,x\Big).$$

The two branch points $x_1 = 2/c$ and $x_2 = 3/c$ are the reciprocals of the per-channel eigenvalues. They coincide at the resonance value $c = -22/5 + 48/(5\kappa_c)$, where the two deformation directions merge.

The holographic datum:

$$\mathcal{H}(\mathcal{W}_{3c}) \;=\; \Big(\mathcal{W}_{3c},\ \mathcal{W}_{3100-c},\ C_c^{W_3},\ r_c^{W_3}(z),\ \Theta_c,\ \nabla_c^{\mathrm{hol}}\Big),$$

with Koszul dual $\mathcal{W}_{3\,100-c}$ ($\kappa_c + \kappa_{100-c} = 250/3$, self-dual at $c = 50$), line-operator category $\mathcal{C}_c^{W_3}$ a module category for the dg-shifted W_3-Yangian, and shadow connection

$$\nabla_c^{\mathrm{hol}} = d - \frac{5c}{6}\,\omega_g - \frac{16}{22+5c}\,\Lambda \cdot \delta_4 - \cdots .$$

7.5. $\beta\gamma$ system

The $\beta\gamma$ system is the contact archetype: the first nonlinearity appears at degree 4 (not 3), and a rank-one abelian rigidity argument kills it, producing shadow depth 4. This is the unique standard family in which the tower terminates for a structural reason distinct from Lie-algebraic cancellation.

Two fields: β of conformal weight λ and γ of conformal weight $1-\lambda$. The OPE is

$$\beta(z)\,\gamma(w) \sim \frac{1}{z-w}, \qquad \beta(z)\,\beta(w) \sim 0, \qquad \gamma(z)\,\gamma(w) \sim 0.$$

A simple pole with no double pole: the contact archetype. Central charge $c_{\beta\gamma} = 3(2\lambda-1)^2 - 1 = 12\lambda^2 - 12\lambda + 2$. At the standard weight $\lambda = 1$, $c_{\beta\gamma} = 2$ (the fermionic bc ghost system has $c_{bc} = -2$).

Shadow obstruction tower. The cubic shadow on the weight line vanishes: $\ell_3^{(0)} = 0$, because β and γ carry different conformal weights and the three-point residue on the diagonal weight sector is zero. The quartic operation is nonzero: $\ell_4^{(0)} \neq 0$, from the four-point function $\langle \beta\gamma\beta\gamma\rangle$. This is the defining property of the contact archetype: the first nonlinearity appears at degree 4, not 3.

The quartic contact invariant $\mu_{\beta\gamma} = \langle \eta, m_3(\eta,\eta,\eta)\rangle = 0$ vanishes by rank-one abelian rigidity: the $\beta\gamma$ system has a $\mathrm{U}(1)$ ghost-number symmetry, and the cyclic pairing on the deformation complex vanishes on the degree-4 component by weight considerations. The quintic obstruction vanishes because the cubic shadow is zero: $o^{(5)} = \{0, Q\}_H = 0$. Shadow depth 4. Contact/quartic archetype.

Koszul dual. The Koszul dual is the bc ghost system: $(\beta\gamma)^! = bc$. Replacing bosonic statistics by fermionic statistics negates the central charge: $c_{bc} = -c_{\beta\gamma}$.

Modular characteristic and complementarity.

$$\kappa(\beta\gamma) = \frac{c_{\beta\gamma}}{2} = 6\lambda^2 - 6\lambda + 1, \qquad \kappa(bc) = -\kappa(\beta\gamma), \qquad \kappa(\beta\gamma) + \kappa(bc) = 0.$$

At $\lambda = 1$: $\kappa(\beta\gamma) = 1$, $\kappa(bc) = -1$.

Genus-two shell profile.

$$\Theta^{(2)}_{\mathrm{loop}\circ\mathrm{loop}} \neq 0, \qquad \Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}} = 0, \qquad \Theta^{(2)}_{\mathrm{pf}} = 0.$$

The sep∘loop shell vanishes because the cubic shadow is zero; the planted-forest shell vanishes because the contact invariant $\mu_{\beta\gamma} = 0$. Only iterated curvature contributes at genus 2: the same shell pattern as Heisenberg, but with a different mechanism (contact termination rather than Gaussian collapse).

7.6. Lattice vertex algebras

Lattice vertex algebras are the arithmetic laboratory: the shadow tower inherits the Gaussian archetype from the Heisenberg factor, but the braiding carries the full arithmetic of the lattice. The curvature sees only the rank; the braiding sees the quadratic form.

Let Λ be an even lattice of rank N with quadratic form q and 2-cocycle ε. The lattice vertex algebra

$$V_\Lambda^{N,q} \;=\; \mathcal{H}_k^{\otimes N} \;\otimes_{\mathbb{C}}\; \mathbb{C}[\Lambda]_\varepsilon$$

is the tensor product of the rank-N Heisenberg algebra with the ε-twisted group algebra of the lattice. The generators include the Heisenberg currents α^i ($i = 1, \ldots, N$) and the exponential vertex operators $e^{\lambda\cdot\phi}$ for $\lambda \in \Lambda$.

Modular characteristic.

$$\kappa(V_\Lambda^{N,q}) \;=\; \mathrm{rank}(\Lambda) \;=\; N,$$

independent of the cocycle ε and the quadratic form q. The curvature is determined entirely by the Heisenberg part: the exponential vertex operators $e^{\lambda\cdot\phi}$ contribute to the braiding but not to the curvature.

Curvature-braiding orthogonality. The modular curvature class $[\kappa \cdot \omega_g] \in H^{1,1}(\overline{\mathcal{M}}_g)$ and the braiding representation $\pi_1(\mathrm{Conf}_n(X)) \longrightarrow \mathrm{Aut}(V_\Lambda^{\otimes n})$ act on complementary factors of the state space. The curvature sees $\mathcal{H}^{\otimes N}$; the braiding sees $\mathbb{C}[\Lambda]_\varepsilon$. The two are orthogonal.

Shadow obstruction tower and shells. The shadow obstruction tower inherits the Gaussian archetype from the Heisenberg factor: shadow depth 2. All cubic and higher obstructions vanish. Genus-two shell profile: loop∘loop only.

Special cases. The E_8 lattice: rank 8, self-dual, $\kappa = 8$. The Leech lattice: rank 24, self-dual, $\kappa = 24$. For a self-dual lattice, V_Λ is a holomorphic vertex algebra: the modular functor is trivial, and the partition function is an honest modular form (the theta function $\Theta_\Lambda(\tau)/\eta(\tau)^N$).

Non-simply-laced algebras. For the non-simply-laced simple Lie algebras B_2, C_2, G_2, F_4: all are class L (shadow depth 3). The short-root correction to the Killing form modifies κ by the lacing number but does not affect the depth classification. The Langlands dual pair (B_N, C_N) has $\kappa(B_N) \neq \kappa(C_N)$ but both are class L: the lacing distinguishes the scalar invariant but not the structural type.

7.7. Yangians and quantum groups

The Yangian is the line-operator algebra: it governs the topological direction of a holomorphic-topological theory, and its R-matrix is the genus-0 binary shadow of the MC element. The Yang–Baxter equation is the degree-3 MC equation; the RTT relations encode all degrees.

Fix a simple Lie algebra $\mathfrak{g}$ of rank ℓ. The Yangian $Y(\mathfrak{g})$ governs the E_1-chiral atom: chiral algebras on ordered configurations in $\mathbb{C} \times \mathbb{R}$, where $\mathbb{C}$ carries the holomorphic direction and $\mathbb{R}$ the topological direction.

The R-matrix as genus-0 binary shadow. The R-matrix

$$R(z) \;=\; 1 \;+\; \frac{r}{z} \;+\; \frac{r^{(2)}}{z^2} \;+\; \cdots \;\in\; Y(\mathfrak{g}) \otimes Y(\mathfrak{g})[[z^{-1}]]$$

is the genus-0 binary shadow of the MC element:

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}).$$

The classical r-matrix $r_{\mathrm{cl}} = k\,\Omega/z$ (where $\Omega = \sum_a J^a \otimes J^a$ is the Casimir tensor and k is the level) satisfies the classical Yang–Baxter equation (CYBE):

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0.$$

This is the Arnold relation on $\mathrm{FM}_3(\mathbb{C})$ evaluated on the Casimir: the three terms correspond to the three boundary divisors of $\overline{C}_3(\mathbb{C})$, and the CYBE is $d^2_{\bar{B}} = 0$ projected to degree 3 and genus 0.

Type-A MC3 reduction. On the type-A Yangian surface, the post-CG problem is reduced to a single remaining compact/completed extension packet. The reduction proceeds via:

(a) *Chromatic filtration.* Filter $K_0(\mathrm{Rep})$ by the support of q-characters. The successive quotients are controlled by products of prefundamental modules.

(b) *Clebsch–Gordan closure.* The tensor product $V_n \otimes L^- = \bigoplus_{j=0}^{n} L^-(n-2j)$ at character level, for all n. Every Verma character lies in the K_0-span of $\{[V_n] \cdot [L^-]\}$.

(c) *Completion formalism* (more precisely, Francis–Gaitsgory pro-nilpotent completion). Evaluation modules are compact, so once an equivalence on the relevant compact core is known, it extends to the corresponding ind-completion. This does not by itself supply that compact-core equivalence.

Extension to arbitrary simple type requires the all-types prefundamental CG package together with the remaining post-CG extension/completion packets.

The dg-shifted Yangian. The modular dg-shifted Yangian $Y_T^{\mathrm{mod}} = \varprojlim_r Y_T^{\mathrm{mod},\le r}$ is the pronilpotent completion of the convolution dg Lie algebra associated to the chiral algebra of a 3d holomorphic-topological theory T. The modular R-matrix

$$R_T^{\mathrm{mod}}(z;\hbar) \;\in\; \mathrm{MC}(Y_T^{\mathrm{mod}})$$

is an MC element. The Yang–Baxter equation is the MC equation projected to degree 3; the higher RTT relations $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$ encode the MC equation at all degrees. The RTT tower is a pro-system of finite-dimensional MC equations, and the inverse limit converges by Mittag-Leffler stabilisation: at each finite truncation, the obstruction group is finite-dimensional and eventually stationary.

Drinfeld–Kohno. The KZ connection on $\mathrm{Conf}_n(\mathbb{C})$ with values in $\mathfrak{g}^{\otimes n}$ is

$$\nabla_{\mathrm{KZ}} \;=\; d \;-\; \hbar \sum_{i<j} \frac{\Omega_{ij}}{z_i - z_j}\, d(z_i - z_j).$$

Its monodromy representation $\pi_1(\mathrm{Conf}_n(\mathbb{C})) \to \mathrm{Aut}(V^{\otimes n})$ is compared by the affine Drinfeld–Kohno theorem with the corresponding quantum-group representation category. In the shadow obstruction tower:

$$\nabla_{\mathrm{KZ}} \;=\; \mathrm{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k}) :$$

on the affine Kac–Moody comparison surface, the KZ connection is identified with the genus-0 shadow connection determined by the universal MC element.

Shadow obstruction tower. The Yangian inherits its shadow structure from the underlying affine algebra. At the E_1-chiral level, the shadow obstruction tower is controlled by the R-matrix expansion: the degree-r shadow at genus 0 is $r^{(r-1)}/z^{r-1}$, and each coefficient satisfies an RTT-type constraint. The genus-0 shadow terminates at depth 3 (for class **L** algebras, the CYBE is the full genus-0 content at degree 3; for class **M**, the degree-3 MC equation carries strictly more, encoding the homotopy Jacobiator); the modular extension to genus $g \geq 1$ requires the completed dg-shifted Yangian and does not terminate.

7.8. Drinfeld–Sokolov reduction as package morphism

DS: $\widehat{\mathfrak{sl}}_{3k} \to \mathcal{W}_{3c(k)}$ with $c(k) = 2 - 24(k+2)^2/(k+3)$ extends to a morphism

$$\mathrm{DS}\colon \mathcal{H}(\widehat{\mathfrak{sl}}_{3k}) \longrightarrow \mathcal{H}(\mathcal{W}_{3c(k)})$$

acting on each component:

	$\mathcal{H}(\widehat{\mathfrak{sl}}_{3k})$	$\mathrm{DS}(\mathcal{H})$
$\mathcal{A}$	$\widehat{\mathfrak{sl}}_{3k}$	$\mathcal{W}_{3c(k)}$
$\mathcal{A}^!$	$\widehat{\mathfrak{sl}}_{3-k-6}$	$\mathcal{W}_{3100-c(k)}$
$r(z)$	$\Omega_{\mathfrak{sl}_3}/z$	$r_c^{W_3}(z)$
κ	$4(k+3)/3$	$5c(k)/6$
depth	3	∞
$\Theta_{\mathrm{pf}}^{(2)}$	0	$\neq 0$

The Casimir r-matrix $k\,\Omega/z$ specialises to $r_c^{W_3}(z)$. The curvature shifts. The shadow depth jumps from 3 to ∞: DS reduction creates the planted-forest shell. The off-diagonal TW-mixing in the genus-1 Hessian is the image of the $\mathfrak{sl}_3$ root structure. At the critical level $k = -3$: $\kappa \to 0$ on the affine side (curvature vanishes), $c \to \infty$ on the $\mathcal{W}_3$ side (Sugawara pole).

The DS-HPL transfer theorem (Volume II, Theorem thm:ds-hpl-transfer) makes the chain-level mechanism precise: homological perturbation through the SDR of the BRST complex transfers the affine dg-shifted Yangian triple (m, r, Δ) to the $\mathcal{W}$-algebra side. The A_∞ products transfer nontrivially (the quartic pole forces infinite degree), and the r-matrix transfers with the expected pole-order shift. The coproduct, however, is annihilated: a ghost-number obstruction kills every HPL tree correction, so $\Delta_z^{\mathcal{W}} = 0$ at all degrees (Principle ??). DS reduction is therefore a functor on modular Koszul triples (Construction ??), and all gravitational dynamics resides in the r-matrix, not in coproduct vertices.

Central charge additivity under DS

The central charge difference $c(\widehat{\mathfrak{sl}}_{3k}) - c(\mathcal{W}_3, k) = 8k/(k+3) - (2 - 24(k+2)^2/(k+3)) = 6(4k+5)$ is k-dependent; it includes both the free ghost bc contribution (which is k-independent) and a stress-tensor improvement from the BRST construction. The modular characteristics involve different anomaly ratios: $\kappa(\widehat{\mathfrak{sl}}_{3k}) = 4(k+3)/3$ (Theorem ??), $\kappa(\mathcal{W}_3, k) = 5c(k)/6$ where $c(k) = 2 - 24(k+2)^2/(k+3)$, and the deficit $\kappa_{\mathrm{aff}} - \kappa_{W_3}$ is a rational function of k (Proposition ??). The DS morphism of holographic data is compatible with the genus expansion at every genus simultaneously, by functoriality of the graph-sum (0.0.7) under DS.

7.9. The geometric localization principle

The algebraic functoriality above has a geometric substrate. Drinfeld–Sokolov reduction is not merely an algebraic construction: it is *Hamiltonian reduction from* $J_\infty(\mathfrak{g}^*)$ *to* $J_\infty(S_f)$, where $S_f = f + \mathfrak{g}^e$ is the

Slodowy transverse slice to the nilpotent orbit of f (here $\mathfrak{g}^e$ is the centralizer of the nilpotent element e in an $\mathfrak{sl}_2$-triple (e, h, f); Definition **??**).

The Slodowy slice is the universal transverse section through a nilpotent orbit: it meets every orbit transversally, carries a Poisson structure from the Kirillov–Kostant bracket, and its quantization is the finite W-algebra (Gan–Ginzburg [?GanGinzburg02], Premet [?Premet02]). The affine/chiral upgrade replaces S_f by its arc space $J_\infty(S_f)$ (the space of formal arcs, i.e. $\mathbb{C}[[t]]$-points of S_f), and $\mathrm{gr}_F\, \mathcal{W}^k(\mathfrak{g}, f) \cong \mathbb{C}[J_\infty(S_f)]$ (Remark **??**). In this reading:

- The PBW–Slodowy collapse (Theorem **??**) is the algebraic expression of *smoothness of the Slodowy slice* (Remark **??**).
- For $\mathcal{W}$-algebras, the shadow depth jump $3 \to \infty$ under DS records the *nonlinearity of the Dirac bracket* on the arc space $J_\infty(S_f)$: the Hamiltonian reduction from the linear Kirillov–Kostant bracket on $J_\infty(\mathfrak{g}^*)$ to the nonlinear Dirac bracket on $J_\infty(S_f)$ creates the infinite shadow obstruction tower (Remark **??**).
- Barbasch–Vogan orbit duality $f \leftrightarrow f^D$, through which the Koszul dual orbit is identified, is *Springer duality of Slodowy slices*: the curvature complementarity $\kappa(\mathcal{W}^k) + \kappa(\mathcal{W}^{k'}) = \mathrm{const}$ is an algebraic identity under the Feigin–Frenkel involution $k + h^\vee \mapsto -(k + h^\vee)$ (Remark **??**).
- The master commutative square (Theorem **??**) encodes all of this: bar complexes on the upper row, Hamiltonian reduction on arc spaces on the lower row, PBW identification on the vertical arrows.

8. The arithmetic of the shadow obstruction tower

The shadow obstruction tower coefficients for the standard families are rational functions of the central charge. Their denominators, zeros, and growth rates carry arithmetic information connecting the shadow calculus to L-functions. What number-theoretic structures does the MC equation force?

The Dirichlet–sewing lift

The genus-1 free energy $F_1^{\mathrm{conn}}(\mathcal{A}; q) = -\log\det(1 - K_q)$, where K_q is the sewing kernel on $\overline{\mathcal{M}}_{1,1}$, encodes connected sewing amplitudes $a_{\mathcal{A}}(N)$ as Fourier coefficients. The *Dirichlet–sewing lift*

$$S_{\mathcal{A}}(u) \;:=\; \sum_{N=1}^{\infty} a_{\mathcal{A}}(N)\, N^{-u} \;=\; \frac{(2\pi)^u}{\Gamma(u)} \int_0^\infty F_{\mathcal{A}}^{\mathrm{conn}}(e^{-2\pi y})\, y^{u-1}\, dy$$

is a Dirichlet series whose analytic structure is controlled by the shadow obstruction tower $\Theta_{\mathcal{A}}$. The sewing amplitudes define an inner product $\langle f, g\rangle_{\mathcal{A}} = \sum_N a_{\mathcal{A}}(N)\, f(N)\, g(N)$ on Dirichlet polynomials. The reproducing kernel $K_{\mathcal{A}}(s,t) = S_{\mathcal{A}}(s+t)$ is the *sewing reproducing kernel*, and the surface moment matrix $\mathscr{M}_{ij} = K_{\mathcal{A}}(\alpha/2+i,\ \alpha/2+j)$ is positive definite for all $\alpha \geq 2$: it is a Gram matrix $V^T D V$ with Vandermonde V (Theorem **??**).

Explicit values:

$$\begin{aligned} S_{\mathcal{H}}(u) &= \zeta(u)\,\zeta(u+1), \\ S_{V_k(\mathfrak{g})}(u) &= \dim(\mathfrak{g}) \cdot \zeta(u)\,\zeta(u+1), \\ S_{\mathrm{Vir}}(u) &= \zeta(u+1)\big(\zeta(u) - 1\big), \\ S_{\mathcal{W}_N}(u) &= \zeta(u+1)\Big((N-1)\zeta(u) - \textstyle\sum_{j=1}^{N-1} H_j(u)\Big), \end{aligned}$$

where $H_j(u) = \sum_{n=1}^{j} n^{-u}$ are partial harmonic sums. The Heisenberg lift is a product of two Riemann zeta functions; the Virasoro lift subtracts the first Fourier coefficient from ζ; the $\mathcal{W}_N$ lifts interpolate via a harmonic sum that stabilises as $N \to \infty$.

The three-tier Euler–Koszul classification

The *Euler defect* $D_{\mathcal{A}}(u) := S_{\mathcal{A}}(u) / (|W(\mathcal{A})|\, \zeta(u)\, \zeta(u+1))$ classifies chiral algebras into three tiers:

Tier	Condition	Families
Exact Euler–Koszul	$D_{\mathcal{A}}(u) \equiv 1$	$\mathcal{H}, V_k(\mathfrak{g}), \beta\gamma$
Finitely defective	$1 - D_{\mathcal{A}}(u)$ is a finite harmonic ratio	Virasoro, $\mathcal{W}_N$
Non-Euler–Koszul	$S_{\mathcal{A}}$ decomposes into Hecke L-functions	lattice VOAs with $h(D) \geq 2$

The Euler–Koszul class $\mathrm{ek}(\mathcal{A}) = \max_i (w_i - 1)$ and the shadow depth $r_{\max}(\mathcal{A})$ are independent invariants (Theorem ??): the first reads the character, the second the OPE. The two axes of difficulty, arithmetic (how the coefficients grow) and algebraic (how the OPE interacts), are logically independent.

The two-variable L-object

The Rankin–Selberg unfolding against the Eisenstein series $E^*(\tau, s)$ produces

$$L_{\mathcal{A}}(s, u) \;:=\; \int_{\mathcal{F}}^{\mathrm{reg}} F_{\mathcal{A}}^{\mathrm{conn}}(\tau)\, E^*(\tau, s)\, y^{u-2}\, dx\, dy.$$

The *sewing–Hecke reciprocity theorem* collapses the two variables to one: $L_{\mathcal{A}}(s, u) = \frac{\Gamma(v)}{(2\pi)^v} S_{\mathcal{A}}(v)$ with $v := s + u - 1$. Special cases: the Dirichlet slice $L_{\mathcal{A}}(1, u) = \frac{\Gamma(u)}{(2\pi)^u} S_{\mathcal{A}}(u)$; the Selberg slice $L_{\mathcal{A}}(s, 0) = \frac{\Gamma(s-1)}{(2\pi)^{s-1}} S_{\mathcal{A}}(s-1)$. The scattering poles of the Eisenstein series at $s = \rho/2$ (for nontrivial zeros ρ of ζ) are *not* poles of $L_{\mathcal{A}}$: the Rankin–Selberg unfolding is holomorphic at these points (Proposition ??). The nontrivial zeros enter as a structural obstruction to the Euler product, not as poles of the L-object.

The shadow-moduli resolution

The shadow obstruction tower enters the arithmetic through the *shadow-moduli map* $t\colon \mathfrak{H} \to \mathbb{C}$, solving

$$\prod_j (1 - \lambda_j\, t(q))^{c_j} \;=\; \det(1 - K_q^{\mathrm{conn}}),$$

where λ_j are the spectral atoms and c_j their multiplicities. The connected free energy factors as $F_1^{\mathrm{conn}}(q; \mathcal{A}) = G_{\mathcal{A}}(t(q))$ with

$$G_{\mathcal{A}}(t) = \int \log(1 - \lambda\, t)\, d\rho_{\mathcal{A}}(\lambda),$$

a function determined by the spectral measure $\rho_{\mathcal{A}}$ alone (Theorem ??). The shadow obstruction tower enters as follows: expanding $G_{\mathcal{A}}(t) = -\sum_{r \geq 1} \frac{\mu_r}{r} t^r$, the moments $\mu_r = \int \lambda^r\, d\rho_{\mathcal{A}}$ are determined by the MC equation degree by degree. At degree $r+1$:

$$\mu_{r+1} \;=\; (r+1)\; \mathrm{Br}_{r+1}(\mu_2, \mu_3, \ldots, \mu_r; \mathcal{A}),$$

where Br_{r+1} is a universal polynomial in the lower moments and the structure constants of $\mathcal{A}$ (Proposition ??). For two Satake-type atoms $\lambda_1 = \alpha$, $\lambda_2 = \beta$, this yields, via the Newton relations, a formal spectral polynomial: the power sums $p_r = e_1\, p_{r-1} - e_2\, p_{r-2}$ encode the MC recursion in spectral coordinates.

The interacting Gram matrix

The shadow obstruction tower provides interacting corrections to the Gram positivity. The *interacting sewing weight*

$$w_{\mathcal{A}}(N;\alpha) \;:=\; a_{\mathcal{A}}(N)\cdot \prod_{j}(1-\lambda_j\, N^{-\alpha/2})^{c_j}$$

is unconditionally positive when all spectral atoms λ_j are negative (Theorem ??(i)). The interacting Gram matrix $\mathscr{M}^{\mathrm{int}}_{ij} = \sum_N w_{\mathcal{A}}(N;\alpha)\, N^{-\alpha/2-i}\, N^{-\alpha/2-j} \succ 0$ in this case. For Virasoro with $c > 0$: $\lambda_{\mathrm{eff}} = -6/c < 0$, so positivity holds for all $\alpha \geq 2$ unconditionally (Corollary ??). The resonance locus $\mathcal{R}$ is empty for $c > 0$ and confined to the non-unitary regime $c < 0$, with the Lee–Yang point $c = -22/5$ as the first entry (Theorem ??).

The Weil analogy

The structural parallel between the MC equation and the Weil conjectures for curves over finite fields organises the arithmetic programme:

Weil proof for $C/\mathbb{F}_q$	**MC equation for $\mathcal{A}$**
Curve $C/\mathbb{F}_q$	chirally Koszul algebra $\mathcal{A}$
Frobenius Frob_q	MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$
$\mathrm{Frob}^2 = q$ (functional equation)	$D^2_{\mathcal{A}} = 0$ (nilpotence)
Étale cohomology $H^i_{\text{ét}}$	cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$
Weight filtration $W_\bullet$	degree filtration $\Theta^{\leq r}_{\mathcal{A}}$
Eigenvalues α_i on H^1	spectral atoms λ_j of $\rho_{\mathcal{A}}$
$\lVert\alpha_i\rVert = q^{1/2}$ (Riemann hypothesis)	$\lvert a_{f_j}(p)\rvert \leq 2p^{(k-1)/2}$ (Ramanujan)
Lefschetz operator L	bracket with $\kappa(\mathcal{A})$

For even unimodular lattice vertex algebras V_Λ with theta series $\Theta_\Lambda = c_E\, E_{r/2} + \sum_j c_j f_j$ (cusp forms f_j), the Hecke–Newton closure (Theorem ??) gives the chain: MC equation $\Rightarrow$ prime-local CG closure $\Rightarrow$ CPS automorphy $\Rightarrow$ $\mathrm{Sym}^{r-1}(f)$ identification $\Rightarrow$ Ramanujan bound. The endpoint recovers the Ramanujan bound for lattice VOAs (Corollary ??). The MC equation supplies the Hecke decomposition and CPS automorphy; the Ramanujan bound itself follows from Deligne's theorem on Hecke eigenvalues of cusp forms [`?Deligne1974`], an external input, not reproved here. The contribution of the MC framework is the systematic derivation of the Hecke decomposition from chiral bar-cobar data, not an independent proof of the bound.

The Polyakov–bar-cobar dictionary

The shadow obstruction tower proves Polyakov's worldsheet functional-integral programme [`?Pol87`] algebraically:

Polyakov object	Bar-cobar counterpart
Conformal anomaly $\langle T^a{}_a \rangle = (c/12)R$	modular characteristic $\kappa(\mathcal{A})$
Functional determinant $\det'_\zeta \Delta$	Fredholm $\det(1-K)$ on Bergman kernel
Critical dimension $c = 26$	dual characteristic $\kappa(\mathrm{Vir}_{26-c}) = 0$
Ghost system bc ($c_{\mathrm{ghost}} = -26$)	Koszul dual $\mathcal{A}^!$ (BRST pairing, not literal)
Genus expansion $Z = \sum_g g_s^{2g-2} Z_g$	MC5 sewing, $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (uniform weight)
BPZ equations (Virasoro degenerate-module surface)	shadow connection ∇^{sh} on that comparison surface
Instanton sectors	Novikov-completed bar $B^\Lambda(\mathcal{A})$
Wilson line $\langle W(C) \rangle$	Swiss-cheese boundary module

The anomaly, the genus expansion, the duality, the critical dimension, and the non-perturbative completions all descend from $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$. The four shadow depth classes (Gaussian, Lie, contact, mixed) correspond to increasing levels of departure from the Gaussian path integral.

The Polyakov dictionary identifies the shadow obstruction tower with the worldsheet programme at the level of individual objects. The following construction upgrades this identification to a connection on the space of Hecke eigenvalues, providing a differential-geometric framework for the prime-by-prime structure of the sewing amplitudes.

The arithmetic packet connection

The Hecke operators T_p (p prime) act on the space of graph amplitudes $\ell_\Gamma^{(g)}(\mathcal{A})$ through their action on modular forms. Let $M_{\mathcal{A}}$ be the $\mathbb{C}$-span of the activated amplitudes, and decompose $M_{\mathcal{A}} = \bigoplus_\chi M_\chi$ into generalised Hecke eigenspaces: on M_χ, the operator T_p acts by a scalar $a_\chi(p)$ plus a nilpotent part N_χ (the Jordan block defect). To each eigenspace attach its *packet function*, the completed L-function of the corresponding eigenform: for an Eisenstein eigenspace, $\Lambda_0(s) = C_0(s)\,\zeta(s)\,\zeta(s-r/2+1)$; for a cusp eigenspace attached to a normalised eigenform f_j, $\Lambda_j(s) = C_j(s)\,L(s, f_j)$.

The *arithmetic packet connection* (Definition **??**) is the flat meromorphic connection

$$\nabla^{\mathrm{arith}}_{\mathcal{A}} \;:=\; d \;-\; \bigoplus_\chi \frac{\Lambda'_\chi(s)}{\Lambda_\chi(s)} \big(\mathrm{id}_{M_\chi} + N_\chi\big)\, ds$$

on the trivial bundle $M_{\mathcal{A}} \otimes \mathcal{O}_{\mathbb{C}}$ over the spectral line $s \in \mathbb{C}$. Flatness is immediate: on each block, the connection form is a scalar 1-form times a constant endomorphism, so both dA_χ and $A_\chi \wedge A_\chi$ vanish for degree reasons (Theorem **??**). The singular divisor $D_{\mathcal{A}} = \bigcup_\chi \mathrm{div}(\Lambda_\chi)$ is the zero-and-pole set of the packet functions; it depends only on the semisimple quotient $M^{\mathrm{ss}}_{\mathcal{A}}$ (the *arithmetic skeleton*), not on the nilpotent parts N_χ. The nilpotent parts produce only unipotent local monodromy $\exp(2\pi i\ \mathrm{ord}_\rho(\Lambda_\chi)\, N_\chi)$ around each component of $D_{\mathcal{A}}$ (Theorem **??**(iii)).

The completed Eisenstein scattering function on $\mathcal{M}_{1,1}$ is $\varphi(s) := \Lambda^*(1-s)/\Lambda^*(s)$, where $\Lambda^*(s) = \pi^{-s}\Gamma(s)\zeta(2s)$. Write $\Lambda_{\mathrm{Eis}}(s) := \Lambda_0(s)$ for the Eisenstein packet. The *frontier defect form* (Definition **??**)

$$\Omega_{\mathcal{A}} \;:=\; d\log \Lambda_{\mathrm{Eis}}(s) \;-\; d\log \varphi(s)$$

vanishes iff the Eisenstein block of ∇^{arith} is gauge-equivalent to the scattering connection $d - d\log\varphi(s)$ (Proposition ??; the additional condition $N_{\mathrm{Eis}} = 0$ is needed for the gauge to be scalar). For even unimodular lattice VOAs at rank r: $\Lambda_{\mathrm{Eis}}(s) = C_0(s)\,\zeta(s)\,\zeta(s-r/2+1)$, while $\varphi(s)$ involves $\zeta(2s)/\zeta(2s-1)$; since $\zeta(s)$ and $\zeta(2s)$ have different zero sets, $\Omega_{V_\Lambda} \neq 0$ in general.

The packet connection treats each algebra uniformly. For $\mathcal{W}_N$ algebras, the DS reduction from affine Kac–Moody provides additional structure: the Miura transformation splits the $\mathcal{W}_N$ character into a Heisenberg core and a finite correction, isolating all arithmetic content beyond Heisenberg in a finite product.

The finite Miura defect

For principal $\mathcal{W}_N$ with generators of spins $2, 3, \ldots, N$, the $\mathcal{W}_N$ character is $\chi_{\mathcal{W}_N}(q) = \prod_{n\geq 2}(1 - q^n)^{-\min(n-1,\, N-1)}$ (Proposition ??). The Heisenberg character is $\chi_{\mathcal{H}}(q) = \prod_{n\geq 1}(1 - q^n)^{-1}$. The ratio is a finite product (Theorem ??):

$$\chi_{\mathcal{W}_N}(q) \;=\; \chi_{\mathcal{H}}(q)^{N-1} \cdot \prod_{m=1}^{N-1}(1 - q^m)^{N-m},$$

since the exponent $(N-1) - \min(n-1, N-1)$ vanishes for $n \geq N$ and equals $N-n$ for $2 \leq n \leq N-1$. The finite product is the *Miura defect*; all genus-1 arithmetic content in $\mathcal{W}_N$ beyond Heisenberg is carried by it. At the Dirichlet level: $D^{\mathrm{prime}}_{\mathcal{W}_N}(u) = -\zeta(u+1)\sum_{m=1}^{N-1}(N-m)\, m^{-u}$ (Proposition ??). The packet connection of $\mathcal{W}_N$ splits as $(N-1)$ copies of the diagonal Heisenberg connection $d - d\log[\zeta(s)\zeta(s+1)]\, ds$ plus a defect connection governing modes $m = 1, \ldots, N-1$ (Proposition ??). The Heisenberg copies contribute only the universal divisor $\mathrm{div}(\zeta(s)) \cup \mathrm{div}(\zeta(s+1))$, and their contribution to $\Omega_{\mathcal{W}_N}$ cancels: the frontier defect form is determined by the defect alone.

The constructions above produce proved arithmetic invariants from the MC element. The final programme of this section reverses the direction: it asks what constraints the nontrivial zeros of L-functions impose on the MC data, through the quartic shadow.

The quartic closure programme

The quartic resonance class $\mathfrak{R}^{\mathrm{mod}}_{4,g,n}(\mathcal{A})$, the first nonlinear shadow of $\Theta_{\mathcal{A}}$ (degree 4 projection of the MC element), satisfies a proved clutching law along every separating boundary stratum $\xi : \overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \to \overline{\mathcal{M}}_{g,n}$:

$$\xi^*\mathfrak{R}^{\mathrm{mod}}_{4,g,n} \;=\; p_1^*\mathfrak{R}^{\mathrm{mod}}_{4,g_1,n_1+1} \;+\; p_2^*\mathfrak{R}^{\mathrm{mod}}_{4,g_2,n_2+1} \;+\; \mathfrak{T}_{3,\xi}$$

(Theorem ??). The 3×3 Schur complement of the mixed (s, u)-transform extracts the quartic contact coupling Q^{ct}_v on the potential side and its reciprocal $(Q^{\mathrm{ct}}_v)^{-1}$ on the Gram side (Theorem ??). For Virasoro: $\Sigma^{\mathrm{Vir}}_2 = 10/[c(5c+22)]$ (Proposition ??).

Three closure conjectures form a strict hierarchy (each implies all predecessors):

(i) *quartic closure* (Conjecture ??): the intersection of quartic Gram-compatibility loci over all families, slices, and regulators is $\{\rho : \mathrm{Re}(\rho) = 1/2\}$;

(ii) *Beilinson closure* (Conjecture ??): the global discrepancy functional $\mathcal{B}^{\mathrm{glob}}(\rho)$ vanishes iff $\mathrm{Re}(\rho) = 1/2$;

(iii) *clutching closure* (Conjecture ??): the residue clutching defect $K_{\mathcal{A},\rho,\xi}$ (Definition ??) vanishes for all families and all boundary strata iff $\mathrm{Re}(\rho) = 1/2$.

The chain of constructions: $\Theta_{\mathcal{A}}$ (proved, Theorem ??) $\to$ residue kernel w_{c,ρ,u_0} (proved, Proposition ??) $\to$ residue Gram matrix (Definition ??) $\to$ Schur complement (proved, Theorem ??) $\to$ quartic resonance divisor (proved, Theorem ??) $\to$ residue clutching defect (Definition ??) $\to$ closure conjecture (conjectural).

For lattice vertex algebras, the arithmetic programme has a concrete incarnation: the bar complex selects a sublattice, and the Epstein zeta function restricted to that sublattice recovers the Riemann zeta in the simplest case.

The constrained Epstein zeta

For a lattice vertex algebra V_Λ of rank r, the *constrained Epstein zeta function*

$$\varepsilon_s^c(V_\Lambda) \;:=\; \sum_{\ell\in\Lambda^{\mathrm{bar}}}{}' \,\|\ell\|^{-2s}$$

restricts the Epstein series to the sublattice $\Lambda^{\mathrm{bar}} \subset \Lambda$ determined by the bar complex: the summation runs over those lattice vectors that contribute nontrivially to the bar differential. For the rank-1 lattice $V_{\mathbb{Z}}$ at self-dual radius $R = 1$, every nonzero integer contributes, giving

$$\varepsilon_s^1(R{=}1) \;=\; 4\,\zeta(2s)\;:$$

the constrained Epstein zeta of the simplest lattice VOA *is* the Riemann zeta function (up to the elementary factor 4).

The Hecke operators decompose the constrained series into L-functions:

$$\varepsilon_s^r(V_\Lambda) \;=\; \sum_j c_j\, L(s,\chi_j)$$

where the sum runs over the Hecke characters χ_j of Λ activated by the bar sublattice. Shadow depth d contributes $d-1$ constituent centres to the decomposition of ε_s^r, one for each degree stratum of the shadow obstruction tower beyond the Gaussian base. The nontrivial zeros of the component L-functions are the scattering resonances of the hyperbolic Laplacian on $\mathcal{M}_{1,1} = \mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$: they are the frequencies at which the genus-1 sewing amplitude of V_Λ fails to decay.

The Böcherer bridge (genus 2). At genus 2, the shadow tautological class for lattice vertex algebras computes a central L-value via the Böcherer conjecture (Furusawa–Morimoto). For the Leech lattice V_Λ with $\Lambda = \Lambda_{24}$: the Siegel modular form $\chi_{12} = \mathrm{SK}(f_{22})$ is the Saito–Kurokawa lift of the unique weight-22 cuspidal eigenform, and the MC-derived second Fourier coefficient $c_2 \approx -1.918\times 10^{-6} \neq 0$ is the first nonzero central L-value produced by the shadow obstruction tower, connecting the MC element $\Theta_{\mathcal{A}}$ to the arithmetic of Siegel modular forms at genus 2.

9. Scalar saturation and the Koszulness programme

The bar construction works for all chiral algebras. But it works *best* (the cobar inverts the bar, the spectral sequence collapses, the resolution is linear) precisely on the Koszul locus. Characterizing this locus is the central foundational question. Two complementary results: scalar saturation (the MC element is determined by a single scalar for all standard families) and the Koszulness meta-theorem (twelve characterizations, of which eight (among (i)–(x)) are genuinely independent bidirectional equivalences, one ((viii)) is a one-way chiral Hochschild consequence, one ((xi)) is conditional on perfectness, and one ((xii)) is a one-directional D-module implication).

9.1. Scalar saturation

For a one-parameter algebraic family $\mathcal{A}_k$ with rational OPE coefficients (meaning the structure constants are rational functions of the level k), the universal MC element satisfies:

$$\Theta_{\mathcal{A}_k} \;=\; \kappa(\mathcal{A}_k) \cdot \eta_{\text{univ}} \otimes \Lambda_{\text{univ}} \;+\; (\text{terms in } F^3\mathfrak{g}^{\text{mod}}_{\mathcal{A}_k}).$$

Scalar saturation universality: the higher-order terms vanish. For every algebraic family with rational OPE coefficients, $\Theta_{\mathcal{A}_k}$ is determined by the single scalar $\kappa(\mathcal{A}_k)$ modulo terms in the degree-≥ 3 filtration.

The proof proceeds in two steps.

Step 1: Whitehead reduction. For a vertex algebra $\mathcal{A}_k$ with reductive symmetry at generic k, the primitive cyclic cohomology $H^2_{\text{cyc,prim}}(\mathfrak{g}^{\text{mod}}/F^3,\ d_2)$ is computed by a Whitehead-type spectral sequence. The E_2 page is

$$E_2^{p,q} \;=\; H^p\big(\mathfrak{g};\ \text{Sym}^q V\big),$$

the Lie algebra cohomology of the symmetry algebra acting on the conformal-weight filtration. For reductive $\mathfrak{g}$ and generic level, $H^2(\mathfrak{g}; \text{Sym}^q V) = 0$ by the Whitehead lemma (vanishing of H^2 for finite-dimensional semisimple Lie algebras acting on finite-dimensional modules). The degree-2 MC class therefore lifts uniquely to the full complex.

Step 2: Algebraic semicontinuity. The vanishing locus

$$E \;=\; \big\{k \in \mathbb{C} : H^2_{\text{cyc,prim}}(\mathcal{A}_k) \neq 0\big\}$$

is Zariski-closed in the parameter space (upper semicontinuity of cohomology dimension as a constructible function). At the free-field point $k = \infty$, the algebra is Heisenberg and $H^2_{\text{cyc,prim}} = 0$. Therefore E is a proper Zariski-closed subset. Explicit computation verifies $E = \varnothing$ for $\mathfrak{sl}_N$ ($N = 2, \ldots, 10$): no exceptional level exists. The result covers the entire standard Lie-theoretic landscape at all non-critical levels, including admissible ones.

Scope. Scalar saturation applies to every family in the standard landscape: affine Kac–Moody (all types, all non-critical levels), Virasoro (all c), $\mathcal{W}_N$ (all N, all non-degenerate levels), $\beta\gamma$ and lattice algebras (all parameters). The residual conjecture, that $E = \varnothing$ for *every* positive-energy chiral algebra with rational OPE coefficients, is open only for non-algebraic-family constructions (no known counterexample).

Free-field exactness. Scalar saturation determines the genus expansion up to the cross-channel correction $\delta F_g^{\text{cross}}$. For interacting multi-weight algebras ($\mathcal{W}_N$ with $N \geq 3$, non-principal $\mathcal{W}$-algebras), this correction is genuinely nonzero at genus ≥ 2 and grows with genus: the scalar formula $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ is an incomplete approximation. The question is whether the cross-channel correction can vanish even when generators carry distinct conformal weights, so that the scalar formula persists beyond its natural uniform-weight habitat.

Free-field algebras provide such an extension. Their genus-0 OPE is purely quadratic ($m_k = 0$ for $k \geq 3$), and this quadratic structure propagates to all genera: every mixed-channel stable graph collapses, not by a single mechanism but by three independent ones, each adapted to a different subclass.

Proposition 0.0.4 (Free-field exactness of the scalar formula). *Let $\mathcal{A}$ be a free-field chiral algebra: a modular Koszul algebra whose genus-0 OPE is purely quadratic ($m_k = 0$ for $k \geq 3$ at genus 0). Then the cross-channel correction vanishes:*

$$\delta F_g^{\mathrm{cross}}(\mathcal{A}) \;=\; 0 \qquad \textit{for all } g \geq 1.$$

The scalar formula $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ holds at all genera, even when the strong generators of $\mathcal{A}$ have distinct conformal weights (all-weight free-field exact case).

The conclusion covers Heisenberg (class **G**), bc and $\beta\gamma$ systems (class **C**), and lattice vertex algebras. Three independent mechanisms force the vanishing: for class **G**, every mixed-channel stable graph has a bivalent vertex with zero cubic shadow; for $bc/\beta\gamma$, the fermionic/bosonic sign rules on mixed-weight propagators cancel pairwise; for lattice algebras, the lattice symmetry group acts transitively on channels. Each mechanism is independently sufficient for its subclass, and together they exhaust the free-field families. Free-field exactness is the borderline: the first interacting multi-weight family ($\mathcal{W}_3$) already has $\delta F_2^{\mathrm{cross}} \neq 0$, and the correction dominates the scalar term by genus 4.

9.2. The Koszulness meta-theorem

The Koszul condition admits twelve formulations, spanning homological algebra (PBW collapse, Ext vanishing), homotopy theory (A_∞ formality, E_2 formality), operadic algebra (Barr–Beck–Lurie comparison, FM boundary acyclicity), modular geometry (FH concentration, tropical Koszulness), and symplectic topology (Lagrangian criterion). Ten of these are unconditionally equivalent; one (the Lagrangian criterion) is conditional on perfectness/nondegeneracy; and one ($\mathcal{D}$-module purity) is one-directional (forward direction proved, converse open).

The PBW criterion detects Koszulness for every standard family but does not characterize it intrinsically. Theorem **??**: ten conditions on $\mathcal{A}$ are mutually equivalent, with one conditional on perfectness and one partial implication.

For $\mathcal{A}$ positive-energy and finitely strongly generated, the following are equivalent:

(i) $\mathcal{A}$ is chirally Koszul (Definition **??**).
(ii) The PBW spectral sequence on $\bar{B}^{(g)}(\mathcal{A})$ collapses at E_2 for every g.
(iii) The minimal A_∞-model of $H^\bullet(\bar{B}(\mathcal{A}))$ is formal: $m_n = 0$ for $n \geq 3$.
(iv) $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ for $p \neq q$ (Ext diagonal vanishing).
(v) The bar-cobar counit $\Omega(\bar{B}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism.
(vi) The Barr–Beck–Lurie comparison for $\bar{B} \dashv \Omega$ is an equivalence.
(vii) $\int_X \mathcal{A}$ is concentrated in degree 0 for every smooth proper X.
(viii) $\mathrm{ChirHoch}^*(\mathcal{A})$ vanishes outside degrees $\{0, 1, 2\}$.
(ix) The Kac–Shapovalov determinant $\det G_h \neq 0$ in the bar-relevant range.
(x) The restriction $i_S^! \, \bar{B}_n(\mathcal{A})$ to every FM boundary stratum S is acyclic outside degree 0.

Under the additional hypothesis that the ambient tangent complex is perfect and nondegenerate:

(xi) $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians in the (-1)-shifted symplectic deformation space $\mathcal{M}_{\mathrm{comp}}$ (Lagrangian criterion).

Finally, (xii) $\Rightarrow$ (x), where:

(xii) Each $\bar{B}_n(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to FM boundary strata ($\mathcal{D}$-module purity).

The converse of (xii) is open.

The proof circuit. The core equivalence chain is (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (v) $\Leftrightarrow$ (viii), proved by PBW concentration $\to$ bar-cobar inversion $\to$ chiral Hochschild polynomial growth $\to$ Koszul-complex acyclicity, with the A_∞ formality branch via HPL transfer and Keller classicality. From this core, the remaining equivalences follow: (iv) by Ext spectral sequence, (vi) by conservativity + log-FM totalization, (vii) by $\bar{B} = \int_X$, (ix) by non-degeneracy forcing PBW injectivity, (x) by stratum-by-stratum PBW concentration. The Lagrangian criterion (xi) uses PTVV derived intersection theory on the perfectness hypothesis. The $\mathcal{D}$-module purity implication (xii)$\Rightarrow$(x) uses weight filtration on the bar complex; the converse would require concentration to force purity. The cycle closes via (viii)$\Rightarrow$(i): Koszul-complex acyclicity implies the defining characterization, completing the equivalence.

Beyond the meta-theorem. Six additional characterizations are proved in the text, complementing but not contained in items (i)–(xii):

(a) *Shadow-formality at low degree.* The shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ equals the L_∞-formality obstruction tower of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ at degrees $r = 2, 3, 4$ (Proposition ??). At degrees 2, 3, 4, the shadow obstruction tower *is* the formality obstruction tower; full equivalence to (iii) requires all degrees.

(b) *E_2-formality of chiral Hochschild cohomology.* Koszulness implies $\mathrm{ChirHoch}^*(\mathcal{A})$ is formal as an E_2-algebra (Deligne conjecture applied to the bar-cobar resolution). A consequence of (v), not an independent equivalence.

(c) *Genus-0 curve independence.* Koszulness on $\mathbb{P}^1$ implies Koszulness on every smooth proper curve (Proposition ??). Proved independently by factorization descent.

(d) *PBW universality* (sufficient condition). Every freely strongly generated vertex algebra is chirally Koszul (Proposition ??). This applies to *universal* algebras: $V_k(\mathfrak{g})$ is Koszul at every level k, including critical and admissible. For simple quotients $L_k(\mathfrak{g})$ at admissible levels, the Koszulness status is rank-dependent: $L_k(\mathfrak{sl}_2)$ is Koszul at all admissible levels (Remark ??), but at $\mathrm{rk}(\mathfrak{g}) \geq 2$ with denominator $q \geq 3$, Koszulness fails via a $\mathrm{rk}(\mathfrak{g})$-dimensional bar obstruction from the abelian Cartan subalgebra (Theorem ?? at $\mathfrak{g} = \mathfrak{sl}_3$; Conjecture ?? at $\mathrm{rk}(\mathfrak{g}) \geq 3$). PBW universality is one-directional: not every Koszul algebra is freely strongly generated.

(e) *Tropical Koszulness.* $\mathcal{A}$ is chirally Koszul iff $\bar{B}^{\mathrm{trop}}(\mathcal{A})$ is acyclic (Theorem ??). Proved by the weight spectral sequence on the real blowup.

(f) *Bifunctor obstruction decomposition* (one-directional). Koszulness implies the RNW one-slot obstruction decomposes into independent bar-side and cobar-side components (Theorem ??). The converse is open.

Status summary.

Meta-theorem (Theorem ??):

(i)–(v), (vii)$\|_{g=0}$, (ix), (x)	**8 genuinely independent bidirectional**	proved
(vi) Barr–Beck–Lurie	consequence of (v)	proved
(viii) chiral Hochschild $\{0, 1, 2\}$	one-way consequence on Koszul locus	proved
(xi) Lagrangian	conditional on perfectness	proved (unconditional for standard landsc
(xii) $\mathcal{D}$-module purity	(xii)$\Rightarrow$(x) only	converse open

Additional characterizations:

(a) Shadow-formality	equivalence at degrees 2, 3, 4	proved
(b) E_2-formality	consequence of Koszulness	proved
(c) Curve independence	independent theorem	proved
(d) PBW universality	sufficient condition	proved
(e) Tropical Koszulness	equivalence	proved
(f) Bifunctor decomposition	one-directional	converse open

9.3. Shadow depth and operadic complexity

The four depth classes G/L/C/M of §6.5 admit a refinement. In general, $r_{\max}$ can take any value in $\{2, 3, 4, 5, \ldots\} \cup \{\infty\}$: the class $\mathbf{F}_r$ (tower terminates at degree r) is structurally permitted for all $r \geq 2$, with $\mathbf{G} = \mathbf{F}_2$, $\mathbf{L} = \mathbf{F}_3$, $\mathbf{C} = \mathbf{F}_4$ naming the standard-landscape representatives.

The operadic complexity theorem. The *operadic complexity theorem* (Theorem ??) asserts:

$$r_{\max}(\mathcal{A}) \;=\; A_\infty\text{-depth}(\mathcal{A}) \;=\; L_\infty\text{-formality level}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}).$$

Here A_∞-depth is the minimal r such that the transferred A_∞-structure on $H^\bullet(\bar{B}(\mathcal{A}))$ has $m_k = 0$ for all $k > r$; and L_∞-formality level is the minimal r such that the L_∞-structure on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is formal through degree r.

The identification has three consequences:

(i) The shadow obstruction tower *is* the L_∞-formality obstruction tower: $\Theta^{\leq r}_{\mathcal{A}}$ encodes precisely the obstruction to L_∞-formality through degree r.

(ii) The depth classification ($\mathbf{F}_r$ / $\mathbf{M}$) becomes a structural invariant of the operadic type of $\mathcal{A}$: $\mathbf{F}_2$ = free, $\mathbf{F}_3$ = Lie, $\mathbf{F}_4$ = quadratic non-Lie, higher $\mathbf{F}_r$ = higher non-Lie, $\mathbf{M}$ = genuinely nonlinear.

(iii) The shadow obstruction tower of a Gaussian algebra is a quadratic form; of a Lie algebra, a Chevalley–Eilenberg complex; of a contact algebra, a cyclic A_∞-structure; of a mixed algebra, an infinite L_∞-tower with no polynomial truncation.

The programme extends to non-principal $\mathcal{W}$-algebras: the Bershadsky–Polyakov algebra BP_k is chirally Koszul at generic level ($k \neq -3$) with dual BP_{-k-6} (Computation ??), and the nilpotent transport from hook-type seeds covers all orbits in $\mathfrak{sl}_2$ through $\mathfrak{sl}_8$ (Proposition ??).

9.4. The PBW spectral sequence

The computational engine for bar cohomology within each fixed genus is the PBW (Poincaré–Birkhoff–Witt) spectral sequence. Filter $\bar{B}^{(g)}(\mathcal{A})$ by the *conformal weight*: assign to each normally ordered monomial in the strong generators its total weight (the sum of the conformal weights of the constituent fields). The associated graded is the bar complex of a free-field algebra, a computation that reduces to combinatorics of symmetric functions.

The E_1 page at genus g decomposes:

$$E_1^{p,q}(g) \;=\; E_1^{p,q}(g{=}0) \;\oplus\; \mathcal{E}_g^{p,q},$$

where the first summand is the genus-0 bar cohomology (determined by Koszulness of the associated graded) and $\mathcal{E}_g^{p,q}$ is the genus-g enrichment from the Hodge bundle. The enrichment at genus g arises from the regular Eisenstein series coupled to the curvature.

Concentration: $E_\infty^{p,q}(g) = 0$ for $q \neq 0$ at all genera $g \geq 1$, for all standard families (Kac–Moody, Virasoro, principal $\mathcal{W}_N$). This is PBW degeneration: the spectral sequence collapses at E_2, and the bar cohomology is concentrated on the diagonal.

The PBW spectral sequence is *not* the genus spectral sequence of section 5.5. The PBW spectral sequence uses conformal weight within a fixed genus to compute the bar cohomology groups; the genus spectral sequence uses the genus filtration on the deformation algebra to control the Maurer–Cartan lift across genera. One computes *what*; the other determines *whether*.

9.5. Chiral Hochschild cohomology (Theorem H)

The chiral Hochschild complex $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ is the derived endomorphism algebra of the identity functor on $\mathcal{A}$-modules, computed by the bar construction with $\mathcal{A}$-bimodule coefficients. At genus 0:

$$\mathrm{ChirHoch}^\bullet(\mathcal{A}) \;\simeq\; R\mathrm{Hom}_{\mathcal{A}\text{-bimod}}(\mathcal{A},\mathcal{A}) \;\simeq\; \mathrm{Ext}^\bullet_{\mathcal{A}^e}(\mathcal{A},\mathcal{A}),$$

where $\mathcal{A}^e = \mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}$ is the enveloping chiral algebra.

For $\mathcal{A} = \mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ a principal $\mathcal{W}$-algebra at generic level, the chiral Hochschild cohomology is concentrated in degrees $\{0,1,2\}$ (Theorem **??**):

$$\mathrm{ChirHoch}^n(\mathcal{W}^k(\mathfrak{g})) = 0 \quad \text{for } n > 2, \qquad P(t) = 1 + t^2.$$

The deformation class in $\mathrm{ChirHoch}^2 = \mathbb{C}$ is the level deformation $k \mapsto k + \epsilon$. The concentration bound is the de Rham amplitude on a curve ($\dim X = 1$); the continuous Lie algebra cohomology $H^*_{\mathrm{cont}}(\mathrm{Lie}(\mathcal{W}), \mathrm{Lie}(\mathcal{W})_0; \mathbb{C}) = \mathbb{C}[\Theta_1, \ldots, \Theta_r]$ (Gel'fand–Fuchs, with generators $\deg \Theta_i = m_i + 1$) is a different, unbounded invariant. For $\mathfrak{g} = \mathfrak{sl}_2$ (Virasoro): $P(t) = 1 + t^2$. For $\mathfrak{g} = \mathfrak{sl}_3$ ($\mathcal{W}_3$): $P(t) = 1 + t^2$.

Three confusions recur in the literature and must be excluded before stating the theorem. The chiral Hochschild complex $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ is a finite-dimensional cohomology theory bounded by $\dim X = 1$; it is *not* the Gel'fand–Fuchs continuous Lie algebra cohomology $H^*_{\mathrm{cont}}(\mathrm{Lie}(\mathcal{A}), \ldots)$ (which is an infinite-dimensional polynomial ring), nor is it the naïve ring $\mathbb{C}[\Theta]$ generated by the MC element. The bound $\dim_{\mathrm{total}} \leq 4$ is on the sum of Betti numbers (cohomological amplitude), not on the virtual dimension or Euler characteristic. The palindromic symmetry in part (c) below is the chiral Hochschild-level shadow of complementarity: it exchanges $\mathcal{A}$ and $\mathcal{A}^!$ through the graded reversal $t \mapsto t^{-1}$, exactly as Verdier duality exchanges $\bar{B}(\mathcal{A})$ and $\bar{B}(\mathcal{A}^!)$ on the Ran space.

Theorem 0.0.5 (Theorem H: polynomial growth of chiral Hochschild cohomology). *Let $\mathcal{A}$ be a chirally Koszul algebra on a smooth projective curve X. Then:*

(a) Concentration. $\mathrm{ChirHoch}^n(\mathcal{A}) = 0$ *for $n < 0$ and $n > 2$. The cohomological amplitude is* $[0, 2]$.

(b) Polynomial growth. *The Hochschild–Hilbert series* $P_{\mathcal{A}}(t) = \sum_{n\geq 0} \dim \mathrm{ChirHoch}^n(\mathcal{A})\, t^n$ *is a polynomial of degree at most 2, with total dimension* $\dim_{\mathrm{total}} \leq 4$.

(c) Koszul palindromy. $P_{\mathcal{A}}(t) = t^2\, P_{\mathcal{A}^!}(t^{-1})$*: the Hilbert polynomial of the Koszul dual is the palindromic reversal.*

The concentration bound $[0, 2]$ is a topological invariant of the complex, determined by $\dim X = 1$ through the de Rham bound on the universal curve; the individual dimensions within that amplitude depend on the specific algebra. The theorem corrects earlier formulations in which the bound was stated as a virtual dimension or in which the chiral Hochschild complex was conflated with the Gel'fand–Fuchs invariant.

On the Koszul locus, $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ carries an E_2-algebra structure that is formal. The Connes periodicity operator $S\colon \mathrm{ChirHoch}^n \to \mathrm{ChirHoch}^{n-2}$ intertwines Koszul duality: $S \circ \mathrm{KD} = \mathrm{KD} \circ S$.

At genus 0, the chiral Hochschild complex identifies with the bulk chiral homology complex of Volume II: $\mathrm{ChirHoch}^\bullet(\mathcal{A}) \simeq \mathrm{ChirHoch}^\bullet_{\mathrm{bulk}}(\mathcal{A})$. The bulk-boundary-line triangle of section 9.6 is the genus-0 shadow of this identification: $\mathcal{A}_{\mathrm{bulk}} \simeq Z_{\mathrm{der}}(\mathcal{B}_\partial) \simeq \mathrm{ChirHoch}^\bullet(\mathcal{A}^!)$.

9.6. The shadow algebra

The *shadow algebra* $\mathcal{A}^{\mathrm{sh}}$ is the cohomology of the modular cyclic deformation complex as a graded-commutative ring:

$$\mathcal{A}^{\mathrm{sh}} \;:=\; H_\bullet\big(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})\big), \qquad \mathcal{A}^{\mathrm{sh}} = \bigoplus_{r\geq 2,\, g\geq 0} \mathcal{A}^{\mathrm{sh}}_{r,g}.$$

The single MC element $\Theta_{\mathcal{A}}$ decomposes into components graded by degree r and genus g. Each named shadow is a projection:

$$\begin{aligned} \kappa(\mathcal{A}) &= \mathcal{A}^{\mathrm{sh}}_{2,0}, \\ \mathfrak{C}(\mathcal{A}) &= \mathcal{A}^{\mathrm{sh}}_{3,0}, \\ \mathfrak{Q}(\mathcal{A}) &= \mathcal{A}^{\mathrm{sh}}_{4,0}, \\ \mathrm{Sh}_r(\mathcal{A}) &= \mathcal{A}^{\mathrm{sh}}_{r,0}, \qquad r \geq 5. \end{aligned}$$

The shadow algebra organises the entire hierarchy. It is a homotopy invariant of $\mathcal{A}$, computed by the Weil homomorphism of the Chern–Weil dictionary (section 5.4).

The all-degree master equation, read in the shadow algebra, becomes: $\nabla_H(\mathrm{Sh}_r) + o^{(r)} = 0$ for all $r \geq 3$. The obstruction class $o^{(r)}$ lies in $\mathcal{A}^{\mathrm{sh}}_{r,0}$ and measures the failure of the truncation $\Theta^{\leq r-1}_{\mathcal{A}}$ to extend to degree r.

9.7. BV/BRST identification at genus 0

At genus 0, the bar differential of a chiral algebra on $\mathbb{C}$ is the BRST differential of the associated holomorphic twist:

$$(d_{\bar{B}})_{g=0} \;=\; Q_{\mathrm{BRST}}.$$

The bar complex $\bar{B}^{(0)}_{\mathbb{C}}(\mathcal{A})$ is the BRST complex of the 2d chiral algebra, and bar-cobar inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (Theorem B) is the statement that the BRST cohomology, processed through the cobar functor, reconstructs the original algebra.

At genus 1, the identification acquires curvature:

$$d_{\text{fib}}^2 \;=\; \kappa(\mathcal{A}) \cdot \omega_1 \;=\; \text{one-loop anomaly},$$

and the curved bar complex is the curved BRST complex of the theory on an elliptic curve. The complementarity $Q_1(\mathcal{A}) \oplus Q_1(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_{1,1}, \mathcal{Z}_{\mathcal{A}})$ is the genus-1 anomaly cancellation condition.

At genus $g \geq 2$, the identification with the Costello–Gwilliam BV formalism requires renormalisation: the higher-loop integrals over $\overline{\mathcal{M}}_{g,n}$ need regularisation beyond the algebraic bar-cobar framework, and the correct receptacle is the coderived category.

For the Heisenberg algebra, the identification is proved at all genera by four independent paths (Theorem ??): zeta-regularisation, Selberg zeta function, Grothendieck–Riemann–Roch, and direct bar complex computation.

The shadow obstruction tower equals the L_∞-formality obstruction tower at all degrees (Theorem ??).

Genus-two shell activation as depth diagnostic. The genus-two shell profile independently confirms the depth classification:

	$\Theta^{(2)}_{\text{loop}^2}$	$\Theta^{(2)}_{\text{sep}\circ\text{loop}}$	$\Theta^{(2)}_{\text{pf}}$
G (Heisenberg, lattice)	✓	--	--
L (affine $\widehat{\mathfrak{g}}_k$)	✓	✓	--
C ($\beta\gamma$)	✓	--	--
M (Virasoro, $\mathcal{W}_N$)	✓	✓	✓

Gaussian and contact algebras activate only the loop∘loop shell; Lie/tree algebras add the sep∘loop shell; only mixed algebras activate all three shells including the planted-forest shell. The planted-forest shell is the diagnostic: its presence forces infinite shadow depth.

Structural refinements. The three bar complexes on a chiral algebra (the ordered tensor bar complex T^c, the symmetric bar complex Sym^c, and the Chevalley–Eilenberg bar complex Lie^c) sit in a chain of inclusions $\mathrm{Lie}^c \hookrightarrow \mathrm{Sym}^c \hookrightarrow T^c$ with distinct coalgebra structures at each level (Theorem ??). The averaging map from ordered to symmetric is surjective but not split: the kernel at degree 3 contains the Drinfeld associator $\Phi_{\mathrm{KZ}}(\mathcal{A})$, and the fibre of splittings is a torsor for GRT_1. The E_1/ordered bar complex is therefore the natural primitive object; the modular/symmetric story is its Σ_n-coinvariant image (Theorem ??). The genus-3 cross-channel correction for $\mathcal{W}_3$ has been computed by summing over all 42 stable graphs of $\overline{\mathcal{M}}_{3,0}$ (Proposition ??): the cross-channel tower accelerates with genus and dominates the scalar part $\kappa \cdot \lambda_g^{\mathrm{FP}}$ by genus 4. On the open/closed side, the Koszul dual cooperad $\mathsf{SC}^{\mathrm{ch,top},!}$ decomposes into three sectors: closed ($\dim = (n-1)!$, Lie cooperad), open ($\dim = m!$, Ass cooperad), and mixed ($\dim = (k-1)!\binom{k+m}{m}$). The mixed sector encodes the bulk-to-boundary module structure by which the chiral derived centre acts on the boundary algebra (Proposition ??, Proposition ??). The BV/BRST = bar identification at higher genus (Conjecture ??): the chain-level obstruction (harmonic propagator correction) is resolved for classes G, L, and C: for G and L by the Heisenberg sewing theorem and its extension to affine algebras, for C by contact decoupling of the harmonic propagator correction. For class M (Virasoro, $\mathcal{W}_N$), the identification holds on bar cohomology but fails at the chain level: the harmonic propagator produces a nonvanishing quartic obstruction, so a coderived reformulation is needed.

10. The open/closed world (Volume II)

The bar complex lives naturally in three dimensions. On the product $\mathbb{C}_z \times \mathbb{R}_t$, the holomorphic direction along $\mathbb{C}$ produces the closed-string bar complex of Sections 1–9: the differential extracts OPE residues from collisions of points in the holomorphic plane. The topological direction along $\mathbb{R}$ produces the open-string coproduct: the ordered splitting of tensor factors at a cut point of the real line. The Swiss-cheese operad encodes the interaction of these two structures: holomorphic operations acting on the fiber above the topological interval, topological operations governing the sequential composition. The resulting two-colour algebraic framework is not an embellishment of the bar complex but its natural habitat: without the topological direction, the coproduct has no operadic home, and the line-operator category has no definition.

The primitive object of three-dimensional holomorphic-topological quantum field theory on $\mathbb{C}_z \times \mathbb{R}_t$ is therefore not the bar complex but the open/closed factorization dg-category C on the bordified curve $\widetilde{X}_D$. Its objects are boundary conditions, its morphisms are open-string states, and a choice of vacuum b yields a boundary algebra $A_b = \mathrm{End}_C(b)$, determined only up to Morita equivalence. The governing operadic structure is the two-coloured Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$, with closed colour $\mathrm{FM}_k(\mathbb{C})$ and open colour $E_1(m)$, and it is homotopy-Koszul (proved).

The bar complex of Volume I is an E_1-chiral coassociative coalgebra over $(\mathrm{ChirAss})^!$: its differential records OPE collision residues along divisors of $\mathrm{FM}_k(\mathbb{C})$, and its deconcatenation coproduct records the ordered splitting of the topological direction. The $\mathsf{SC}^{\mathrm{ch,top}}$ structure emerges on the derived center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\mathcal{A})$, not on the bar complex itself. The bar complex classifies *twisting morphisms*, universal couplings between $\mathcal{A}$ and its Koszul dual $\mathcal{A}^!$, and is not the bulk.

Three independent objects are constructed from C, in distinct ambients:

(i) the *bar complex* $\bar{B}(A_b)$ classifies couplings between A_b and $A^!_b$;
(ii) the *chiral derived center* $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(C) = R\mathrm{Hom}_{\mathrm{Fun}(C,C)}(\mathrm{Id},\mathrm{Id})$ is the universal closed sector, computed in any chart by chiral Hochschild cochains $C^\bullet_{\mathrm{ch}}(A_b, A_b)$;
(iii) the *Koszul dual* $A^!_b$ governs the line-operator category.

The primitive hierarchy is:

$$C_{\mathrm{op}} \xrightarrow{\mathrm{End}(b)} A_b \xrightarrow{\bar{B}} \bar{B}(A_b) \xrightarrow{\Theta} \Theta_{\mathcal{A}} \xrightarrow{+\mu^{M_j}} \Theta^{\mathrm{oc}}_{\mathcal{A}}.$$

Each arrow is a lossy functor: C_{op} is Morita-invariant; A_b depends on the choice of boundary condition b; $\bar{B}$ forgets the bulk; $\Theta_{\mathcal{A}}$ is the closed-sector MC element; $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ restores the full open/closed package. Modularity is determined by the open sector: a cyclic trace on $\mathrm{End}(b)$ seeds the Calabi–Yau structure; the annulus identification $\int_{S^1} C \simeq \mathrm{HH}_\bullet(C)$ is the first modular shadow; clutching (composing along boundary circles via nodal degeneration) raises genus. The genus expansion of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ yields the perturbative free energies F_g; the shadow archetypes $\mathsf{G}/\mathsf{L}/\mathsf{C}/\mathsf{M}$ classify the nonlinear complexity of the Swiss-cheese operations.

Volume II constructs this architecture in four stages: *Stage 1* (local one-colour theorem: Swiss-cheese universality of the derived center) $\Rightarrow$ *Stage 2* (open primitive: C with Morita invariance) $\Rightarrow$ *Stage 3* (globalization on tangential log curves (X, D, τ)) $\Rightarrow$ *Stage 4* (modularization: trace plus clutching producing Θ^{oc}). Stages 1–3 are proved. In Stage 4, the open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ is constructed, the modular MC equation holds at all genera (Theorem **??**), and the projection principle recovers both the closed-sector genus tower and the open-sector module structures (Theorem **??**). The annulus trace is proved (Theorem **??**). The independent open-sector derivation of higher-genus amplitudes and the chain-level cooperad compatibilities remain open.

10.1. The holomorphic-topological Swiss-cheese operad

The *holomorphic-topological Swiss-cheese operad* $\mathsf{SC}^{\mathrm{ch,top}}$ is a two-colored operad with color set $\{\mathrm{ch}, \mathrm{top}\}$. The color ch is *closed* (holomorphic); the color top is *open* (topological). The operation spaces are:

$$\begin{aligned}\mathsf{SC}(\mathrm{ch}^k; \mathrm{ch}) &= \mathrm{FM}_k(\mathbb{C}),\\ \mathsf{SC}(\mathrm{ch}^k, \mathrm{top}^m; \mathrm{top}) &= \mathrm{FM}_k(\mathbb{C}) \times E_1(m),\\ \mathsf{SC}(\dots, \mathrm{top}, \dots; \mathrm{ch}) &= \varnothing.\end{aligned}$$

Closed-to-closed is the Fulton–MacPherson compactification of configuration spaces in $\mathbb{C}$. The mixed direction combines holomorphic configurations with the E_1-operad: k holomorphic and m topological insertion points, with holomorphic insertions acting on the fiber above the interval. There is no operation with topological input and holomorphic output.

Operadic composition is determined by the boundary stratification of FM spaces in the holomorphic direction and by nested interval inclusions in the topological direction. If σ_S denotes the boundary stratum of $\mathrm{FM}_k(\mathbb{C})$ corresponding to the simultaneous collision of the subset $S \subset \{1, \dots, k\}$, then the operadic composition map γ_S is the inclusion of the fiber product

$$\gamma_S\colon\ \mathrm{FM}_{|S|}(\mathbb{C}) \times \mathrm{FM}_{k-|S|+1}(\mathbb{C}) \ \hookrightarrow\ \mathrm{FM}_k(\mathbb{C})$$

as the normal bundle of σ_S. In the topological direction, E_1-composition is the standard operadic substitution of intervals into intervals.

A bar element of tensor degree k lives on the product $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$. The bar differential extracts OPE residues along collision divisors of the first factor: when points z_i and z_j collide, the residue of the propagator $\eta_{ij} = d\log(z_i - z_j)$ against the collision divisor $\sigma_{\{i,j\}}$ yields the OPE mode $a_{(n)}b$. The coproduct splits an ordered sequence of tensor factors at a point of the second factor: an element $a_1 \otimes \cdots \otimes a_k$ with ordering $t_1 < \cdots < t_k$ on $\mathbb{R}$ is decomposed at a cut point $t_p < c < t_{p+1}$ into $(a_1 \otimes \cdots \otimes a_p) \otimes (a_{p+1} \otimes \cdots \otimes a_k)$. These two operations make the ordered bar complex an E_1-chiral coassociative coalgebra. The $\mathsf{SC}^{\mathrm{ch,top}}$ structure appears only on the derived-center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}), \mathcal{A})$.

10.2. Homotopy-Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$

The classical Swiss-cheese operad SC is Koszul (Livernet, Voronov; Ginzburg–Kapranov). The proof: the Koszul dual cooperad $\mathsf{SC}^{\text{!`}}$ has a quadratic presentation, and the bar-cobar resolution is a quasi-isomorphism by the combinatorial criterion of distributive lattices applied to the poset of nested disks. Kontsevich formality supplies a quasi-isomorphism

$$\Phi\colon\ \mathsf{SC}^{\mathrm{ch,top}} \ \xrightarrow{\ \sim\ }\ \mathsf{SC}$$

constructed from Kontsevich's formality morphism for the E_2-operad (configuration space integrals on FM spaces with propagators), extended to the Swiss-cheese setting by including the half-plane $\mathbb{H}$ and its boundary $\mathbb{R}$. The bar-cobar adjunction preserves quasi-isomorphisms by the standard filtration-spectral-sequence argument on the cobar side; the two-out-of-three property in the model category of operads then gives

$$\Omega\mathbf{B}(\mathsf{SC}^{\mathrm{ch,top}}) \xrightarrow{\sim} \mathsf{SC}^{\mathrm{ch,top}}.$$

The bar-cobar adjunction on $\mathsf{SC}^{\mathrm{ch,top}}$-algebras is therefore a *Quillen equivalence*: the derived unit and counit are both weak equivalences. This has three consequences:

(i) Every $\mathsf{SC}^{\mathrm{ch,top}}$-algebra admits a cofibrant replacement by a bar-cobar resolution.
(ii) The homotopy category of $\mathsf{SC}^{\mathrm{ch,top}}$-algebras is equivalent to the homotopy category of coalgebras over the Koszul dual cooperad.
(iii) The line-operator category $\mathcal{A}^!\text{-}\mathsf{mod}$ is unconditionally defined as a module category over the bar resolution, without any finiteness hypothesis on $\mathcal{A}$.

SC-formality characterises class G. The homotopy-Koszulity of the Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$, established in Section 10.2, is a property of the operad itself: it holds universally and places no constraint on the algebra. A strictly finer condition is formality of a specific *algebra* with respect to that operad. The question is which algebras in the standard landscape are Swiss-cheese formal; that is, for which $\mathcal{A}$ do the transferred Swiss-cheese operations m_k^{SC} vanish for all $k \geq 3$?

The answer singles out exactly one depth class.

Proposition 0.0.6 (SC-formality iff class G). *Let $\mathcal{A}$ be a chiral algebra in the standard landscape. Then $\mathcal{A}$ is Swiss-cheese formal if and only if $\mathcal{A}$ belongs to class* **G**.

The forward direction is immediate: class **G** algebras (Heisenberg, free fermion) have purely quadratic OPE, so all nested brackets vanish and $m_k^{\mathrm{SC}} = 0$ for $k \geq 3$. For the converse: SC-formality forces the cubic shadow $S_3 = 0$, which by nondegeneracy of the invariant bilinear form on the standard landscape implies the current algebra bracket is trivial; an abelian current algebra at level k is Heisenberg, hence class **G**.

The distinction between Koszulness and SC-formality is sharp: all four depth classes contain chirally Koszul algebras, but only class **G** is SC-formal. Affine Kac–Moody (class **L**) has $m_3^{\mathrm{SC}} \neq 0$ from the Lie bracket; $\beta\gamma$ (class **C**) has $m_4^{\mathrm{SC}} \neq 0$ from the quartic contact invariant; Virasoro (class **M**) has $m_k^{\mathrm{SC}} \neq 0$ for all $k \geq 3$. SC-formality therefore refines the classification: Koszulness is the cobar-inversion condition (Section 9.2), SC-formality is the vanishing of all open/closed higher homotopies, and the gap between them is measured by the shadow depth.

10.3. The A_∞ chiral algebra structure

The full Swiss-cheese MC element $\alpha_T \in \mathrm{MC}(\mathfrak{g}_T^{\mathrm{SC}})$ lives in the Swiss-cheese convolution algebra $\mathfrak{g}_T^{\mathrm{SC}} := \mathrm{Hom}_\Sigma\big(\mathbf{B}(\mathsf{SC}^{\mathrm{ch,top}}), \mathrm{End}_{(B_\partial, C_{\mathrm{line}})}\big)$. Its closed-colour projection $(m_k)_{k\geq 1}$ encodes the A_∞ chiral algebra on $\mathcal{A}$. Each component

$$m_k \colon \mathcal{A}^{\otimes k} \longrightarrow \mathcal{A}((\lambda_1))\cdots((\lambda_{k-1}))$$

is a configuration-space integral on $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$. Write ω_I for the form on $\mathrm{FM}_k(\mathbb{C})$ obtained by wedging the propagators $\eta_{ij} = d\log(z_i - z_j)$ for all pairs $(i,j) \in I$. The holomorphic factor produces the OPE singularity; the topological factor produces the ordering. Explicitly, m_k at degree k is the sum over trees T with k leaves of

$$m_T(a_1, \dots, a_k) = \int_{\mathrm{FM}_k(\mathbb{C})} \omega_T \cdot \prod_{v \in V(T)} a_{(n_v)} b_v\,,$$

where the integration is against the compactified configuration space volume form and the product is over internal vertices of T with the OPE modes prescribed by the edge labels.

The Maurer–Cartan equation $\partial\alpha + \alpha \star \alpha = 0$ is the *spectral Stasheff identity*. Integration over the boundary strata of $\mathrm{FM}_k(\mathbb{C})$ produces the A_∞ relations by Stokes' theorem:

$$\sum_{\substack{p+q=k+1\\ 1\le i\le p}} \pm\, m_p(a_1,\ldots,a_{i-1},\, m_q(a_i,\ldots,a_{i+q-1}),\, a_{i+q},\ldots,a_k) \;=\; 0.$$

Each term in this identity corresponds to a codimension-one boundary stratum of $\mathrm{FM}_k(\mathbb{C})$ where a subset of points collides: the inner m_q extracts the OPE from the collision, and the outer m_p evaluates the remaining configuration.

10.4. PVA descent: Theorem G

On cohomology $H^\bullet(\mathcal{A}, Q)$, the operations simplify by a dimension count.

The commutative product. The regular part of m_2, the constant term as the two points separate, is a commutative associative product μ:

$$\mu(a,b) \;=\; \mathrm{Reg}_{z_1\to z_2}\; m_2(a(z_1), b(z_2)).$$

Commutativity: $\mathrm{FM}_2(\mathbb{C})$ has a $\mathbb{Z}/2$ involution exchanging the two points; the regular part is invariant. Associativity: the three boundary strata of $\mathrm{FM}_3(\mathbb{C})$ corresponding to pairwise collisions contribute only to the singular part; the regular boundary term is associativity.

The λ-bracket. The singular part of m_2 defines the λ-bracket:

$$\{a_\lambda b\} \;=\; \mathrm{Res}_{z_1\to z_2}\; e^{\lambda(z_1-z_2)}\; m_2(a(z_1), b(z_2)).$$

The exponential extracts all pole orders simultaneously: $\{a_\lambda b\} = \sum_{n\ge 0} \lambda^n a_{(n)} b/n!$.

Vanishing of higher operations. All $m_{k\ge 3} = 0$ on $H^\bullet$: the topological factor $\mathrm{Conf}_k(\mathbb{R})$ is contractible for $k \ge 3$ (it retracts to a point by pulling all k ordered points together), so the cohomological representative of m_k is $d_{\mathbb{R}}$-exact and therefore trivial on $H^\bullet$.

Jacobi identity from the Arnold relation. On $\mathrm{FM}_3(\mathbb{C})$, the three propagators $\eta_{12}, \eta_{23}, \eta_{31}$ satisfy the Arnold relation:

$$\eta_{12}\wedge\eta_{23} \;+\; \eta_{23}\wedge\eta_{31} \;+\; \eta_{31}\wedge\eta_{12} \;=\; 0.$$

This identity on differential forms is equivalent to the three codimension-one boundary divisors of $\overline{M}_{0,4}$ being homologous to zero in sum. Taking the residue of this relation against $a\otimes b\otimes c$ produces the three terms of the Jacobi identity for $\{-_\lambda-\}$:

$$\{a_\lambda\{b_\mu c\}\} \;-\; \{b_\mu\{a_\lambda c\}\} \;=\; \{\{a_\lambda b\}_{\lambda+\mu} c\}.$$

Leibniz rule from three-face Stokes. The boundary of $\mathrm{FM}_3(\mathbb{C})$ decomposes into three codimension-one faces, each corresponding to a pairwise collision. The Stokes identity

$$\int_{\partial\,\mathrm{FM}_3(\mathbb{C})} \omega \;=\; \int_{\sigma_{12}} \omega \;+\; \int_{\sigma_{23}} \omega \;+\; \int_{\sigma_{13}} \omega \;=\; 0$$

applied to the form $\omega = \eta_{12} \wedge \eta_{23} \cdot a(z_1)b(z_2)c(z_3)$ gives: on σ_{12}, the bracket $\{a_\lambda b\}$ is formed first and then multiplied by c; on σ_{23}, a multiplied by $\{b_\lambda c\}$; on σ_{13}, $\{a_\lambda c\} \cdot b$. The cancellation is the Leibniz rule:

$$\{a_\lambda (b \cdot c)\} \ = \ \{a_\lambda b\} \cdot c \ + \ b \cdot \{a_\lambda c\}.$$

Skew-symmetry from the exchange cylinder. The exchange involution on $\mathrm{FM}_2(\mathbb{C}) \times E_1(2)$ swaps the two insertion points. The monodromy of η_{12} around the exchange loop in the E_1 direction produces a sign and a shift: $\{a_\lambda b\} = -\{b_{-\lambda-\partial} a\}$. The derivative ∂ appears because the exchange path on $E_1(2)$ winds once around the difference coordinate $z_1 - z_2$.

Thus $(\mu, \{-_\lambda-\})$ is a (-1)-shifted Poisson vertex algebra, each axiom following in one line from the geometry of $\mathrm{FM}_k(\mathbb{C})$.

10.5. The Heisenberg algebra in Swiss-cheese coordinates

For the Heisenberg algebra $\mathcal{H}_k$, generated by a single field J of conformal weight 1 with $J(z)J(w) \sim k/(z-w)^2$:

$$m_2(J, J) = k \cdot \mathbf{1}, \qquad m_{k\geq 3} = 0, \qquad \{J_\lambda J\} = k\lambda.$$

All higher operations vanish because the OPE has no first-order pole: the only nontrivial residue is $J_{(1)}J = k$, and configuration-space integrals at degree $k \geq 3$ factor through this single datum, yielding zero by degree considerations (the form $\eta_{12} \wedge \cdots$ has too high a degree to pair nontrivially with the rank-one OPE data). The spectral R-matrix is

$$R(z) \ = \ \exp(k/z).$$

The Yang–Baxter equation is trivially satisfied: the R-matrix is scalar (acting on $\mathbb{C} \otimes \mathbb{C}$), so both sides of the YBE are $\exp(k/z_{12} + k/z_{13} + k/z_{23})$. The PVA $\{J_\lambda J\} = k\lambda$ is the free bosonic PVA, the universal enveloping PVA of the rank-one abelian Lie conformal algebra.

10.6. The Koszul triangle

The open factorization dg-category $\mathcal{C}$ on $\widetilde{X}_D$ produces three independent algebraic objects, each by a different functor:

(i) *Boundary.* $A_b = \mathrm{End}_{\mathcal{C}}(b)$ for a chosen vacuum object b: an algebra over the open colour of $\mathsf{SC}^{\mathrm{ch,top}}$, encoding the E_1-composition. The boundary algebra is a *chart* on $\mathcal{C}$, determined up to Morita equivalence by the choice of b.

(ii) *Bulk.* $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}) = R\mathrm{Hom}_{\mathrm{Fun}(\mathcal{C},\mathcal{C})}(\mathrm{Id}, \mathrm{Id})$: the chiral derived center, the universal algebra of bulk operators. It is Morita-invariant, independent of the choice of b. In any chart it is computed by chiral Hochschild cochains $C^\bullet_{\mathrm{ch}}(A_b, A_b)$.

(iii) *Lines.* $A^!_{b\infty} = D_{\mathrm{Ran}}(\bar{B}(A_b))$: the *homotopy* Koszul dual, obtained by Verdier duality on the bar coalgebra. On the Koszul locus, its cohomology recovers the strict Koszul dual $A^!_b = (H^*(\bar{B}(A_b)))^\vee$ (Theorem A). The module category $A^!_b\text{-}\mathsf{mod}$, equipped with the spectral R-matrix $R(z)$, is the braided tensor category of line operators.

The three objects are related by a triangle of identifications

$$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}) \ \simeq \ \mathrm{HH}^\bullet_{\mathrm{alg}}(A_b) \ \simeq \ \mathrm{ChirHoch}^\bullet(A^!_b), \qquad A^!_b\text{-}\mathsf{mod} \ = \ \{\text{line operators}\},$$

but the three functors (endomorphisms, Ran-space Verdier duality, identity-bimodule endomorphisms) must never be conflated. In particular: the bar complex $\bar{B}(A_b)$ classifies twisting morphisms (universal couplings between A_b and $A_b^!$) and is not the bulk. The derived center classifies bulk operators acting on the boundary.

The spectral braiding. The R-matrix $R(z)\colon L_1 \otimes L_2 \xrightarrow{\sim} L_2 \otimes L_1$ is the spectral braiding on $\mathcal{A}^!$-modules. Stokes' theorem on $\mathrm{FM}_3(\mathbb{C})$ applied to the triple convolution $\alpha \star \alpha \star \alpha$ decomposes the boundary into three pairwise collision divisors, one for each side of the Yang–Baxter equation:

$$R_{12}(z_1-z_2)\, R_{13}(z_1-z_3)\, R_{23}(z_2-z_3) \;=\; R_{23}(z_2-z_3)\, R_{13}(z_1-z_3)\, R_{12}(z_1-z_2).$$

The three boundary strata of $\mathrm{FM}_3(\mathbb{C})$ are in bijection with the three channels $(12|3)$, $(13|2)$, $(23|1)$; the Stokes cancellation $\sum_\sigma \int_\sigma \omega = 0$ is precisely the YBE.

Closed sector = derived center. Let $\mathcal{B}_\partial$ be the boundary factorization algebra on $\mathbb{R}$ (the open/topological algebra of the SC structure). Its derived center $Z_{\mathrm{der}}(\mathcal{B}_\partial) := R\,\mathrm{Hom}_{\mathcal{B}_\partial\text{-bimod}}(\mathcal{B}_\partial, \mathcal{B}_\partial)$ computes endomorphisms of the identity bimodule. The E_1-structure on $\mathcal{B}_\partial$ gives $Z_{\mathrm{der}}(\mathcal{B}_\partial) \simeq \mathrm{HH}^\bullet(\mathcal{B}_\partial)$ as an E_2-algebra. The Swiss-cheese structure maps the holomorphic bulk algebra to this E_2-algebra, and the Quillen equivalence makes the map an equivalence.

Lines = $\mathcal{A}^!$-modules. An $\mathcal{A}^!$-module M defines a line operator: the module structure $\mathcal{A}^! \otimes M \to M$ is a topological boundary condition parametrised by a line in $\mathbb{R}$, and the spectral braiding $R(z)$ provides the crossing data when two lines intersect. The category $C := \mathcal{A}^!\text{-}\mathsf{mod}$ with braiding $R(z)$ is the braided tensor category of line operators of the 3d holomorphic-topological theory.

10.7. Curved Swiss-cheese at genus $g \geq 1$

At genus $g \geq 1$, the bar complex $(\bar{B}^{\mathrm{ch},(g)}(\mathcal{A}),\ d_{\mathrm{fib}},\ \Delta)$ carries curvature:

$$d_{\mathrm{fib}}^2 \;=\; \kappa(\mathcal{A}) \cdot \omega_g.$$

The curvature ω_g is the Arakelov $(1,1)$-form on the fiber of the universal curve $\pi\colon C_g \to \overline{\mathcal{M}}_g$, representing the first Chern class of the relative dualising sheaf. The coproduct Δ remains strictly coassociative: the topological E_1-structure on $\mathbb{R}$ sees no curvature (the interval has no nontrivial topology). The curvature lives entirely in the holomorphic direction.

The total corrected differential

$$D_g \;=\; d_{\mathrm{fib}} + \nabla_g^{\mathrm{GM}}$$

restores flatness over $\overline{\mathcal{M}}_g$ by coupling the $\mathbb{R}$-ordering to the Hodge bundle $\mathbb{E}$. The Gauss–Manin connection ∇^{GM} absorbs the $(1,1)$-curvature: $D_g^2 = 0$ because $d_{\mathrm{fib}}^2 + [d_{\mathrm{fib}}, \nabla^{\mathrm{GM}}] + (\nabla^{\mathrm{GM}})^2 = \kappa\omega_g - \kappa\omega_g = 0$.

The derived category of modules must be replaced by the *coderived* category $\mathsf{D}^{\mathrm{co}}(\mathcal{A}^{(g)})$ (or the contraderived category $\mathsf{D}^{\mathrm{ctr}}(\mathcal{A}^{(g)})$, depending on the side): curvature forces $d^2 \neq 0$ on fibers, so the ordinary derived category is empty (every object is acyclic), while the coderived category, defined via totalisation of the curvature bicomplex, retains the correct homotopy theory. The passage from derived to coderived is the categorified form of the passage from $d^2 = 0$ to the curved Maurer–Cartan equation $d\alpha + \alpha^2 + \kappa\omega = 0$.

10.8. The Lagrangian self-intersection

Volume II identifies the geometric substrate of the bar construction: the bar complex is the structure sheaf of the Lagrangian self-intersection $\mathfrak{S} = \mathcal{L} \times_{\mathcal{M}} \mathcal{L}$ in a (-2)-shifted symplectic stack, and every theorem of this volume is a face of that geometry. The bar-cobar adjunction (Theorem A) is the groupoid comodule-module adjunction for $\mathfrak{S}$; Koszul inversion (Theorem B) reconstructs the Lagrangian from its clean self-intersection; complementarity (Theorem C) is two Lagrangians whose intersection carries a (-1)-shifted symplectic structure; the modular characteristic (Theorem D) is the Kodaira–Spencer class of the Lagrangian family over the Hodge bundle; and the chiral Hochschild polynomial growth (Theorem H) is the HKR theorem for the Lagrangian embedding.

10.9. The factorization-primary hierarchy

The Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ is a *local model*: the formal completion of a factorization structure at collision points. The global truth is a factorizable $\mathcal{D}$-module on $\mathrm{Ran}(\Sigma_g)$ (the Ran space of finite non-empty subsets of the curve), varying in families over $\overline{\mathcal{M}}_g$. The operad is extracted from this by restricting to formal neighborhoods of the small diagonal, where $\Sigma_g \simeq \mathbb{C}$ and the factorization pattern reduces to FM-space operations. Six layers organise the hierarchy:

(i) *Geometric substrate:* smooth and stable nodal curves, Ran space, $\overline{\mathcal{M}}_{g,n}$.

(ii) *Factorizable $\mathcal{D}$-modules* (Beilinson–Drinfeld): a chiral algebra is a factorizable $\mathcal{D}$-module on $\mathrm{Ran}(X)$, not an algebra over an operad.

(iii) *HT product geometry:* $\Sigma_g \times \mathbb{R}$ with holomorphic factorization in Σ_g and locally constant factorization in $\mathbb{R}$.

(iv) *Bar as factorization coalgebra:* the bar complex is the factorization coalgebra on $\mathrm{Ran}(X)$ Koszul-dual to the chiral algebra. At genus 0: honest dg coalgebra. At genus $g \geq 1$: curved factorization coalgebra.

(v) *Koszul duality as factorization duality:* bar-cobar is factorization algebra $\leftrightarrow$ coalgebra duality. Feynman involution $\mathrm{FT}^2 \simeq \mathrm{id}$ is the involutivity of this duality.

(vi) *Three models as three gauges:* the flat associated graded ($D_0^2 = 0$, operadic/local), the corrected holomorphic ($D^{(g)\,2} = 0$, Fay trisecant, still local), and the curved geometric ($d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$, Arakelov, global). The corrected model is visible to the local operad; the curved model requires the factorization framework.

The curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ is the measure of what the local extraction loses: it is the monodromy of the $\mathcal{D}$-module connection around the B-cycles of Σ_g, invisible to the formal-disc model. The modular bar complex $\bar{B}_{\mathrm{mod}}(\mathcal{A})$ is flat ($D^2 = 0$) because it couples all genera simultaneously via the non-separating operator D_1; the genus-g bar complex on a fixed surface is curved because restricting to a single genus decouples this connection.

The relative Feynman transform $\mathrm{FT}_{\mathsf{Com}_{\mathrm{mod}}/\mathsf{SC}^{\mathrm{ch,top}}}$ mediates between the factorization (global) and operadic (local) viewpoints. The modular bar coalgebra is naturally an algebra over this relative transform. Relative homotopy-involutivity plus factorization Koszul duality gives the flat-derived / curved-coderived comparison at all genera: the two flat models share an ordinary derived package, while the genus-fixed curved model is its coderived avatar rather than a third ordinary chain complex.

10.10. The four-stage architecture

Four stages, each depending on its predecessor.

Stage 1: Local one-colour theorem. The Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ is homotopy-Koszul; the resulting bar-cobar adjunction is a Quillen equivalence. The chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}) \simeq C^{\bullet}_{\mathrm{ch}}(A_b, A_b)$ is the universal closed-sector algebra. Proved.

Stage 2: Open primitive. The primitive object is the open factorization dg-category $\mathcal{C}$. The boundary algebra $A_b = \mathrm{End}_{\mathcal{C}}(b)$ is a chart, not an essence; the bulk $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C})$ is Morita-invariant. Proved.

Stage 3: Globalization. Tangential log curves (X, D, τ) provide the geometric substrate for bordered Fulton–MacPherson compactification. The local-global bridge lifts the operad-level Swiss-cheese structure to a factorizable $\mathcal{D}$-module on $\mathrm{Ran}(X)$. Proved locally; modest globally.

Stage 4: Modularization. Trace and clutching on the open sector produce the full modular Maurer–Cartan element $\Theta^{\mathrm{oc}} = \Theta_{\mathcal{A}} + \sum \mu^{M_j}$. The annulus trace $\int_{S^1} \mathcal{C} \simeq \mathrm{HH}_{\bullet}(\mathcal{C})$ is the first modular shadow of the open sector; clutching along boundary circles via nodal degeneration raises genus. Annulus trace proved; full chain-level clutching is programme.

Five negative principles constrain the architecture:

N1. *Bar ≠ bulk.* The bar complex $\bar{B}(A_b)$ is a factorization coalgebra classifying twisting morphisms. The bulk $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C})$ is the chiral derived center classifying bulk operators. Different functors, different outputs.

N2. *Boundary algebra = chart.* $A_b = \mathrm{End}_{\mathcal{C}}(b)$ depends on the choice of vacuum b. Only $\mathcal{C}$ itself and its derived center are intrinsic.

N3. *Tangential log geometry required.* The bordified curve $\widetilde{X}_D$ is a manifold with corners on an snc pair (X, D). Ordinary Fulton–MacPherson compactification is the $D = \varnothing$ special case.

N4. *Modularity = trace + clutching on the open sector.* The genus expansion is not an additional axiom on the closed algebra; it is determined by the cyclic trace on $\mathrm{End}_{\mathcal{C}}(b)$ and the gluing along boundary circles.

N5. *Honest scope.* The dg-shifted Yangian is proved for affine lineage (Kac–Moody and its standard reductions). Extension to non-standard families is programme, not theorem.

11. Modular PVA quantization

The open/closed world of Section 10 produces a genus-zero datum: the Swiss-cheese MC element α_T, the PVA $(\mu, \{-_\lambda-\})$ on cohomology, and the KZ/BPZ shadow connections at degree n. All of these are tree-level structures. The question that forces the present section is: how does this genus-zero structure extend across the genus filtration? The answer is the modular PVA quantization programme: a genus-by-genus obstruction theory whose input is the genus-zero PVA and whose output is the full modular MC element. The nonseparating degeneration operator D_1 is the primitive genus-raising differential; it squares to zero, anticommutes with the tree-level differential D_0, and defines a spectral sequence whose obstructions are the quantization anomalies.

11.1. The one-edge principle

To verify a modular operad structure, it suffices to check compatibility with a single edge contraction: all higher-edge relations follow from codimension-two cancellation.

If a complete filtered dg Lie algebra L carries a genus filtration and its differential preserves genus exactly, then any genus-zero Maurer–Cartan element $\Theta_0 \in \mathrm{MC}(\mathrm{gr}^0_F L)$ extends tautologically to a full MC element: genus by genus, the obstruction class $[D_1 \Theta_0] \in H^2(\mathrm{gr}^1 L)$ vanishes because no differential raises genus. Therefore every nontrivial modular obstruction theory must contain a primitive genus-raising operator.

The one-edge expansion of the stable-graph differential furnishes this primitive. Every one-edge degeneration of a stable graph is either *separating* (two vertices joined by one edge, total genus preserved) or *nonseparating* (one vertex acquiring one loop, total genus raised by 1). Set

$$D_{\mathrm{exp}} \;=\; D_{\mathrm{sep}} + D_{\mathrm{nsep}}, \qquad D_0 \;:=\; D_{\mathrm{int}} + D_{\mathrm{sep}}, \qquad D_1 \;:=\; D_{\mathrm{nsep}}.$$

Then D_0 preserves total genus, D_1 raises total genus by one, and $(D_0 + D_1)^2 = 0$ stratifies by genus shift into three identities:

$$D_0^2 = 0, \qquad D_0 D_1 + D_1 D_0 = 0, \qquad D_1^2 = 0.$$

The first: D_0 is a differential. The second: D_1 is a D_0-derivation (it anticommutes with D_0, so it descends to a map on D_0-cohomology). The third: D_1 squares to zero, so the genus-one obstruction is a well-defined cohomology class. The three identities are the codimension-two cancellation axiom stratified by genus.

Trees govern chirality; loops govern modularity. D_0 (tree-level degenerations) encodes OPE residues and factorization. D_1 (loop-creating degenerations) encodes genus raising, trace maps, the odd Laplacian. The full modular bar theory is their interaction.

11.2. The modular bar datum

A *modular bar datum* on a reduced, complete, finite-type graded collection $F \mapsto C(F)$ consists of:

(i) An internal differential $d\colon C(F) \to C(F)$ of degree $+1$, with $d^2 = 0$.

(ii) For each one-edge expansion $\epsilon \in \mathrm{OE}(F)$ (an edge e adjoined to the flag set F, either separating (producing two flag sets $F_1 \sqcup F_2 = F \setminus \{f_1, f_2\} \sqcup \{f_1', f_2'\}$) or nonseparating (the flag set F acquires two new flags $\{h, h'\}$ identified as half-edges of e, raising genus by 1)), a degree-$+1$ map

$$\Delta_\epsilon\colon sC(F) \;\longrightarrow\; \mathfrak{o}(\epsilon) \otimes \bigotimes_{u \in V(\epsilon)} sC(\mathrm{Fl}_\epsilon(u)),$$

where $\mathfrak{o}(\epsilon)$ is the orientation line of the new edge.

(iii) The *codimension-two cancellation axiom*: for every connected stable graph with exactly two internal edges, the signed sum of the two ways of producing it by successive one-edge expansions is zero:

$$\sum_{\substack{\epsilon_1, \epsilon_2 \\ \epsilon_1 \circ \epsilon_2 = \Gamma}} \mathrm{sgn}(\epsilon_1, \epsilon_2) \cdot \Delta_{\epsilon_2} \circ \Delta_{\epsilon_1} \;=\; 0.$$

(iv) The anticommutation relation $d \circ \Delta_\epsilon + \Delta_\epsilon \circ d = 0$.

The total differential $D = D_{\mathrm{int}} + D_{\mathrm{exp}} = d + \sum_\epsilon \Delta_\epsilon$ satisfies $D^2 = 0$ by three mechanisms: $d^2 = 0$ on vertex labels; anticommutation of d with one-edge expansions; and the codimension-two cancellation axiom for two-edge graphs. This is the algebraic incarnation of the cell decomposition of $\overline{\mathcal{M}}_{g,n}$ into stable-graph strata: the codimension-two axiom is the statement that every codimension-two stratum arises from two distinct codimension-one degenerations, with opposite signs.

11.3. The complete modular bar coalgebra

For a connected stable graph Γ of genus g with vertex set $V(\Gamma)$, edge set $E(\Gamma)$, and leg set $L(\Gamma)$, set

$$C[\Gamma] \;:=\; \mathfrak{o}_(\Gamma) \otimes \bigotimes_{v \in V(\Gamma)} sC(\mathrm{Fl}(v)),$$

where $\mathfrak{o}_(\Gamma) = \bigotimes_{e \in E(\Gamma)} \mathfrak{o}_(e)$ is the total orientation line and $\mathrm{Fl}(v)$ is the flag set at vertex v. The *complete modular bar coalgebra* is

$$\mathrm{B}_{\mathrm{mod}}(C) \;=\; \prod_{[\Gamma] \in \mathsf{StGraph}^{\mathrm{conn}}} \big(C[\Gamma]\big)_{\mathrm{Aut}(\Gamma)},$$

completed over total genus $g(\Gamma) = b_1(\Gamma) + \sum_v g(v)$. The reduced coproduct is edge contraction: contracting an internal edge of Γ strictly decreases the number of internal edges, so the coalgebra is conilpotent. The differential on $\mathrm{B}_{\mathrm{mod}}(C)$ is the sum of the internal differential (acting on vertex labels) and the expansion differential (summing over all possible one-edge insertions into each graph).

11.4. The strict modular deformation algebra

The coderivation dg Lie algebra

$$L_{\mathrm{mod}}(C) \;:=\; \mathrm{Coder}\big(\mathrm{B}_{\mathrm{mod}}(C)\big)[-1]$$

is the *strict modular deformation algebra*. Its elements are coderivations of the modular bar coalgebra, shifted by -1 to place the Lie bracket in degree 0. The convolution description: the universal property of the cofree coalgebra gives

$$L_{\mathrm{mod}}(C) \;\cong\; \mathrm{Hom}\big(\mathrm{B}_{\mathrm{mod}}(C),\, sC\big)[-1]$$

with the commutator bracket $[f, g] = f \star g - (-1)^{|f||g|} g \star f$, where $f \star g$ is the convolution product (compose f with the coproduct, apply g to one tensor factor, compose back).

The algebra $L_{\mathrm{mod}}(C)$ is complete filtered by genus: $F^g L_{\mathrm{mod}}(C)$ consists of coderivations supported on graphs of genus $\geq g$. The completion is pro-nilpotent: $F^1 L_{\mathrm{mod}}$ is a pro-nilpotent dg Lie algebra, and $\mathrm{gr}^0 L_{\mathrm{mod}}$ is the tree-level deformation algebra. The full algebra is Lie-multiplicative: the bracket $[f, g]$ respects the genus filtration, and the associated graded inherits a dg Lie structure at each genus.

11.5. The genus-one obstruction theorem

For a genus-zero MC element $\Theta_0 \in \mathrm{MC}(\mathrm{gr}^0_F L)$ satisfying $D_0\Theta_0 + \frac{1}{2}[\Theta_0, \Theta_0]_0 = 0$, the genus-one obstruction is

$$\mathrm{Ob}_1(\Theta_0) \;=\; [D_1\Theta_0] \;\in\; H^2\big((\mathrm{gr}^1_F L)^{\Theta_0},\, d_0^{\Theta_0}\big),$$

where $d_0^{\Theta_0} = D_0 + [\Theta_0, -]$ is the Θ_0-twisted differential on the genus-one piece. The class $\mathrm{Ob}_1(\Theta_0)$ is well-defined: $D_1\Theta_0$ is a $d_0^{\Theta_0}$-cocycle because of the identity $D_0D_1 + D_1D_0 = 0$ and the MC equation for Θ_0. It vanishes if and only if Θ_0 lifts to a genus-one MC element $\Theta_0 + \hbar\Theta_1$ satisfying

$$(D_0 + D_1)(\Theta_0 + \hbar\Theta_1) + \tfrac{1}{2}[\Theta_0 + \hbar\Theta_1, \Theta_0 + \hbar\Theta_1] \;=\; 0 \qquad \mod \hbar^2.$$

In PVA coordinates, the nonseparating one-edge operator acts as the reduced odd Laplacian. For a cyclic A_∞-algebra $(A, m_k, \langle -, - \rangle)$ with basis $\{e^i\}$ and dual basis $\{e_i\}$:

$$\ell_1^{(1)}(a) \;=\; \hbar\Delta_{\mathrm{cyc}}(a) \;:=\; \hbar \sum_i m_2(e^i, m_2(a, e_i)) \;\in\; A,$$

the cyclic contraction over the single loop. This is the genus-raising operator: it takes a genus-0 vertex label and produces a genus-1 correction by contracting one pair of insertions through the cyclic pairing. The full genus-one obstruction Ob_1 is the cohomology class of $\Delta_{\mathrm{cyc}}(\Theta_0)$ in the twisted complex.

11.6. The $\mathcal{W}_3$ computation

A *filtered quadratic PVA* is a differential commutative algebra freely generated by fields $u^1, \ldots, u^N$ and their derivatives, with a λ-bracket encoded by a skew-adjoint matrix differential operator $\Pi^{ab}(\partial) = \sum_{m\geq 0} \Pi^{ab}_m \partial^m$ whose coefficients have total generator degree at most 2. Classical $\mathcal{W}_3$ fits this pattern: two generators T (weight 2) and W (weight 3), with the quasi-primary composite

$$\Lambda \;=\; {:}TT{:} \;-\; \tfrac{3}{10}\,\partial^2 T$$

controlling the nonlinear channel. The WW OPE is

$$\begin{aligned} W(z)W(w) \;\sim\; & \frac{c/3}{(z-w)^6} \;+\; \frac{2T}{(z-w)^4} \;+\; \frac{\partial T}{(z-w)^3} \\ & +\; \frac{1}{(z-w)^2}\left(\tfrac{3}{10}\partial^2 T + \frac{16}{22+5c}\Lambda\right) \;+\; \frac{1}{z-w}\left(\cdots\right). \end{aligned}$$

At $c = -22/5$, the denominator $22 + 5c$ vanishes, Λ becomes null, and $\mathcal{W}_3$ is no longer finitely generated by T and W alone.

The *triangular W-normal form vanishing theorem* asserts: in the local, translation-invariant, filtration-preserving sector, $\mathrm{Ob}_1(\Theta_0) = 0$ for classical $\mathcal{W}_3$. The computation: the relevant cohomology group is

$$H^1\big((\mathrm{gr}^1 L)^{\Theta_0},\, d_0^{\Theta_0}\big) \;\cong\; \mathbb{C}\cdot \partial_c P_c\,,$$

one-dimensional, spanned by the central-parameter direction $\partial_c P_c$, where P_c is the λ-bracket matrix viewed as a function of c. Concretely: every genus-one lift of the classical $\mathcal{W}_3$ PVA is gauge equivalent to a renormalization of the central charge $c \mapsto c + \hbar c_1$. The genus-one quantum $\mathcal{W}_3$ algebra is the unique quantization in the direction ∂_c.

11.7. Curved extension via S-tails

In the inhomogeneous quadratic setting (Gui–Li–Zeng), the correct input is a *twisted pair* (B, B°, S): a CDG-algebra B, a CDG-coalgebra B°, and a curvature element S. An *S-tail expansion* attaches a new univalent S-decorated vertex to a pre-existing corolla: formally, a unary expansion map

$$\Delta_S\colon sC(F) \;\longrightarrow\; sC(F \sqcup \{\bullet, \star\})$$

followed by contraction of the new internal edge, with the vertex $\star$ carrying the curvature label S. Adjoining S-tail expansions to the one-edge expansion set, with the same codimension-two compatibilities, produces a *curved complete modular bar object*.

The Maurer–Cartan equation in the curved setting is:

$$\not{\mu}\big((S+\alpha)\boxtimes(S+\alpha)\big) \;=\; 0,$$

the modular extension of the Gui–Li–Zeng curved equation. The element $S+\alpha$ combines the curvature S (a genus-zero scalar, the analogue of m_0) with the deformation parameter α. The equation unpacks genus by genus: at genus 0, it is the curved A_∞ MC equation $\sum_k m_k(\alpha^{\otimes k}) = 0$ with $m_0 = S$; at genus 1, it is $\Delta_{\mathrm{cyc}}(S+\alpha) = 0$, the vanishing of the curved odd Laplacian; at genus g, it is the BV quantum master equation $\hbar\Delta(e^{(S+\alpha)/\hbar}) = 0$.

11.8. Genus-zero comparison with Gui–Li–Zeng

At tree level, the modular MC equation reduces to the classical twisting morphism equation. For a Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ with $\mathcal{A}^!$ a CDG algebra with curvature S, the genus-zero MC space is

$$\mathrm{MC}_0(L_{\mathrm{mod}}) \;\cong\; \mathrm{Hom}_{\mathsf{dg}}(\mathcal{A}, B^\circ) \;\hookrightarrow\; \mathrm{MC}(\mathcal{A}^!\widehat{\otimes}\mathcal{B}),$$

where $\mathcal{B}$ is a test algebra. This recovers the Gui–Li–Zeng correspondence: tree-level MC elements of the modular deformation algebra are twisting morphisms $\mathcal{A} \to B^\circ$, which are in bijection with CDG-algebra morphisms $B \to \mathcal{A}^!\widehat{\otimes}\mathcal{B}$. The modular theory is the genus extension of this classical picture.

12. Holographic modular Koszul duality

The open/closed world (Section 10) supplies the three-dimensional geometric framework; the modular PVA quantization programme (Section 11) supplies the genus-by-genus obstruction theory. The holographic modular Koszul datum packages both into a single algebraic object: a six-component structure that records the boundary algebra, its Koszul dual, the line-operator category, the spectral R-matrix, the universal MC element, and the modular shadow connection. Every ingredient constructed so far (the bar complex, the shadow obstruction tower, the Swiss-cheese structure, the PVA descent, the genus expansion) appears as a projection of this datum. The datum is not a summary; it is the natural unit of holographic data, the smallest package from which the full correspondence can be reconstructed.

12.1. The holographic modular Koszul datum

Every three-dimensional holomorphic-topological system T is controlled by a *holographic modular Koszul datum*

$$\mathcal{H}(T) \;=\; \big(\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}\big),$$

defined as follows.

(i) $\mathcal{A}$: the *boundary chiral algebra*, a chiral algebra on a complex curve X. It is the algebra of boundary local operators of T restricted to the holomorphic boundary $\{0\}\times X \subset \mathbb{R}_+ \times X$.

(ii) $\mathcal{A}^!$: the *strict chiral Koszul dual* of $\mathcal{A}$, obtained by linear duality of bar cohomology: $\mathcal{A}^! = (\mathcal{A}^{\mathrm{i}})^\vee$ where $\mathcal{A}^{\mathrm{i}} = H^*(\bar{B}(\mathcal{A}))$. In the holographic setting, $\mathcal{A}^!$ is a *dg-shifted Yangian*. The bar complex

$\bar{B}(A_b)$ is an E_1 chiral coassociative coalgebra over $(\mathrm{ChirAss})^!$; the two-coloured $\mathrm{SC}^{\mathrm{ch,top}}$ structure emerges on the derived center pair $(C^\bullet_{\mathrm{ch}}(A_b, A_b),\ A_b)$, not on $\bar{B}(A_b)$ itself. The Koszul dual $\mathcal{A}^!$ is the E_1-projection of the Verdier dual; it carries an E_1-algebra structure from the topological direction.

(iii) $\mathcal{C} \simeq \mathcal{A}^!\text{-}\mathsf{mod}$: the *line-operator category*, a braided tensor category. Objects are line defects of T (topological lines in $\mathbb{R}_+$); morphisms are junction operators; the braiding comes from crossing in the holomorphic plane.

(iv) $r(z) \in \mathcal{A}^! \otimes \mathcal{A}^!((z^{-1}))$: the *dg-shifted Yangian R-matrix*, a meromorphic endomorphism with a pole at $z = 0$. It satisfies the classical Yang–Baxter equation

$$[r_{12}(z_1-z_2),\ r_{13}(z_1-z_3)] + [r_{12}(z_1-z_2),\ r_{23}(z_2-z_3)] + [r_{13}(z_1-z_3),\ r_{23}(z_2-z_3)] \ = \ 0.$$

The R-matrix is the binary genus-0 shadow of $\Theta_{\mathcal{A}}$:

$$r(z) \ = \ \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}).$$

(v) $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$: the *universal Maurer–Cartan element* of the modular convolution L_∞-algebra, existing unconditionally by the bar-intrinsic construction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$. All lower structures ($\kappa$, $r(z)$, the shadow connection) are projections of $\Theta_{\mathcal{A}}$ to specific genus and degree components.

(vi) $\nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})$: the *modular shadow connection*, a connection on the modular shadow bundle over $\overline{\mathcal{M}}_{g,n}$. Flatness of ∇^{hol}:

$$(\nabla^{\mathrm{hol}}_{g,n})^2 \ = \ d\,\mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}}) + \tfrac{1}{2}[\mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}}), \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})] \ = \ 0$$

is the MC equation projected to genus g and degree n.

12.2. The collision filtration

The modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ carries a *collision filtration* $F^0_{\mathrm{coll}} \supset F^1_{\mathrm{coll}} \supset \cdots$ indexed by the minimum pairwise distance on $\mathrm{FM}_n(\mathbb{C})$: depth 0 is the free-propagation regime (all points well-separated), depth 1 is the first collision stratum (one pairwise collision), depth k is the k-fold nested collision locus (a sequence of k nested screens in the FM compactification).

The associated spectral sequence has

$$E_1^{p,*} \ = \ H^*\big(\mathrm{gr}^p_{\mathrm{coll}}\ \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}},\ d_0\big),$$

with $E_1^{0,*}$ the free-field contribution (no collisions, d_0 the free differential) and $E_1^{1,*}$ the collision residues (single pairwise collisions, extracting the leading OPE singularity). The first differentials $d_1\colon E_1^{p,*} \to E_1^{p+1,*}$ are Arnold-type cancellations: the Arnold relation on $\mathrm{FM}_3(\mathbb{C})$ forces $d_1^2 = 0$ on the E_1 page. The spectral sequence converges to $H^*(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$, with the collapse at the E_r page governed by the shadow depth: for class-**G** algebras (shadow depth 2), the spectral sequence collapses at E_2; for class-**M** algebras (shadow depth ∞), it does not collapse at any finite page.

12.3. Four proved recovery theorems

The collision filtration recovers four structures from $\Theta_{\mathcal{A}}$.

(R1) Collision residue = twisting. The depth-1 collision stratum at degree 2 extracts the binary shadow:

$$\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \;=\; \pi_{\mathcal{A}},$$

the universal twisting morphism producing the dg-shifted Yangian $r_{\mathcal{A}}(z)$. The collision residue extracts the leading OPE pole $a_{(0)}b/(z_1 - z_2)$; the MC equation at this depth is the twisting morphism equation $\partial\pi + \pi \star \pi = 0$.

(R2) CYBE from Arnold. The three boundary divisors of $\overline{M}_{0,4}$ correspond to the three channels (12|34), (13|24), (14|23). Evaluating the Arnold relation $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$ on the Casimir tensor $\Omega \otimes \Omega \in (\mathcal{A}^!)^{\otimes 3}$ produces the three terms of the CYBE:

$$[r_{12}(z_{12}),\, r_{13}(z_{13})] \;+\; [r_{12}(z_{12}),\, r_{23}(z_{23})] \;+\; [r_{13}(z_{13}),\, r_{23}(z_{23})] \;=\; 0,$$

where $z_{ij} = z_i - z_j$. Each bracket corresponds to a boundary divisor, and the Arnold relation is the vanishing of the boundary integral.

(R3) Affine shadow/KZ comparison on the affine comparison surface. For the affine algebra $\mathcal{A} = \widehat{\mathfrak{g}}_k$, on the affine Kac–Moody comparison surface the genus-0 shadow one-form at degree n is the Knizhnik–Zamolodchikov connection form:

$$\mathrm{Sh}_{0,n}(\Theta_{\widehat{\mathfrak{g}}_k}) \;=\; \sum_{i<j} r_{ij}(z_i - z_j)\, dz_{ij} \;=\; \frac{1}{h^\vee + k} \sum_{i<j} \frac{\Omega_{ij}\, dz_{ij}}{z_{ij}}\,,$$

where $r_{ij}(z) = \Omega_{ij}/((h^\vee + k)z)$ denotes the connection coefficient, not the Arnold form. Flatness of $\nabla_{\mathrm{KZ}} = d - \mathrm{Sh}_{0,n}$ is the MC equation at genus 0 and degree n. For $\mathcal{A} = \mathrm{Vir}_c$, on the Virasoro degenerate-module comparison surface the analogous Virasoro conformal Ward extraction, together with higher collision residues, yields the BPZ differential equations. The affine Drinfeld–Kohno theorem compares the braid-group monodromy of the genus-0 shadow of $\Theta_{\widehat{\mathfrak{g}}_k}$ with the corresponding quantum-group representation category.

(R4) Quartic = L_∞ obstruction. The quartic resonance class

$$[\mathfrak{R}^{\mathrm{mod}}_4(\mathcal{A})] \;=\; [o_4(\mathcal{A})] \;\in\; H^2(\mathcal{A}^{\mathrm{sh}}_{4,0})$$

is the obstruction to extending the Yangian seed $r(z)$ from degree 2 to degree 4. It vanishes for classes **G**, **L**, **C** (the R-matrix extends to all degrees) and is nonzero for class **M** (the nonlinear OPE forces a genuine quartic obstruction: $Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c + 22)]$).

12.4. Five conjectural theorem targets

(i) *Boundary-defect realization.* The boundary chiral algebra $\mathcal{A}_\partial(T)$ and its Koszul dual $\mathcal{A}^!_\partial(T)$ are the two algebras appearing in the holographic datum for every 3d HT system T, with $C \simeq \mathcal{A}^!$-mod as braided tensor category of line operators. The physical content: boundary local operators and bulk-to-boundary maps determine the full holographic datum up to quasi-isomorphism.

(ii) *Yangian-shadow.* In full derived generality: for every HT system T, the collision residue $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ recovers the dg-shifted Yangian $r(z)$ attached to T, not merely at the classical level but at the chain level, including all higher homotopies of the A_∞ structure.

(iii) *Sphere reconstruction.* The Gaiotto–Zhang (2026) commuting differentials on the non-planar sphere algebra are the scalar shadow $\mathrm{Sh}_{0,n}(\Theta_{\mathcal{A}})$: the 2026 sphere algebra is the binary genus-0 shadow of the modular MC class. The n-point sphere amplitude $\langle O_1(z_1)\cdots O_n(z_n)\rangle_{S^2}$ is, on the established genus-0 comparison surfaces, a flat section of $\nabla^{\mathrm{hol}}_{0,n}$.

(iv) *Quartic resonance obstruction.* The quartic resonance class $\mathfrak{R}_4^{\mathrm{mod}}$, the first global nonlinear invariant, is the first test of the modular datum beyond the spectral determinant: its non-vanishing for class-**M** algebras is the quantitative obstruction to planarity of the holographic system, measuring the failure of the large-N expansion to truncate at tree level.

(v) *Singular-fiber descent.* At degenerate complex structure (boundary of $\overline{\mathcal{M}}_{g,n}$), the modular datum degenerates to the boundary of the Mok compactification: $\nabla^{\mathrm{hol}}|_{\partial\overline{\mathcal{M}}} \sim \nabla^{\mathrm{clutch}}$ with residues controlled by the clutching law of $\Theta_{\mathcal{A}}$. The clutching law: $\Theta_{\mathcal{A}}|_{\sigma_\Gamma} = \prod_{v\in V(\Gamma)}\Theta_v \cdot \prod_{e\in E(\Gamma)} P_e$, where σ_Γ is a boundary stratum of $\overline{\mathcal{M}}_{g,n}$ labelled by a stable graph Γ, Θ_v is the vertex MC element, and P_e is the edge propagator.

12.5. The colored modular deformation object

A holomorphic-topological theory T produces three observation algebras:

$$\mathrm{Obs}^{\mathrm{hol}}\ (\text{bulk}), \qquad \mathrm{Obs}^{\partial}\ (\text{boundary}), \qquad \mathrm{Obs}^{\ell}\ (\text{line}).$$

The colored Swiss-cheese FM space $\mathrm{FM}^{\log}_{\bullet,\mathrm{SC}}$ encodes the collision geometry of all three simultaneously: bulk points in $\mathbb{C}$, boundary points on $\mathbb{R}\subset\mathbb{C}$, and line points on individual copies of $\mathbb{R}_+$. The three-color modular deformation object is

$$\mathfrak{g}^{\mathrm{HT,log}}_{\mathrm{mod}} \;=\; \hom^{\alpha_{\mathrm{HT}}}\big(C_\bullet(\mathrm{FM}^{\log}_{\bullet,\mathrm{SC}}),\ \mathrm{End}_{\mathrm{Ch}_\infty}(\mathrm{Obs}^{\mathrm{hol}}_\infty,\ \mathrm{Obs}^{\partial}_\infty,\ \mathrm{Obs}^{\ell}_\infty)\big),$$

the convolution sL_∞-algebra between the chains on the colored log-FM space and the endomorphism operad of the three observation algebras. The three colors interact through the Swiss-cheese operadic structure: bulk fields propagate to the boundary (closed-to-open maps), boundary fields propagate to lines (open-to-open composition), and lines cannot produce bulk fields (the $\varnothing$-prohibition of SC).

The modular effective action is the graph-sum

$$S^{\mathrm{mod}}_{\mathrm{HT}} \;=\; \sum_{\Gamma\in\mathsf{StGraph}_{\mathrm{SC}}} \frac{\hbar^{b_1(\Gamma)}}{|\mathrm{Aut}(\Gamma)|}\, W^{\log}_\Gamma\, O_\Gamma\,,$$

where the sum is over connected stable graphs Γ decorated by the three colors, $b_1(\Gamma)$ is the first Betti number, $W^{\log}_\Gamma$ is the Feynman weight (a period integral on the log-FM stratum σ_Γ), and O_Γ is the vertex operator (the tensor product of observation-algebra elements at each vertex). The BV quantum master equation

$$\big(d_{\mathrm{BV}} + \hbar\Delta_{\mathrm{odd}}\big)\exp\!\big(S^{\mathrm{mod}}_{\mathrm{HT}}/\hbar\big) \;=\; 0$$

is the single equation governing the 3d HT system T. At genus 0 ($\hbar^0$): the classical BV master equation $\{S_0, S_0\} = 0$. At genus 1 ($\hbar^1$): the one-loop anomaly cancellation $\Delta_{\mathrm{odd}}S_0 + \{S_0, S_1\} = 0$. At genus g ($\hbar^g$): the g-loop Ward identity.

12.6. The Khan–Zeng bridge

Khan and Zeng construct a three-dimensional holomorphic-topological Poisson sigma model on the half-space $\mathbb{R}_+ \times \mathbb{C}$ attached to a freely generated PVA. The field content: a $\mathfrak{g}$-valued $(0,1)$-form $A_{\bar{z}}$ on $\mathbb{C}$, a $\mathfrak{g}^*$-valued scalar ϕ, and a $\mathfrak{g}$-valued 1-form η along $\mathbb{R}_+$. The action is

$$S_{\mathrm{KZ}} = \int_{\mathbb{R}_+\times\mathbb{C}} \big(\langle\phi,\, \bar{\partial}A\rangle + \langle\phi,\, \tfrac{1}{2}\{A_\lambda A\}|_{\lambda=0}\rangle + \langle\eta,\, d_t\phi\rangle\big)\, dt\wedge dz\wedge d\bar{z}.$$

The results:

(i) Gauge invariance of S_{KZ} holds if and only if the λ-bracket $\{-_\lambda-\}$ satisfies the PVA Jacobi identity. The gauge algebra is the Lie conformal algebra underlying the PVA.

(ii) A Virasoro element T in the PVA upgrades the symmetry: the stress tensor provides a coupling to worldsheet gravity, promoting holomorphic-topological to fully topological (diffeomorphism-invariant on the boundary $\{0\}\times\mathbb{C}$).

(iii) The boundary theory on $\{0\}\times\mathbb{C}$ carries a (-1)-shifted Poisson bracket $\{-,-\}_{-1}$ and produces the genus-zero quantization problem: deforming $\{-,-\}_{-1}$ to a differential $Q = Q_0 + \hbar Q_1 + \cdots$ with $Q^2 = 0$.

(iv) The modular extension, coupling the bulk theory to stable curves at genus $g \geq 1$ via the fibration $C_g \to \overline{\mathcal{M}}_g$, controls the full genus tower of the quantum Koszul datum.

The deformation-quantization pipeline:

$$\text{classical coisson shadow} \xrightarrow{\text{KZ quantization}} \text{genus-0 quantum Koszul} \xrightarrow{\text{modular lift}} \mathcal{H}(T).$$

The first arrow is perturbative quantization of the Khan–Zeng action on $\mathbb{P}^1$ (tree-level Feynman diagrams). The second arrow is the genus-by-genus obstruction theory of §§10–11, extending the genus-0 MC element across the genus filtration.

The bridge is completed in Volume II on the affine lane by the one-loop-exact half-space BV theorem. For finite-type nonlinear Khan–Zeng PVAs, the classical BV action and the equivalence between the CME and PVA Jacobi are constructed; the all-loop QME, reflected weight identity, and boundary vertex algebra require the KZ analytic SDR package, including the image-choice compactification and orientation-compatible Stokes pushforward. The affine Poisson–vertex sigma model satisfies the hypotheses of the physics bridge theorem (BV data, one-loop finiteness, polynomial interactions), so it is a logarithmic $\mathsf{SC}^{\mathrm{ch,top}}$-algebra; Drinfeld–Sokolov reduction preserves this structure. The proved standard landscape gates (Kac–Moody, Virasoro, $\mathcal{W}_N$) enter through these two gates. The six-component holographic datum

$$\big(B_\partial,\ B^!_\partial,\ C_{\mathrm{line}},\ r(z),\ \Theta_{B_\partial},\ \nabla^{\mathrm{hol}}\big)$$

exists and is consistent for every logarithmic $\mathsf{SC}^{\mathrm{ch,top}}$-algebra: bulk $\simeq$ derived center of boundary, line operators $\simeq$ Koszul-dual modules, spectral R-matrix from the Laplace-transformed λ-bracket.

The genus expansion of the modular bar differential, $D = \sum_{g\geq 0}\hbar^g D^{(g)}$ with $D^{(g)} = \sum_{\Gamma:\, g(\Gamma)=g} |\mathrm{Aut}(\Gamma)|^{-1} D_\Gamma$, is the Feynman transform of the modular operad; the canonical MC element $\delta_{\mathcal{A}} = D - D^{(0)}$ computes the positive-genus corrections. The combinatorial structure (stable graphs, automorphism weights, genus weights) is the mathematical shadow of the 't Hooft expansion. The identification $\hbar \leftrightarrow 1/N$ for a specific holographic model remains the principal open bridge from the algebraic machine to gauge-theory holography.

12.7. The Heisenberg holographic datum

The base case: shadow class $\mathbf{G}$, shadow depth 2, trivial line-operator category, scalar R-matrix. For $\mathcal{A} = \mathcal{H}_k$:

$$\mathcal{A} = \mathcal{H}_k, \qquad \mathcal{A}^! = \mathrm{Sym}^{\mathrm{ch}}(V^*), \qquad \mathcal{C} = \mathsf{Vect},$$
$$r(z) = k/z, \qquad R(z) = \exp(k/z), \qquad \Theta_{\mathcal{H}_k} = k \cdot \eta \otimes \Lambda.$$

The shadow connection is trivial: $\nabla^{\mathrm{hol}}_{g,n} = d$ for all (g,n), because the MC element is scalar (it lives in the one-dimensional space $\mathbb{C} \cdot \eta \otimes \Lambda$, and the shadow $\mathrm{Sh}_{g,n}(\Theta_{\mathcal{H}_k})$ is a scalar multiple of the identity, producing a trivial connection).

All five conjectural targets are trivially verified:

(i) The boundary algebra $\mathcal{H}_k$ is free (one generator, no nonlinear OPE).
(ii) The Koszul dual $\mathrm{Sym}^{\mathrm{ch}}(V^*)$ is the curved symmetric chiral algebra with curvature $m_0 = -k\omega$ (Theorem **??**; note: $\mathcal{H}_k^! \neq \mathcal{H}_{-k}$).
(iii) The line-operator category is Vect: no nontrivial lines (the R-matrix is scalar, so the braiding is symmetric, and $\mathcal{H}_k^!$-modules over Vect are just vector spaces).
(iv) The quartic obstruction vanishes: shadow class $\mathbf{G}$, $o_4(\mathcal{H}_k) = 0$.
(v) Singular-fiber descent is trivial: $\Theta_{\mathcal{H}_k}$ has no dependence on moduli.

12.8. The affine $\widehat{\mathfrak{sl}}_2$ holographic datum

The first nontrivial holographic example: class $\mathbf{L}$, depth 3, $r(z) = k\,\Omega/z$ (Yang's solution of the CYBE), line-operator category $Y(\mathfrak{sl}_2)$-mod. The braiding is nonsymmetric, the shadow connection is KZ, and quantum-group representation theory emerges from genus-0 monodromy.

The Koszul dual is $\widehat{\mathfrak{sl}}_{2\,-k-4}$ (Feigin–Frenkel involution). The holographic datum is 3d Chern–Simons at level k; Drinfeld–Kohno (KZ monodromy = quantum-group braiding) follows from the genus-0 shadow connection.

The datum (3d Chern–Simons at level k):

$$\mathcal{H}(\widehat{\mathfrak{sl}}_{2k}) \;=\; \Big(\widehat{\mathfrak{sl}}_{2k},\; \widehat{\mathfrak{sl}}_{2-k-4},\; \mathcal{O}_q^{\mathrm{sh}},\; \frac{k\,\Omega}{z},\; \Theta_k,\; d - \kappa_k\,\omega_g\Big),$$

$r(z) = k\,\Omega/z$ (Casimir), $q = e^{i\pi/(k+2)}$, $\kappa_k = 3(k+2)/4$. At $k = -2$: $\kappa = 0$ (uncurved), the center $\mathfrak{z}(\widehat{\mathfrak{g}}_{-h^\vee})$ is the Feigin–Frenkel algebra of $\mathfrak{sl}_2$-opers. The monodromy of the KZ connection $\nabla_{\mathrm{KZ}} = d - (\hbar\,\Omega/z)\,dz$ (with $\hbar = 1/(k+2)$) is compared with the quantum-group R-matrix of $U_q(\mathfrak{sl}_2)$ by the affine Drinfeld–Kohno package; on the reduced evaluation comparison surface, $\mathcal{C}^{\mathrm{red}}_{\mathrm{line}}|_{\mathrm{eval}}$ is identified with the corresponding quantum-group side (Volume II), and the colored Jones polynomial is recovered from the monodromy representation classified by the bar complex, the Steinberg convolution of the chiral self-intersection on $\mathrm{FM}_k(\mathbb{C})$.

12.9. The Virasoro holographic datum

The Virasoro algebra is the first example of shadow class **M** (mixed, infinite shadow depth): the R-matrix $r_c(z) = (c/2)/z^3 + 2T/z$ has both a third-order pole (gravitational) and a first-order pole (stress-tensor exchange), and the quartic contact invariant $Q^{\text{contact}} = 10/[c(5c + 22)]$ is nonzero. The holographic datum packages three-dimensional quantum gravity on the half-space $\mathbb{R}_+ \times \mathbb{C}$: the Koszul dual is Vir_{26-c} (self-dual at $c = 13$, not $c = 26$), the shadow connection carries curvature proportional to $\kappa_c = c/2$, and the transferred coproduct is strictly primitive at all degrees: gravity does not produce particles, only braids them.

Virasoro at central charge c (3d gravity on $\mathbb{R}_+ \times \mathbb{C}$):

$$\begin{aligned}
\mathcal{H}(\mathrm{Vir}_c) &= \Big(\mathrm{Vir}_c,\ \mathrm{Vir}_{26-c},\ C_c,\ r_c(z),\ \Theta_c,\ \nabla_c^{\mathrm{hol}}\Big),\\
r_c(z) &= \frac{c/2}{z^3} + \frac{2T}{z}, \qquad \kappa_c = \frac{c}{2},\\
\nabla_{1,1}^{\mathrm{hol}} &= d - \kappa_c\,\omega_1 - \frac{10}{c(5c+22)}\,\delta_4 - \cdots .
\end{aligned}$$

The shadow connection has curvature $\kappa_c\,\omega_g$ at leading order. The quartic contact correction $\mathfrak{Q} = 10/[c(5c + 22)]$ is the first term sensitive to the planted-forest boundary of $\overline{\mathcal{M}}_{g,n}$. The genus tower has the formal structure of a perturbative expansion of 3d quantum gravity. The transferred coproduct Δ_z^{Vir} is strictly primitive at all degrees (Principle **??**): the ghost-number obstruction intrinsic to DS reduction kills every HPL tree correction. All gravitational dynamics resides in the collision residue $r_c(z) = (c/2)/z^3 + 2T/z$ and the infinite A_∞ product tower; the coproduct channel is identically zero. Gravity does not produce particles, only braids them.

12.10. The M2 brane example

The first genuinely interacting holographic example: boundary algebra $\mathcal{W}_{1+\infty}$ (infinitely many generators), Koszul dual a dg-shifted Yangian for $\mathfrak{psl}(2|2)$, line-operator category a quantum affine superalgebra. All three genus-2 shells are active, with the planted-forest shell receiving its first nontrivial contribution from Mok's log-FM strata.

The system arises from 3d $\mathcal{N} = 4$ ADHM gauge theory: N M2 branes probing $\mathbb{C}^2/\mathbb{Z}_k$.

The boundary algebra. The boundary chiral algebra $\mathcal{A}_{M2,N}$ at finite N is the VOA of the 3d $\mathcal{N} = 4$ theory on $\mathbb{R}_+ \times \mathbb{C}$. At large N, the boundary VOA $\mathcal{A}_{M2,\infty}$ is a $\mathcal{W}$-algebra of type $\mathcal{W}_{1+\infty}[c]$ at $c \sim N$: it has infinitely many strong generators of weights $1, 2, 3, \ldots$, with OPE coefficients that are rational functions of N.

The OPE structure contains both simple and higher poles. The stress tensor T (weight 2) has the Virasoro OPE with $c = c(N)$; the weight-3 primary W has a nonlinear WW OPE involving the quasi-primary $\Lambda = {:}TT{:} - \frac{3}{10}\partial^2 T$ at order $(z - w)^{-2}$; and the weight-s generators for $s \geq 4$ produce nested composites that make the shadow obstruction tower infinite.

The Koszul dual. The chiral Koszul dual $\mathcal{A}^!_{M2,\infty}$ is a dg-shifted Yangian for the Lie superalgebra $\mathfrak{psl}(2|2)$:

$$\mathcal{A}^!_{M2,\infty} \simeq Y_\hbar^{\mathrm{dg}}(\mathfrak{psl}(2|2)).$$

The R-matrix $R_{M2}(z) \in Y_\hbar^{\mathrm{dg}}(\mathfrak{psl}(2|2)) \otimes Y_\hbar^{\mathrm{dg}}(\mathfrak{psl}(2|2))[[z^{-1}]]$ is the collision residue $R_{M2}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{M2})$. The classical r-matrix $r_{M2}(z) = k\,\Omega_{\mathfrak{psl}(2|2)}/z$ satisfies the classical Yang–Baxter equation, which is the Arnold relation on $\mathrm{FM}_3(\mathbb{C})$ evaluated on the $\mathfrak{psl}(2|2)$-Casimir.

The line-operator category.

$$C_{M2} \simeq \mathrm{Rep}\big(U_q(\widehat{\mathfrak{psl}}(2|2))\big),$$

where $q = e^{2\pi i/(k+2)}$. The line operators are modules for the quantum affine superalgebra. The spectral braiding $R_{M2}(z)\colon L_1 \otimes L_2 \xrightarrow{\sim} L_2 \otimes L_1$ on pairs of line operators satisfies the Yang–Baxter equation: Stokes' theorem on $\mathrm{FM}_3(\mathbb{C})$ decomposes the boundary into three collision divisors, one for each crossing in the YBE.

The holographic datum. The full six-component holographic modular Koszul datum is:

$$\begin{aligned}
\mathcal{A} &= \mathcal{A}_{M2,\infty} \text{ (boundary } \mathcal{W}_{1+\infty}), & \mathcal{A}^! &= Y_\hbar^{\mathrm{dg}}(\mathfrak{psl}(2|2)),\\
C &= \mathrm{Rep}(U_q(\widehat{\mathfrak{psl}}(2|2))), & r(z) &= k\,\Omega_{\mathfrak{psl}(2|2)}/z,\\
\Theta_{M2} &= \sum_\Gamma \frac{W_\Gamma^{\log \mathrm{FM}}}{|\mathrm{Aut}(\Gamma)|} \otimes \Phi_\Gamma^{M2}, & \nabla^{\mathrm{hol}}_{g,n} &= d - \mathrm{Sh}_{g,n}(\Theta_{M2}).
\end{aligned}$$

Shadow obstruction tower analysis. The MC element Θ_{M2} has infinite shadow depth (class **M**: mixed modular):

- The cubic shadow $\mathfrak{C}_{M2} \neq 0$ because $\mathcal{A}_{M2,\infty}$ has a nonabelian current subalgebra.
- The quartic contact shadow $Q^{\mathrm{contact}}_{M2} \neq 0$ because $:TT: - \frac{3}{10}\partial^2 T$ appears in the WW OPE with coefficient $16/(22+5c(N))$.
- The quintic obstruction $o^{(5)}_{M2} = \{\mathfrak{C}_{M2}, Q^{\mathrm{contact}}_{M2}\}_H \neq 0$: the tower is infinite.

At genus 2, all three shells contribute: $\Theta^{(2)}_{\mathrm{loop}\circ\mathrm{loop}} \neq 0$, $\Theta^{(2)}_{\mathrm{sep}\circ\mathrm{loop}} \neq 0$, $\Theta^{(2)}_{\mathrm{pf}} \neq 0$. The planted-forest shell $\Theta^{(2)}_{\mathrm{pf}}$ is the first contribution from Mok's log-FM strata to the holographic genus expansion.

The modular shadow connection. The shadow connection $\nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{M2})$ is nontrivial: the shadow $\mathrm{Sh}_{g,n}$ is a matrix-valued (g,n)-dependent differential operator on the moduli space, encoding the genus-by-genus quantum corrections to the holographic correspondence. The genus-0 degree-n shadow $\mathrm{Sh}_{0,n}(\Theta_{M2})$ has the expected KZ/BPZ-type leading behavior for the $\mathfrak{psl}(2|2)$ benchmark surface; the genus-1 shadow carries the one-loop anomaly $\kappa(\mathcal{A}_{M2,\infty}) \cdot \omega_1$; higher-genus shadows encode the multi-loop holographic amplitudes.

The quartic contact invariant $Q^{\mathrm{contact}}_{M2}$ is the first obstruction to planarity in the holographic genus expansion: it is the first invariant that distinguishes the full string-theoretic amplitude from its planar $(N \to \infty)$ limit. The modular Koszul datum $\mathcal{H}(M2)$ packages the complete holographic correspondence (boundary algebra, bulk reconstruction, line operators, braiding, genus tower, and anomaly cancellation) into a single MC problem.

Summary. Sections 1–2 constructed the bar complex and its genus tower from the logarithmic propagator and the Arnold relation. Sections 3–4 identified these as projections of a single chain-level object: the modular convolution L_∞-algebra and its universal MC element $\Theta_{\mathcal{A}}$. Sections 5–6 extracted the full hierarchy of characteristic invariants from $\Theta_{\mathcal{A}}$ and classified chiral algebras by shadow depth. Section 7 tested the machine on every standard family. Section 8 lifted the shadow obstruction tower into number theory: Dirichlet–sewing series, Euler–Koszul classification, the two-variable L-object, and the arithmetic packet connection. Section 9 proved one-channel line concentration on the simple-Lie-symmetry locus, the scalar package on the proved scalar lane, and assembled the Koszulness programme. Sections 10–11 transported the machine to three dimensions: the bar differential became holomorphic factorization on the closed colour, the deconcatenation coproduct became topological factorization on the open colour, and the derived center pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ acquired an $\mathrm{SC}^{\mathrm{ch,top}}$-algebra structure whose PVA shadow governs the deformation quantization of Poisson vertex algebras to vertex algebras. Section 12 packaged the entire holographic correspondence into a six-component modular Koszul datum controlled by a single MC equation.

13. Completion and the frontier

The holographic datum of Section 12 packages a chiral algebra, its Koszul dual, the line-operator category, the spectral R-matrix, the universal MC element, and the shadow connection into a single algebraic object. The frontier consists of structural questions the packaging forces: the completed bar-cobar problem by infinite-type algebras, the analytic sewing problem by genus $g \geq 1$ curvature, the factorization-envelope problem by the passage from Lie conformal input to modular output, and the non-principal duality problem by Drinfeld–Sokolov reduction.

13.1. The Maurer–Cartan frontier

Master conjectures MC1 through MC4 are proved; MC5 is partially proved, with the analytic HS-sewing package established at all genera and the genuswise BV/BRST/bar identification conjectural. MC3 holds for all simple types on the evaluation-generated core via multiplicity-free ℓ-weights (Theorem **??**, Corollary **??**). The residual problem DK-4/5 (extension beyond evaluation modules) is downstream.

MC1 (PBW concentration). For every standard family (Kac–Moody, Virasoro, principal $\mathcal{W}_N$, $\beta\gamma$, lattice) and every genus $g \geq 1$, the PBW spectral sequence for the genus-g bar complex has concentrated E_∞ page: $E_\infty^{p,q}(g) = 0$ for $q \neq 0$.

The proof has two steps. First, genus-0 Koszulness supplies concentration of $E_1^{p,q}(g{=}0)$: the bar complex at genus 0 is formal, and the bar-cobar spectral sequence collapses at E_2 (this is the definition of chirally Koszul). Second, the genus-g enrichment $\mathcal{E}_g^{p,q}$ from the Hodge bundle is d_0^{PBW}-exact because it arises from the regular Eisenstein series, which is exact under the Poincaré residue map $\mathrm{Res}\colon H^{q+1}(\overline{\mathcal{M}}_{g,n}, \Omega^{q+1}_{\log}) \to H^q(\sigma_\Gamma, \Omega^q_{\log})$. The concentration is uniform in the level parameter k: the PBW filtration is independent of k, and the spectral sequence differentials are k-linear.

MC2 (universal $\Theta_{\mathcal{A}}$). The bar-intrinsic construction:

$$\Theta_{\mathcal{A}} \;:=\; D_{\mathcal{A}} - d_0 \;\in\; \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$$

is automatically MC because $D_{\mathcal{A}}^2 = 0$ (the total bar differential squares to zero). The MC equation $d_0\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ unpacks to $D_{\mathcal{A}}^2 - d_0^2 = 0$, which holds because both $D_{\mathcal{A}}^2 = 0$ and $d_0^2 = 0$.

Scalar saturation universality: for every one-parameter algebraic family $\mathcal{A}_k$ with rational OPE coefficients,

$$\Theta_{\mathcal{A}_k} \;=\; \kappa(\mathcal{A}_k) \cdot \eta_{\mathrm{univ}} \otimes \Lambda_{\mathrm{univ}}.$$

The higher-order terms in $H^2_{\mathrm{cyc,prim}}(\mathcal{A}_k)$ vanish. Proof: *Whitehead reduction* computes $H^2_{\mathrm{cyc,prim}}$ via a spectral sequence whose E_2 page is $H^2(\mathfrak{g}; \mathrm{Sym}^n V) = 0$ by the Whitehead lemma for reductive $\mathfrak{g}$ (the second cohomology of a semisimple Lie algebra with coefficients in a finite-dimensional module vanishes). *Algebraic semicontinuity*: the dimension function $k \mapsto \dim H^2_{\mathrm{cyc,prim}}(\mathcal{A}_k)$ is upper-semicontinuous, vanishes at $k = \infty$ (the free-field point), and therefore vanishes on a Zariski-dense open set containing all non-critical levels including admissible ones.

Ambient $D^2 = 0$: the full differential on the ambient planted-forest complex (including log-FM strata) satisfies $D^2 = 0$ by the normal-crossings result of Mok: the log-FM compactification $\mathrm{FM}_n(X|D)$ is a manifold with normal-crossings boundary, and the differential respects the stratification.

MC3 (thick generation). On the type-A Yangian module surface, the four-package MC3 problem reduces to a single remaining compact/completed extension packet, by the following four steps.

(a) *Chromatic filtration.* Stratify the category by the support of the q-character on the Grothendieck ring. The successive quotients $\mathrm{gr}^p_{\mathrm{chrom}}$ are controlled by products of prefundamental modules at chromatic level p.

(b) *Prefundamental Clebsch–Gordan.* The tensor product $V_n \otimes L^-$ decomposes as $\bigoplus_{j=0}^n L^-(n-2j)$ at the character level, a formal series identity $\mathrm{ch}(V_n) \cdot \mathrm{ch}(L^-) = \sum_{j=0}^n \mathrm{ch}(L^-(n-2j))$, verified to arbitrary depth. Every Verma character $\mathrm{ch}(M(\lambda))$ lies in the K_0-span of $\{[V_n] \cdot [L^-]\}$.

(c) *Completion formalism* (more precisely, Francis–Gaitsgory pro-nilpotent completion). Evaluation modules are compact objects (finite composition series in $\mathcal{O}^{\mathrm{sh}}$), so once an equivalence on the relevant compact core is known, it extends to the corresponding ind-completion.

(d) *Remaining DK compact/completion packet.* One still has to extend the Koszul duality functor from the evaluation-generated core to the compact shifted-prefundamental core, and then compare that compact-core equivalence with the relevant ind/pro-nilpotent completion.

The CG decomposition part is extended to all simple types (Theorem **??**): the multiplicity-free ℓ-weight property (Chari–Moura) replaces the minuscule hypothesis, making the block-separation argument uniform across all Dynkin types. The remaining post-CG completion packets are open beyond type A (Corollary **??**).

MC4 (completed bar-cobar). The *strong completion-tower theorem*: the strong filtration axiom

$$\mu_r(F^{i_1}, \dots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$$

yields an automatic *degree cutoff*: the r-ary operation maps $F^{i_1} \otimes \cdots \otimes F^{i_r}$ into $F^{i_1+\cdots+i_r}$, so operations with sufficiently many inputs land in arbitrarily deep filtration. For a fixed filtration degree N, only finitely many degrees $r \le N$ contribute to the F^N-piece of the bar-cobar resolution. This makes continuity and the Mittag-Leffler condition automatic (no $\lim^1$ obstruction).

The *completion closure* $\mathrm{CompCl}(\mathcal{F}_{\mathrm{ft}})$, the full subcategory of complete filtered objects with finite-type associated graded, carries a quasi-inverse bar-cobar homotopy equivalence. The key results:

(i) *Coefficient-stability criterion*: convergence of the completed bar-cobar reduces to finite matrix stabilisation on each graded piece.

(ii) *MC twisting closure*: the completion closure is stable under MC twisting (Θ-twisted objects remain in CompCl).

(iii) *Completed twisting representability*: twisting morphisms in the completed setting are representable by the completed bar-cobar resolution.

(iv) *Uniform PBW bridge*: MC1 (PBW concentration) implies MC4 (completed bar-cobar) for all standard families via the uniform filtration.

MC4 splits into two sub-problems:

- *MC4*$^{+}$ (positive towers: $\mathcal{W}_{1+\infty}$, affine Yangians, RTT algebras): solved by weight stabilisation. The weight grading provides a natural Cauchy criterion, and the completed bar-cobar converges because the bar complex has bounded weight in each filtration degree.
- *MC4*$^{\circ}$ (resonant towers: Virasoro, non-quadratic $\mathcal{W}_N$): reduced to a finite resonance problem by the resonance-filtered bar-cobar theorem. The *resonance rank* $\rho(\mathcal{A}) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ measures the depth of the resonance: $\rho(\mathcal{H}_k) = 0$, $\rho(\widehat{\mathfrak{g}}_k) = 0$, $\rho(\beta\gamma) = 0$, $\rho(\mathrm{Vir}_c) = 1$ (the weight-0 subspace of the bar complex has a one-dimensional resonance at the zero-mode of L_0).

The *resonance completion theorem* (Theorem ??): every positive-energy chiral algebra has $\rho < \infty$. The Virasoro case: Vir_{26-c} is the depth-zero resonance shadow, the image of the finite-dimensional resonance truncation R_{Vir}, not the final Koszul dual. The genuine $\mathcal{W}_\infty$-type dual requires the full resonance-filtered completion.

MC5 (genus expansion). Partially proved. The analytic sewing package is established at all genera, together with the algebraic genus tower; the full genuswise BV/BRST/bar identification remains conjectural. At genus 0, the algebraic BRST/bar comparison is proved (Theorem ??), while the tree-level amplitude pairing is conditional on Corollary ??; bar-cobar inversion $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is Theorem B. At genus 1, the curved bar complex with $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_1$ supplies the algebraic one-loop package, and the complementarity splitting $Q_1(\mathcal{A}) \oplus Q_1(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_{1,n}, Z(\mathcal{A}))$ is the genus-1 modular-invariance constraint.

At genus $g \geq 2$ the resolution proceeds in five steps. First, *inductive genus determination* (Theorem ??): the genus-g component $\Theta_{\mathcal{A}}^{(g)}$ is determined by $\Theta_{\mathcal{A}}^{(<g)}$ via the graph-sum recursion on $\overline{\mathcal{M}}_{g,n}$. Second, *2D convergence* (Proposition ??): there is no UV renormalization on curves: the Feynman integrals $W_\Gamma = \int_{\sigma_\Gamma} \omega_\Gamma$ converge absolutely on each stratum $\sigma_\Gamma \subset \overline{\mathcal{M}}_{g,n}$ without counterterm subtraction. Third, the *analytic-algebraic comparison* (Theorem ??): $D^{\mathrm{an}} = D^{\mathrm{alg}}$ on the algebraic core, so the formal graph-sum differential coincides with the analytic one. Fourth, *general HS-sewing* (Theorem ??): polynomial OPE growth together with subexponential sector growth implies the Hilbert–Schmidt sewing condition, yielding trace-class closed amplitudes at every genus. Fifth, the *Heisenberg one-particle sewing theorem* (Theorem ??) identifies the Heisenberg sewing amplitude with a Fredholm determinant, providing the decisive first-case verification. These results resolve the algebraic recursion and analytic convergence lanes, but not the full genuswise BV/BRST/bar identification. The correct receptacle remains the coderived category: curvature at $g \geq 1$ forces coderived/contraderived passage, not ordinary derived.

The completion programme: q-windows and completion entropy. The principal $\mathcal{W}_N$-algebra has $N-1$ generators. The inverse limit $\mathcal{W}_\infty = \varprojlim_N \mathcal{W}_N$ has infinitely many, one in each conformal weight $2, 3, 4, \ldots$

The bar-cobar quasi-isomorphism $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is proved for each finite-type stage $\mathcal{W}_N$. The passage to the inverse limit requires completed bar-cobar: a filtration $F^0\mathcal{A} \supset F^1\mathcal{A} \supset \cdots$ that makes the bar differential continuous and the Mittag–Leffler condition automatic. The strong filtration axiom $\mu_r(F^{i_1}, \ldots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$ achieves this in one stroke: on each quotient $\mathcal{A}_{\leq N}$ only degrees $r \leq N$ survive, so the completed differential is quotientwise finite and the quasi-isomorphism passes to the limit (Theorem ??). The completion closure $\mathrm{CompCl}(\mathcal{F}_{\mathrm{ft}})$, the class of all such inverse limits, carries a quasi-inverse bar-cobar equivalence stable under Maurer–Cartan twisting.

The completed bar complex decomposes into *finite windows*. Define the *reduced weight* $\rho(a) := \mathrm{wt}(a) - 1$ for a field of conformal weight $\mathrm{wt}(a)$, and extend additively to bar words: $\rho(sa_1|\cdots|sa_\ell) = \sum_i \rho(a_i)$. The *q-window* $K_q(\mathcal{A})$ is the span of bar words of reduced weight $\leq q$.

For Virasoro at $q = 2$: the window $K_2(\mathrm{Vir})$ has basis $\{sT,\ sT|sT\}$, one letter of reduced weight 1 and one two-letter word of total reduced weight 2. The bar differential on this two-dimensional complex is $d(sT|sT) = s(T \cdot T)$. The product $T \cdot T$ decomposes into the quasi-primary composite $\Lambda =: (TT) : -\frac{3}{10}\partial^2 T$, derivatives $\partial^2 T$, and the central term $\frac{c}{2} \cdot \mathbf{1}$. But Λ has conformal weight 4 (reduced weight $3 > 2$) and $\partial^2 T$ has reduced weight $3 > 2$, so both project to zero in K_2. Only the central term survives:

$$d(sT|sT)\big|_{K_2} \;=\; \frac{c}{2} \cdot s\mathbf{1}.$$

This is $\kappa(\mathrm{Vir}_c) = c/2$ emerging from a 2×1 matrix computation. The curvature is not an abstract scalar extracted from a theorem; it is the sole surviving coefficient of the bar differential on the smallest nontrivial window.

Each window $K_q(\mathcal{A})$ depends only on fields of conformal weight $\leq q + 1$: a letter in the window has $\mathrm{wt}(a) \leq q + 1$, and the bar differential cannot create higher-weight letters because the OPE estimate $\mathrm{wt}\,\mu_r(a_1, \ldots, a_r) \leq \sum_i \mathrm{wt}(a_i) - (r - 1)$ prevents weight increase. For the $\mathcal{W}_N$ tower, $K_q(\mathcal{W}_N)$ stabilises at $N = q + 1$: the infinite-generator problem is killed exactly, one window at a time.

The window dimensions grow exponentially. Define the *completion entropy* $h_K(\mathcal{A}) := \log(\rho_{\mathcal{A}}^{-1})$, where $\rho_{\mathcal{A}}$ is the radius of convergence of the primitive generating series $G_{\mathcal{A}}(t) = \frac{1-t}{t}(\chi_{\mathcal{A}}(t) - 1)$ extracted from the vacuum character $\chi_{\mathcal{A}}(q) = \mathrm{tr}_{\mathcal{A}}(q^{L_0})$ (the graded trace of the Virasoro zero mode). The entropy measures kinematic completion complexity: it depends on the character alone, not on OPE coefficients or curvature. Its values form the monotone ladder

$$h_K(\mathcal{W}_2) \approx 0.567 \;<\; h_K(\mathcal{W}_3) \approx 0.772 \;<\; h_K(\mathcal{W}_4) \approx 0.835 \;<\; \cdots \;<\; h_K(\mathcal{W}_\infty) \approx 0.872,$$

governed at the limit by the MacMahon plane-partition generating function $\chi_{\mathcal{W}_\infty}(q) = \prod_{n\geq 2}(1 - q^n)^{-(n-1)}$.

13.2. The analytic frontier

Curvature forces this direction. At genus $g \geq 1$ the bar complex satisfies $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g \neq 0$: the ordinary derived category is empty, and the correct receptacle is the coderived category. But the coderived passage requires completed, Hilbert-space-valued structures, not the bare algebraic OPE data of genus zero. The analytic frontier asks: when does the algebraic skeleton extend to a convergent sewing amplitude?

The bare VOA is a dense algebraic skeleton. The actual object for partition functions and sewing is the *sewing envelope* $\mathcal{A}^{\mathrm{sew}}$: the Hausdorff completion of $\mathcal{A}$ for the topology defined by all-amplitude seminorms

$$\|a\|_{\mathrm{amp}} \;:=\; \sup_{v\in\mathcal{V}} \langle v, \pi(a)v\rangle$$

over suitable families of states $\mathcal{V}$. The algebraic OPE data embed densely in $\mathcal{A}^{\mathrm{sew}}$; the closure provides the convergent sewing amplitudes needed for genus $g \geq 1$.

The *HS-sewing condition*: if the multiplication operators $m^c_{a,b}$ are Hilbert–Schmidt in the sense that

$$\sum_{a,b,c\geq 0} q^{a+b+c}\|m^c_{a,b}\|^2_{\mathrm{HS}} \;<\; \infty$$

for $|q| < 1$, then the closed amplitudes $\mathrm{Tr}(\pi(a) \cdot q^{L_0})$ are trace-class and the sewing operation $\sum_n \langle v_n, \pi(a) w_n\rangle$ converges absolutely. The condition is a Hilbert–Schmidt bound on the OPE coefficients graded by conformal weight.

The *analytic bar coalgebra* $B^{\mathrm{an}}(\mathcal{A})$ is the graph-norm closure of the algebraic bar coalgebra: each bar element $s^{-1}a_1|\cdots|s^{-1}a_n$ is assigned the norm $\prod_i \|a_i\|_{\mathrm{amp}}$, and B^{an} is the completion in this norm. An *analytic Koszul pair* is a pair $(\mathcal{A}, \mathcal{A}^!)$ such that the analytic bar-cobar adjunction restricts to a Quillen equivalence on the sewing-complete categories.

Four targets (two proved, two conjectural):

(a) *Heisenberg sewing* (the decisive first case): the sewing envelope of $\mathcal{H}_k$ is the ind-Hilbert completion $\mathrm{Sym}\ A_2(D) \cong \text{ind-Hilb}(\mathcal{H}_k)$ (Moriwaki 2026). The ind-Hilbert structure is the conformally flat factorization homology of $\mathcal{H}_k$ valued in $\mathsf{IndHilb}$.

(b) *Lattice sewing*: the sewing envelope of V_Λ is the vertex-algebraic completion by lattice theta functions, with the theta-function series converging absolutely in the sewing-amplitude topology.

(c) *Analytic realisation criterion*: a criterion for when the algebraic Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ extends to an analytic one. The expected criterion: the HS-sewing condition on $\mathcal{A}$ plus the dual HS condition on $\mathcal{A}^!$ should suffice.

(d) *Boundary bar duality*: the analytic bar of the boundary chiral algebra equals the topological bar of the bulk factorization algebra: $B^{\mathrm{an}}(\mathcal{A}_\partial) \simeq B^{\mathrm{top}}(\mathrm{Fact}_T^{\mathrm{bulk}})$, identifying the boundary sewing data with the bulk topological data.

13.3. Factorization envelopes and the modular Koszul datum

The bar construction starts from a chiral algebra, but the natural input in every geometric construction is a Lie conformal algebra. The factorization envelope bridges this gap: it is the functor that produces a chiral algebra from Lie conformal data, and the programme asks whether the entire modular Koszul package can be constructed functorially at the Lie conformal level, without passing through the vertex algebra. The target is a universal modular factorization envelope $U_X^{\mathrm{mod}}(L)$ that is left adjoint to extracting primitive currents.

A *cyclically admissible* Lie conformal algebra L (a $\mathbb{C}[\partial]$-module equipped with a λ-bracket $\{a_\lambda b\}$ satisfying sesquilinearity, skew-symmetry, and the Jacobi identity) has four properties: a conformal grading $L = \bigoplus_{\Delta \geq 0} L_\Delta$ by conformal weight, a decreasing filtration $F^0 L \supset F^1 L \supset \cdots$ compatible with the λ-bracket, bounded poles (the λ-bracket $\{a_\lambda b\}$ is polynomial in λ of degree bounded by conformal weights), and an invariant symmetric bilinear pairing $\langle -, - \rangle \colon L \otimes L \to \mathbb{C}$ compatible with the conformal structure.

From a cyclically admissible L, one constructs the *modular Koszul datum*:

$$\Pi_X(L) \;=\; \big(\mathrm{Fact}_X(L),\ \bar{B}_X(L),\ \Theta_L,\ \mathcal{L}_L,\ (V_L^{\mathrm{br}}, T_L^{\mathrm{br}}),\ R_4^{\mathrm{mod}}(L)\big):$$

the factorization algebra (the Nishinaka factorization envelope), the bar coalgebra, the universal MC class, the line-operator datum, the branch-line data, and the quartic resonance class. Each component is functorially determined by L.

The target: the *universal modular factorization envelope* $U_X^{\mathrm{mod}}(L)$ as left adjoint of the modular primitive-current functor $\mathrm{Prim}^{\mathrm{mod}}$:

$$\mathrm{Hom}_{\mathsf{Fact}^{\mathrm{mod}}}\big(U_X^{\mathrm{mod}}(L),\ \mathcal{F}\big) \;\cong\; \mathrm{Hom}_{\mathsf{LCA}^{\mathrm{cyc}}}\big(L,\ \mathrm{Prim}^{\mathrm{mod}}(\mathcal{F})\big).$$

The envelope-shadow functor $\Theta^{\mathrm{env}}_{\leq r}(R)$ extracts finite-order shadows from the envelope data: the universal property of U^{mod}_X implies that $\Theta_L = \Theta^{\mathrm{env}}_{\leq\infty}(L)$ is the universal MC class of the modular Koszul datum.

For direct sums $L = L_1 \oplus L_2$ with vanishing mixed OPE: $\kappa(L) = \kappa(L_1) + \kappa(L_2)$ (additivity), $T^{\mathrm{br}}_L = T^{\mathrm{br}}_{L_1} \oplus T^{\mathrm{br}}_{L_2}$ (direct sum), $\Delta_L = \Delta_{L_1} \cdot \Delta_{L_2}$ (multiplicativity), $R^{\mathrm{mod}}_4(L) = R^{\mathrm{mod}}_4(L_1) + R^{\mathrm{mod}}_4(L_2)$ (additivity). BRST reduction on total modular Koszul datums: the current-plus-ghost package descends to $\Pi_X(\mathcal{W}_N)$, hence $\Theta_{\mathcal{W}_N} = H^{\circ}_{Q_{\mathrm{DS}}}(\Theta_{V_k(\mathfrak{g})\otimes F_{\mathrm{gh}}})$: the shadow obstruction tower of the W-algebra is the Drinfeld–Sokolov reduction of the full current-plus-ghost tower, not of the bare affine tower alone.

13.4. Non-principal W-algebra duality

Drinfeld–Sokolov reduction connects the affine landscape to the W-algebra landscape: it takes $\widehat{\mathfrak{g}}_k$ and a nilpotent element f and produces the W-algebra $\mathcal{W}_k(\mathfrak{g}, f)$. For principal f, the reduction functor commutes with bar-cobar duality by the Feigin–Frenkel theorem, and the modular Koszul datum descends. For non-principal f, this commutation is unproved, and the frontier begins. The theory demands this extension because the non-principal W-algebras include the physically central cases ($\mathcal{W}_k(\mathfrak{sl}_N, f_{\mathrm{sub}})$ controls $\mathcal{N} = 2$ superconformal theories) and the mathematically central ones (the Feigin–Tipunin algebras at admissible levels).

The hook-type nilpotent orbit in type A is the first non-principal corridor for Koszul duality, conditional on DS-bar compatibility (Theorem ??). For a partition $\lambda = (n - p, 1^p)$ (hook type) in $\mathfrak{sl}_N$, the W-algebra $\mathcal{W}_k(\mathfrak{sl}_N, f_\lambda)$ is defined by quantum Drinfeld–Sokolov reduction with respect to the nilpotent element f_λ. The duality:

$$\big(\mathcal{W}_k(\mathfrak{sl}_N, f_\lambda)\big)^! \;\simeq\; \mathcal{W}_{k^\vee_\lambda}(\mathfrak{sl}_N, f_{\lambda^t}),$$

where λ^t is the transpose partition and $k^\vee_\lambda$ is the dual level (a rational function of k depending on λ). The proof: for hook-type partitions, the DS reduction functor commutes with bar-cobar duality by the results of Fehily and Creutzig–Linshaw–Nakatsuka–Sato (2023–2025).

The *transport propagation lemma*: hook-type seeds combined with edge-compatibility (the property that the DS reduction functor preserves the collision structure at each FM boundary stratum) propagate duality results along edges of the nilpotent Hasse diagram in type A. The *type-A transport-to-transpose conjecture*: for every partition λ of N, $(\mathcal{W}_k(\mathfrak{sl}_N, f_\lambda))^! \simeq \mathcal{W}_{k^\vee_\lambda}(\mathfrak{sl}_N, f_{\lambda^t})$.

The general obstruction is not defining the DS reduction (which exists for arbitrary nilpotent orbits by Kac–Roan–Wakimoto) but proving that DS reduction *commutes with the bar-cobar adjunction*: the natural map $\mathrm{DS}(\bar{B}(\widehat{\mathfrak{g}}_k)) \to \bar{B}(\mathrm{DS}(\widehat{\mathfrak{g}}_k))$ must be a quasi-isomorphism. For principal reduction, this follows from the Feigin–Frenkel theorem; for hook-type in type A, it is conditional on hook-wide DS-bar compatibility (Theorem ??); for arbitrary nilpotent orbits, it is the frontier.

13.5. The boxed machine

The entire construction can be stated in a single pipeline.

$$\boxed{\text{homotopy chiral input } \mathcal{A} \;+\; \text{log-FM cutting cooperad } C^!_{\mathrm{ch}} \;+\; \text{modular convolution } L_\infty \;\Longrightarrow\; \Theta_{\mathcal{A}},\; \mathcal{T}^{\mathrm{mod}}_{\Theta_{\mathcal{A}}},\; \kappa,\; \Delta_{\mathcal{A}},\; \mathfrak{R}^{\mathrm{mo}}_4}$$

The modular convolution L_∞-algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is the single organising structure. It carries a natural modular L_∞ structure from the Feynman transform of the modular operad; the dg Lie algebra $\mathrm{Conv}_{\mathrm{str}}(C^!_{\mathrm{ch}}, \mathcal{P}^{\mathrm{ch}})$ is its strict model. The universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$, satisfying

$$D \cdot \Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] \;=\; 0,$$

is proved: it is the bar-intrinsic extraction $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$, which is MC because $D^2_{\mathcal{A}} = 0$.

The shadow obstruction tower consists of finite-order projections:

$$\begin{array}{lll}
\Theta^{\le 2}_{\mathcal{A}} : & \kappa(\mathcal{A}) & \text{(modular characteristic, scalar curvature),}\\
\Theta^{\le 3}_{\mathcal{A}} : & \mathfrak{C}(\mathcal{A}) & \text{(cubic shadow, Lie structure),}\\
\Theta^{\le 4}_{\mathcal{A}} : & \mathfrak{Q}(\mathcal{A}) & \text{(quartic resonance class, contact structure),}\\
\Theta^{\le r}_{\mathcal{A}} : & \mathrm{Sh}_r(\mathcal{A}) & \text{(degree-}r\text{ shadow),}\\
\Theta_{\mathcal{A}} : & \varprojlim_r \Theta^{\le r}_{\mathcal{A}} & \text{(the universal MC element).}
\end{array}$$

Theorems A–D+H and the genus expansion are proved projections of the scalar level κ. The holographic datum $\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^!, C, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$ is the packaging of $\Theta_{\mathcal{A}}$ and its projections into the data of a 3d holomorphic-topological field theory. The weight filtration on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ controls the extension tower; each finite truncation $\Theta^{\le r}_{\mathcal{A}}$ is constructive and does not require the full all-genera modular envelope. The modular tangent complex extracts Chern–Weil invariants. The genus spectral sequence controls the lift across genera. The standard landscape is computed.

Principle 0.0.7 (The four-test interface)**.** *The modular Koszul machine has a complete algebraic-geometric interface with $\overline{\mathcal{M}}_{g,n}$, consisting of four independent proved tests:*

(i) $D^2_{\mathcal{A}} = 0$*: the bar differential squares to zero at all genera and degrees* (*Theorems* **??**, **??**).

(ii) $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ *for uniform-weight algebras at all genera, and unconditionally at genus* 1 *for all families: the genus-g obstruction factors as modular characteristic times Hodge class* (*Theorem* **??**).

(iii) $Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) \;=\; H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$*: complementarity assembles the Koszul pair into a Lagrangian decomposition* (*Theorem* **??**).

(iv) *Sewing: genus-g amplitudes converge from lower-genus data* (*Theorem* **??**).

Each test captures a distinct aspect: (i) *is purely algebraic and requires no Koszulness hypothesis;* (ii) *identifies the tautological class but does not involve $\mathcal{A}^!$;* (iii) *requires the Koszul pair but is genus-by-genus;* (iv) *is analytic and independent of the Hodge-class identification. Any three can hold with the fourth failing. Three regions lie outside: multi-weight algebras at $g \ge 2$* (*Open Problem* **??***, resolved negatively: cross-channel correction nonzero*)*, BV/BRST = bar at higher genus* (*Conjecture* **??**)*, and non-perturbative completion. See Remark* **??** *for the full discussion.*

The holographic frontier. The holographic datum $\mathcal{H}(T)$ is a boundary object. Its lift to a genuine 3d bulk theory is controlled by the *relative holographic deformation complex* $\mathfrak{h}^{\Theta}_T := \mathrm{fib}(\mathfrak{g}^{\mathrm{SC}}_T \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})_{\alpha_T}$ (Definition **??**): lifts of $\Theta_{\mathcal{A}}$ to full bulk/boundary/line data are exactly MC elements of the twisted fibre (Proposition **??**). The graded obstruction tower $[\mathfrak{o}_r(\beta)] \in H^2(\mathrm{gr}^F_r\, \mathfrak{h}^{\Theta}_T)$ turns the holographic frontier into a computable cohomology calculus. At depth 4, the *relative quartic obstruction* $\mathfrak{R}^{\mathrm{rel}}_4(T)$ is the first invariant detecting when a boundary datum cannot arise from a planar holographic realization (Conjecture **??**). The *Loop–Connes principle* $\beta^*_{\mathrm{der}}(\Delta_{\mathrm{open}}) = \Lambda_P$ (Conjecture **??**): closed genus creation is the image of the open BV operator under the derived-center map; loops in the bulk are traces on the boundary.

Cyclic trace, ribbon structure, and $1/N$. A bare modular graph sum does not define a literal 't Hooft expansion (Lemma ??). Cyclic trace structure on the open E_1-chiral algebra resolves this: cyclicity enables ribbon thickening (Theorem ??), and a matrix realisation gives the exact weight $N^{\chi(\Sigma_\Gamma)}$ (Theorem ??). The three levels of language (plain modular E_1 (genus expansion), cyclic traced modular E_1 ('t Hooft expansion), ribbonised modular Swiss-cheese (holographic open/closed 't Hooft language)) are made precise in the E_1 modular Koszul chapter (Chapter ??).

13.6. Admissible levels and logarithmic CFT

At admissible levels $k = -h^\vee + p/q$ (coprime $p \geq h^\vee$, $q \geq 1$), the Zhu algebra $A(\mathcal{V}_k(\mathfrak{g}))$ is semisimple but the vertex algebra itself is not simple: there is a maximal proper ideal J_k with $L_k(\mathfrak{g}) = \mathcal{V}_k(\mathfrak{g})/J_k$ the simple quotient. The bar complex of $\mathcal{V}_k$ acquires a periodic CDG structure indexed by the q-th roots of unity: the curvature $m_0 = \kappa \cdot \omega_g$ factors through the cyclotomic polynomial $\Phi_q(\zeta)$, and the weight filtration has a finite plateau at level $N = q$. This is the algebraic door to logarithmic conformal field theory on the degenerate admissible lane, where the simple quotient $L_k(\mathfrak{g})$ has a logarithmic (non-semisimple) module category. On the non-degenerate admissible lane the simple quotient is rational, while the bar-side admissible analysis records vacuum null vectors and nilradical data as the obstruction surfaces across the broader admissible picture. One still has weightwise finite-page character-theoretic degeneration on the non-degenerate lane, whereas an identification of the resulting bar cohomology with logarithmic extensions is not proved here.

13.7. The relative Feynman transform

The modular bar coalgebra $\bar{B}_{\mathrm{mod}}(\mathcal{A})$ is naturally an algebra over the relative Feynman transform $\mathrm{FT}_{\mathsf{Com}_{\mathrm{mod}}/\mathsf{SC}^{\mathrm{ch,top}}}$. This is the algebraic skeleton shared by the factorization approach (§10.8) and the operadic approach (§3): the factorization structure on $\mathrm{Ran}(\Sigma_g) \times \mathrm{Ran}(\mathbb{R})$ over $\overline{\mathcal{M}}_g$ produces it via families of factorizable $\mathcal{D}$-modules; the operadic structure on $\mathsf{SC}^{\mathrm{ch,top}}_{\mathrm{mod}}$ produces it via formal completion at collision points.

Recognition. $\bar{B}_{\mathrm{mod}}(\mathcal{A})$ is naturally a $\mathrm{FT}_{\mathsf{Com}_{\mathrm{mod}}/\mathsf{SC}^{\mathrm{ch,top}}}$-algebra: the tree-level part D_0 is the E_1-chiral coassociative coalgebra structure, and the genus-raising part D_1 is the $\mathsf{Com}_{\mathrm{mod}}$-modular structure.

Homotopy-involutivity. $\mathrm{FT}^2_{\mathsf{Com}_{\mathrm{mod}}/\mathsf{SC}^{\mathrm{ch,top}}} \simeq \mathrm{id}$. The proof proceeds in three steps: genus-0 involutivity from operadic homotopy-Koszulity (the homotopy-Koszulity theorem, Vol II), closed-colour modular involutivity from the Feynman involution (Theorem ??), and assembly via the genus-filtration spectral sequence.

Three routes unified. Route B (factorization, §10.8) provides the global truth: factorization on $\mathrm{Ran}(\Sigma_g) \times \mathrm{Ran}(\mathbb{R})$ over $\overline{\mathcal{M}}_g$. Route A (operadic, §3) provides the local approximation: $\mathsf{SC}^{\mathrm{ch,top}}_{\mathrm{mod}}$ at formal completions. Route C (this section) provides the algebraic skeleton: $\mathrm{FT}_{\mathsf{Com}_{\mathrm{mod}}/\mathsf{SC}^{\mathrm{ch,top}}}$ as the common abstraction. The factorization structure projects to the relative Feynman transform by forgetting $\mathcal{D}$-module data; the operadic structure injects into it by formal completion. The relative Feynman transform is the categorical image.

13.8. The computational infrastructure

Every formula in this preface has been verified by an independent symbolic computation layer comprising tens of thousands of tests across several hundred modules, covering the shadow obstruction tower through degree 32, the standard landscape for all seven families, the genus-2 and genus-3 planted-forest formulas, the DS-shadow cascade (degrees 2–8 for $N = 2, \dots, 5$), the W_3 multi-generator shadow metric on the (T, W)-plane, theBöcherer bridge at genus 2, the Pixton shadow bridge, the entanglement programme (Page curve, holographic codes), the non-simply-laced bar computations ($B_2/C_2/G_2/F_4$), minimal model bar and fusion rules, the dressed propagator engine (scalar complementarity, R-matrix $R_1 = 1/12$), and the constrained Epstein zeta for lattice VOAs. The computation is adversarial: tests encode cross-family consistency checks (additivity, complementarity, anti-symmetry) that cannot be fooled by a single hardcoded value. Volume III extends the engine to Calabi–Yau geometry: forty-two CY/BKM compute engines with 5,700+ tests cover the $K3 \times E$ programme (shadow–Siegel gap, BKM root multiplicities, moonshine multiplier, Schottky barrier), toric CY_3 CoHA integrality, and the scalar bar-Euler face of the automorphic correction. In the toric point $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$; the $\mathcal{W}_{1+\infty}$ algebra enters after Drinfeld double/centre/Fock/evaluation.

Volume II develops the 3d theory in three new chapters: the factorization Swiss-cheese algebra (the global truth on $\mathrm{Ran}(\Sigma_g) \times \mathrm{Ran}(\mathbb{R})$), the modular Swiss-cheese operad (the local approximation at formal completions), and the relative Feynman transform (the algebraic skeleton connecting both). These chapters, together with the 3d gravity chapter (six movements: from the Koszul triangle to anomaly-completed MC recursion at all genera), the affine half-space BV theorem (from linear to nonlinear via the doubling principle), and eleven additional computational engines, constitute the Volume II programme.

13.9. The organizing ambient problem

Everything proved in this monograph is a finite-order projection of the universal MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$. Everything open is a question about its ambient:

$$\boxed{\textit{Characterise the modular convolution algebra } \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \textit{ as a geometric object.}}$$

Precisely: the modular convolution algebra is the endomorphism algebra of the Lagrangian embedding $\mathcal{L}_{\mathcal{A}} \hookrightarrow \mathcal{M}_{\mathrm{vac}}(\mathcal{A})$ in the (-2)-shifted symplectic stack of vacua. Volume I computes the Taylor jets of this embedding (the shadow obstruction tower). Volume II identifies the local composition law (the Swiss-cheese operad). What remains is the global geometry of the stack itself: the derived symplectic structure, the moduli of Lagrangians, and the higher-categorical descent that would make the construction independent of choices.

The ~150 open conjectures in these two volumes are projections of this single question:

- the D-module purity conjecture asks whether the Lagrangian is pure (characterisation (xii) of Koszulness);
- the non-principal W-duality programme asks how the Lagrangian transforms under Drinfeld–Sokolov reduction at arbitrary nilpotent orbits;
- the analytic completion programme asks when the algebraic Lagrangian extends to a convergent one (the sewing envelope);
- the factorization envelope programme asks for the universal construction of the Lagrangian from Lie conformal data alone;

- the holographic completion question asks whether the genus-0 Lagrangian extends over $\overline{\mathcal{M}}_{g,n}$ (the modular programme of Volume II);
- the Calabi–Yau quantum group programme asks how the automorphic correction of a generalized Borcherds–Kac–Moody superalgebra is recovered from the chiral shadow of a CY category: the CY_2 lane is E_2-chiral, the CY_3 BPS/CoHA lane is E_1-chiral before passing to centre constructions, and the chain-level $\mathbb{S}^3$-framing remains the explicit CY_3 proof obligation (Volume III).

Each is a different facet of the geometry of $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$, and none requires going beyond the framework already established.

The entire structure unfolds from a single datum: the logarithmic 1-form $\eta = d\log(z_1 - z_2)$. At n points it became a differential on the Fulton–MacPherson compactification; across all genera it became the total bar differential $D_{\mathcal{A}}$; in the modular convolution algebra it became the Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$. One 1-form, one relation (Arnold), one extraction (residues): a universal invariant of chiral algebras that is algebraic, computable, and complete.